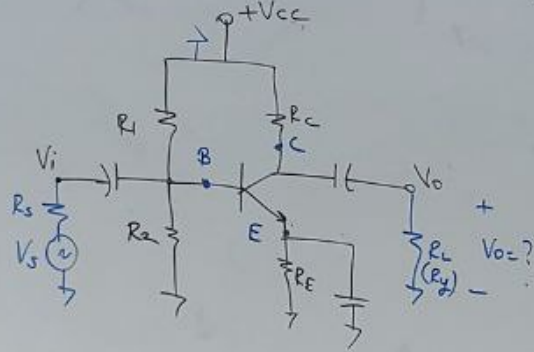
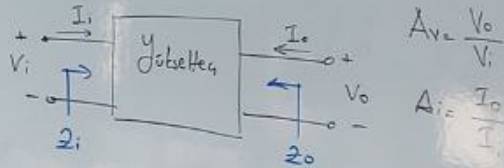
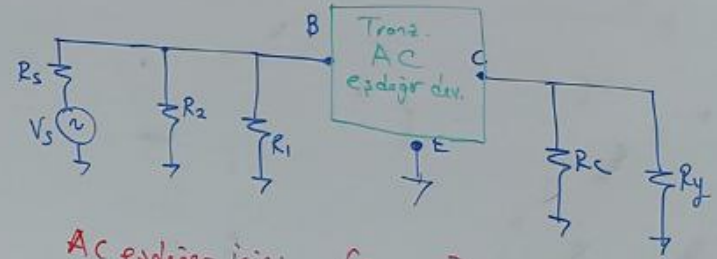


TRANSİSTÖRLERİN AC EŞDEĞER DEVRE MODELLERİ

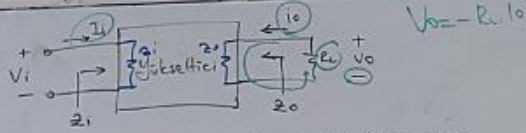
1. BJT Küçük Sinyal Analizi



Orta frk. (~kHz) C, kısa devre
Ger.kayn. kısa devre (toprak pot.)
Akım kayn. açık devre



AC eşdeğer için;
 r_e
+hibrit
 π } modelleri kullanılır.



$$V_{ce} = -R_c I_c$$

$$A_i \left(\frac{I_o}{I_i}, \text{ akım kazancı} \right):$$

$$I_i = \frac{V_i}{Z_i} \quad k = \frac{-V_o}{R_c}$$

$$A_i = \frac{I_o}{I_i} = \frac{V_o}{R_c} \cdot \frac{Z_i}{V_i}$$

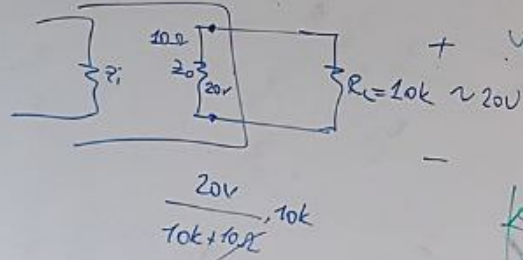
$$A_i = -A_v \cdot \frac{Z_i}{R_c}$$

$$A_i = -A_v \cdot \frac{Z_i}{R_c}$$

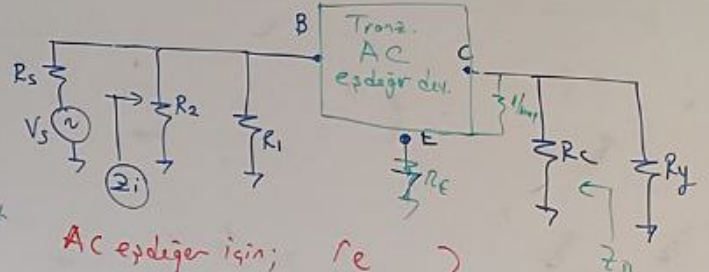
$$A_v = -3$$

$$A_v = +15$$

Faz farkı: 0 veya 180° dir.



Orta frk. (~1kHz) C, kısa devre
Ger. kayn. kısa devre (toprak pot.)
Akım kayn. açık devre



AC eşdeğer için;

re
Hibrit
π

modelleri kullanılır.

Z_i (giriş empedansı):
* Bir kez belirlendikten sonra değişmez.
* $\leq 100\text{kHz}$ için resistiftir.
* Ω ile $M\Omega$ arasıdır.

Z_o (çıkış empedansı):
* Giriş sinyali sıfırlanmış durumda çıkış portundan içeriye bakıldığında görülen empedanstır.
* $\leq 100\text{kHz}$ için resistif.
* Ω ile $M\Omega$ arasıdır.

$$A_v \left(\frac{V_o}{V_i}, \text{ gerilim kazancı} \right):$$

$$A_{v_{AC}} = \frac{V_o}{V_i}$$

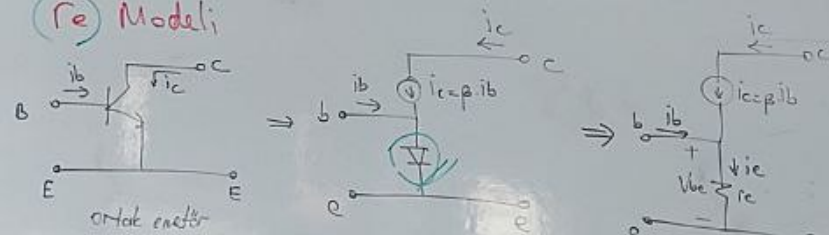
* Kısa ya yük bağlı değilse çıkış gerilim olarak ifade edilir $A_{v_{AC}}$

$$A_{v_{AC}} > A_v$$



BJT Küçük Sinyal Modelleri

r_e Modeli



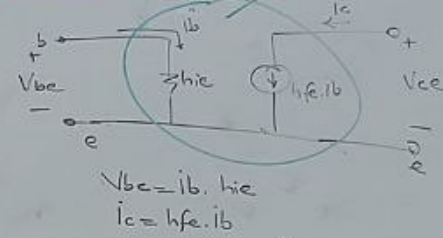
$$r_e = \frac{26 \text{ mV}}{I_c}$$

$$Z_i = \beta r_e \quad Z_o = \infty$$

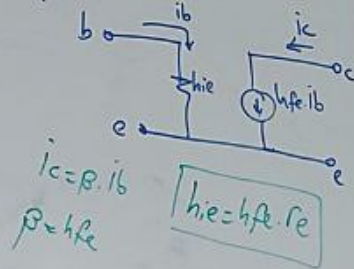
$$A_v = -\frac{R_c}{r_e} \quad A_i = \beta$$

r_e Hibrid Modeli

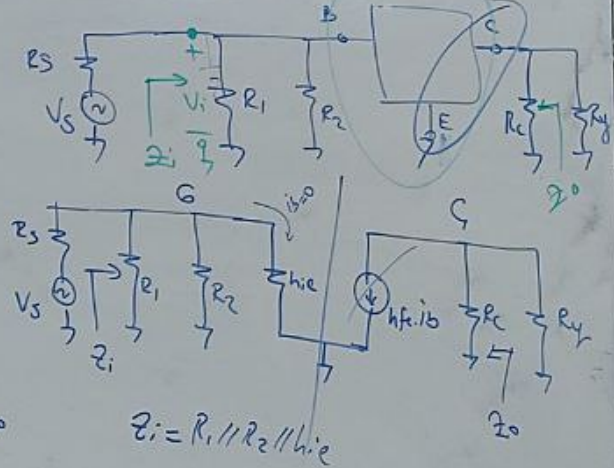
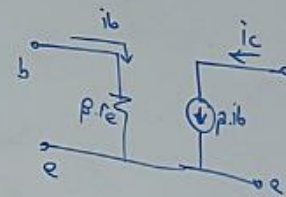
Yaklaşık Model



Yaklaşık Hibrid Model

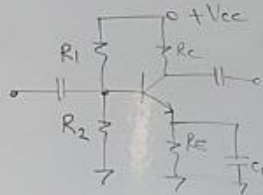


r_e Modeli



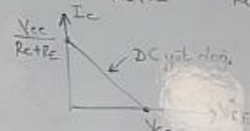


DC Yük Doğrusu;

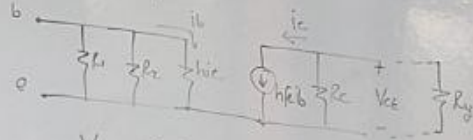


$$I_c = \frac{V_{cc} - V_{ce}}{R_c + R_e}$$

$$= -\frac{1}{R_c + R_e} V_{ce} + \frac{V_{cc}}{R_c + R_e}$$



AC Yük Doğrusu;

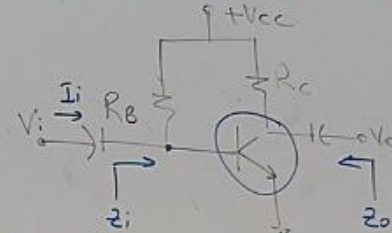


$$V_{ce} = -R_c i_c$$

$$i_c = -\frac{1}{R_c} V_{ce}$$

2. BJT'li Yükseltilen Devrelerinin Analizi

2.1 Ortak Emitörli Sabit Kutuplanmalı Yüks.



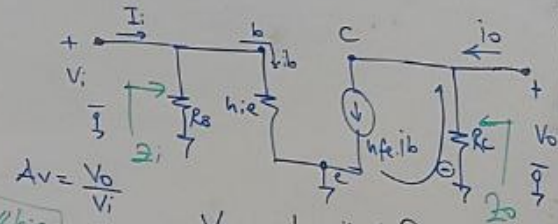
$$I_o = h_{fe} i_b$$

$$i_b = I_i \cdot \frac{R_8}{R_8 + h_{ie}}$$

$$\Rightarrow I_o = h_{fe} \cdot I_i \cdot \frac{R_8}{R_8 + h_{ie}}$$

$$A_i = \frac{I_o}{I_i} = \frac{h_{fe} \cdot R_8}{R_8 + h_{ie}}$$

AC



$$A_v = \frac{V_o}{V_i}$$

$$V_o = -h_{fe} i_b \cdot R_c$$

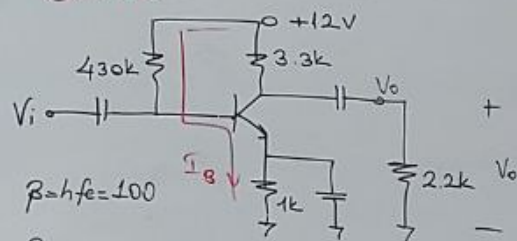
$$V_i = i_b \cdot h_{ie}$$

$$A_v = \frac{V_o}{V_i} = -\frac{h_{fe} i_b \cdot R_c}{i_b \cdot h_{ie}}$$

$$A_v = -\frac{h_{fe} \cdot R_c}{h_{ie}}$$



Örnek:



$A_i, A_v, Z_i, Z_o = ?$

$$h_{ie} = h_{fe} r_e \quad r_e = \frac{26mV}{I_E}$$

h_{ie} yoksa (verilmemişse)
DC analizi yap!

$$12V = 430k \cdot I_B + 0.7 + 10k \cdot I_B \cdot 1k$$

$$I_B = 2.28 \mu A$$

$$I_E = 2.15 mA$$

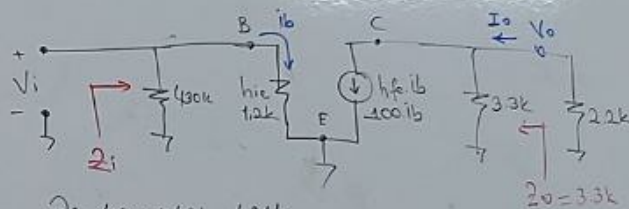
$$I_E = I_B + I_C$$

$$I_E = (\beta + 1) I_B$$

$$r_e = \frac{26mV}{2.15mA} \rightarrow r_e = 12.1 \Omega$$

$$h_{ie} = h_{fe} \cdot r_e = 100 \cdot (12.1 \Omega)$$

$$h_{ie} = 1.2k \Omega$$



$$Z_i = 430k \parallel 1.2k = 1.21k$$

$$\left. \begin{aligned} V_o &= -h_{fe} \cdot i_b (R_C \parallel R_L) \\ V_i &= i_b \cdot h_{ie} \end{aligned} \right\} \Rightarrow A_v = \frac{-h_{fe} (R_C \parallel R_L)}{h_{ie}} = -\frac{100 \cdot (3.3k \parallel 2.2k)}{1.2k}$$

$$A_v = -108.20$$

$$A_i = -A_v \cdot \frac{Z_i}{R_L} = -(-108.20) \cdot \frac{1.21k}{2.2k} \Rightarrow A_i = 59.50$$

Alternatif olarak;

$$I_o = h_{fe} \cdot i_b + \frac{V_o}{R_L}$$

$$I_o = h_{fe} \cdot i_b - \frac{h_{fe} \cdot i_b (R_C \parallel R_L)}{R_L}$$

$$I_i = i_b + \frac{V_i}{R_s} = i_b + \frac{i_b \cdot h_{ie}}{R_s}$$

$$A_i = \frac{I_o}{I_i} = \frac{h_{fe} \cdot R_C \cdot R_L}{(R_C \parallel R_L) \cdot (R_s + h_{ie})}$$

Bu formül ile A_i hesaplanırsa;

$$A_i = \frac{100 \cdot (3.3k) \cdot (430k)}{(3.3k \parallel 2.2k) \cdot (430k + 1.2k)}$$

$$A_i = 59.83$$

b) $h_{oe} = 20 \mu S$ olarak verilseydi;

$$r_o = \frac{1}{h_{oe}} = \frac{1}{20 \mu S} = 50 k$$

$$Z_o = 50 k // 3.3 k = 3.09 k$$

$$A_v = \frac{V_o}{V_i} \quad \left\{ \begin{array}{l} V_o = -h_{fe} \cdot i_b \cdot (R_c // R_y // 1/h_{oe}) \\ V_i = h_{ie} \cdot i_b \end{array} \right.$$

$$A_v = \frac{V_o}{V_i} = - \frac{h_{fe} (R_c // R_y // 1/h_{oe})}{h_{ie}} = -105.33$$

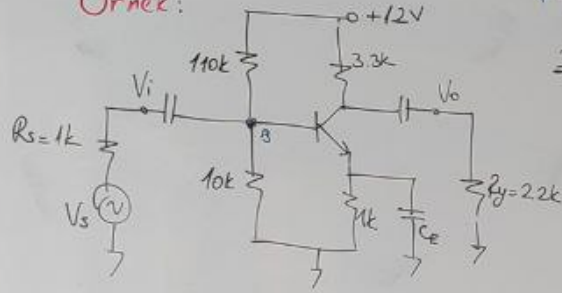
$$A_i = -A_v \cdot \frac{Z_i}{R_y} = -(-105.33) \cdot \frac{1.2 k}{2.2 k} \Rightarrow \boxed{A_i = 57.93}$$



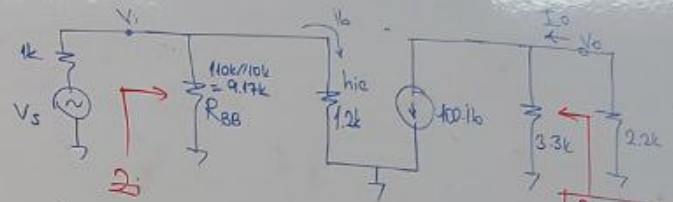
2. Common Emitter Yükselteç

Sabit kut. devresiyse benzer şekilde analiz edilir.
 R_B yerine $R_{BB} = R_1 // R_2$ yer alır.

Örnek:



* $h_{ie} = 1.2k$
 $h_{fe} = 100$
 $Z_i, Z_o, A_v, A_i, A_{v_s} = ?$



$$Z_i = R_{BB} // h_{ie} = 9.17k // 1.2k$$

$$Z_i = 1.06k$$

$$A_v = - \frac{h_{fe} (R_C // R_L)}{h_{ie}} = - \frac{100 (3.3k // 2.2k)}{1.2k}$$

$$A_v = -110$$

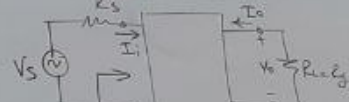
$$A_i = \frac{I_o}{I_i} \quad I_o = 100I_b + \frac{V_o}{3.3k} = 100I_b - \frac{100I_b (3.3k // 2.2k)}{3.3k}$$

$$I_o = 60I_b$$

$$I_i = I_b + \frac{V_i}{9.17k} = I_b + \frac{1.2k I_b}{9.17k} = 1.13I_b$$

$$A_i = \frac{I_o}{I_i} = \frac{60I_b}{1.13I_b} = 53.09 \approx 53$$

Kaynak ve Yük Direncinin Etkisi



$$V_i = \frac{Z_i}{Z_i + R_s} V_s$$

$$A_v = \frac{V_o}{V_i} = \frac{V_o}{V_s} \cdot \frac{Z_i + R_s}{Z_i}$$

$$A_{v_s} = A_v \cdot \frac{Z_i}{Z_i + R_s}$$

$$A_{v_s} = A_v \cdot \frac{Z_i}{Z_i + R_s}$$

$$A_{v_s} = A_v \cdot \frac{Z_i}{Z_i + R_s}$$

$$A_{v_s} = -110 \cdot \left(\frac{1.06k}{1.06k + 1k} \right)$$

$$A_{v_s} = -56.6$$

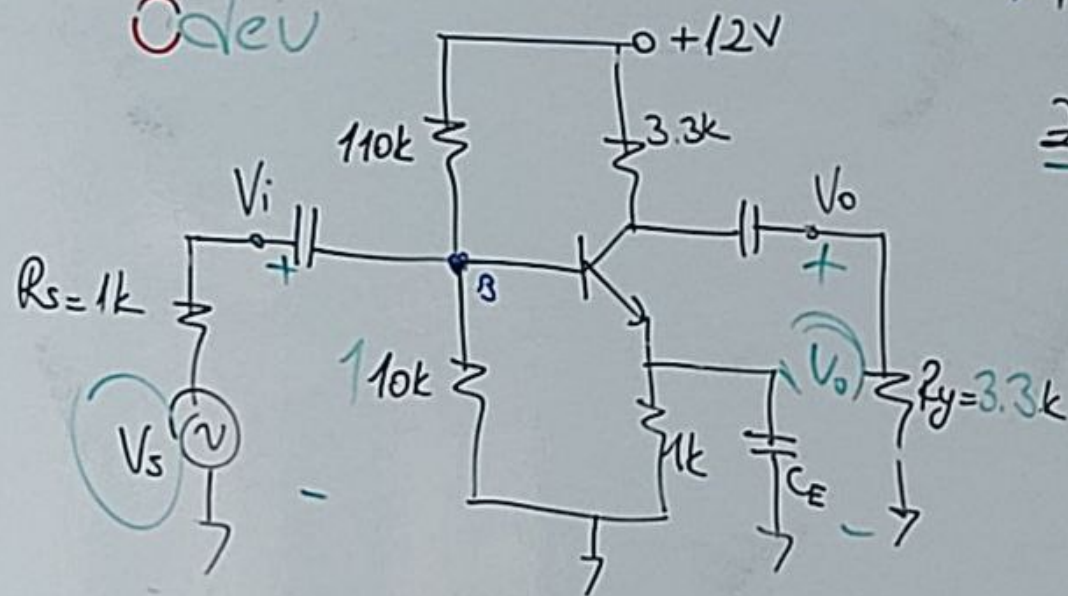
$$A_i = \frac{I_o}{I_i} = \frac{V_o}{V_i} \cdot \frac{Z_i}{R_L} = \frac{V_o}{V_i} \cdot \frac{Z_i}{R_L} \cdot \frac{Z_i}{Z_i} = \frac{V_o}{V_i} \cdot \frac{Z_i}{R_L} \cdot \frac{Z_i}{Z_i}$$

$$A_i = A_v \cdot \frac{Z_i}{R_L}$$

2. Gerilim Bölücü Yükselteç

Sabit kut. devresiyse benzer şekilde analiz edilir.
 R_B yerine $R_{BB} = R_1 // R_2$ yer alır.

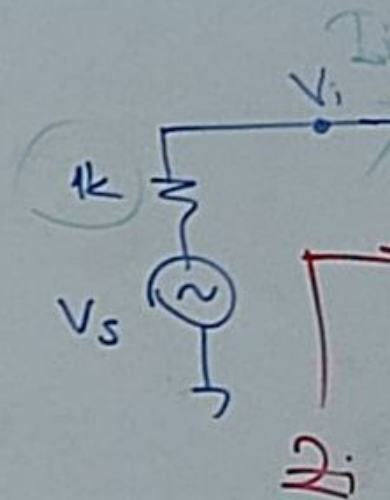
Örnek



* $h_{ie} = 1.2k$
 $h_{fe} = 100$

$Z_i, Z_o, A_v, A_i, A_{V_s} = ?$

$1/h_{oe} = 40k$



$Z_i = R_{BB} //$
 $Z_i = 1.06k$

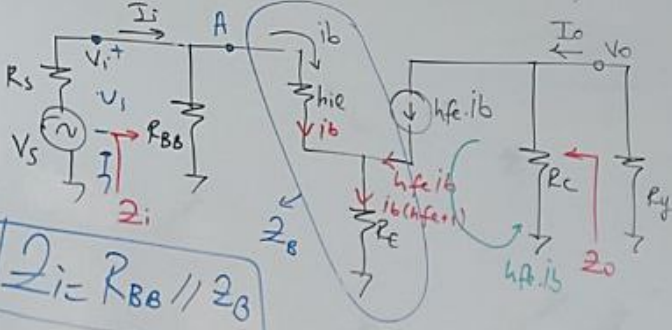
$A_v = - \frac{h_{fe}}{h_{ie}}$

$A_v = -110$

$A_i = \frac{I_o}{I_i}$

3. Emetör Köprülenmemiş Yükselteç (h_{oe} yok)

Gerilim böl. yükselteç devresinden;



$$Z_i = R_{BB} // Z_B$$

$$Z_B = \frac{V_s}{I_B} = \frac{h_{ie} \cdot I_B + R_E (h_{fe} + 1) \cdot I_B}{I_B}$$

$$Z_B = h_{ie} + (h_{fe} + 1) R_E$$

ÖRNEK: 1. Sadece G_e yokken tekrar göz.

$$A_v = \frac{V_o}{V_i}$$

$$V_i = I_B \cdot Z_B$$

$$V_o = -h_{fe} I_B (R_C // R_L)$$

$$A_v = \frac{-h_{fe} (R_C // R_L)}{Z_B}$$

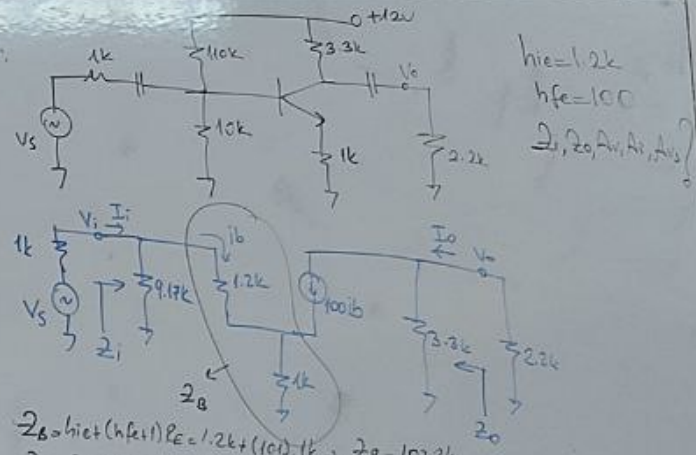
$$A_i = -A_v \cdot \frac{Z_i}{R_L}$$

Z_o 'ı bulurken, $V_s = 0$ yapılırsa;
 $(R_S // R_{B1} // R_{B2}) I_B + h_{ie} I_B + (1 + h_{fe}) I_B R_E = 0$
 Bu eşitlik $I_B = 0$ olursa 0 olur.

Dolayısıyla $h_{fe} I_B$ bağımlı akım kay. AKIK DEVRE

$$Z_o = R_C$$

ÖRNEK:



$$Z_B = h_{ie} + (h_{fe} + 1) R_E = 1.2k + (100 + 1) \cdot 1k \Rightarrow Z_B = 102.2k$$

$$Z_i = 9.17k // 102.2k \Rightarrow Z_i = 8.41k$$

$$Z_o = 3.3k$$

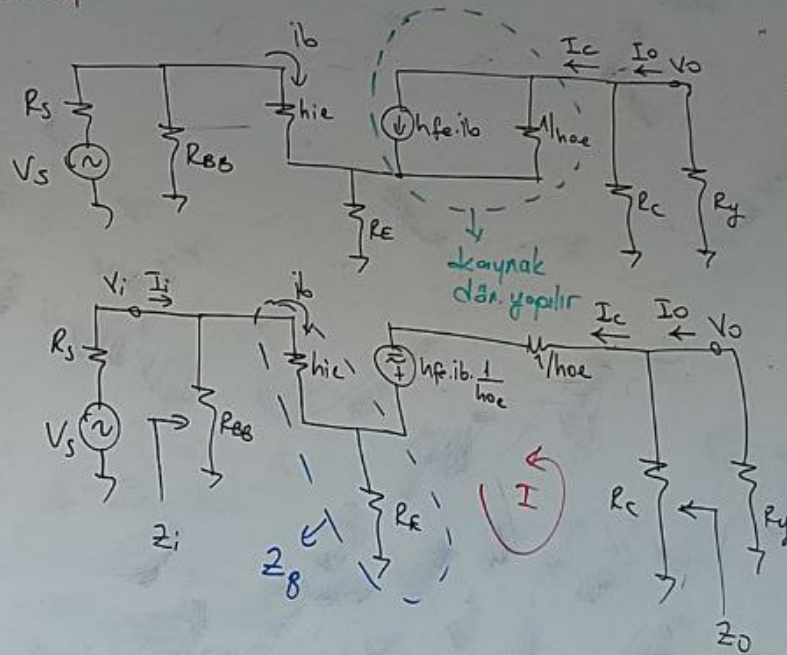
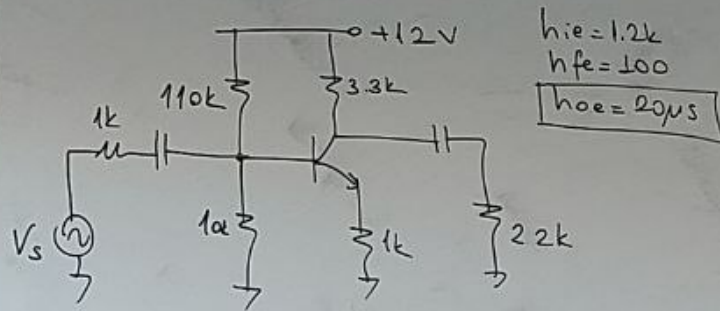
$$A_v = -\frac{100 (3.3k // 2.2k)}{102.2k} \Rightarrow A_v = -1.29$$

$$A_{v_s} = (-1.29) \cdot \frac{8.41k}{8.41k + 1k} \Rightarrow A_{v_s} = -1.15$$

$$A_i = -A_v \cdot \frac{Z_i}{R_L} = 1.29 \cdot \frac{8.41k}{2.2k}$$

$$A_i = 4.93$$

3. Ematör Köprülenmemiş Yükseltici (hoe VAR)



Z_i bulmak için;

$$Z_B = \frac{V_i}{i_b} = \frac{h_{ie} i_b + R_E (i_b + i_c)}{i_b}$$

i_c 'nin i_b cinsinden yazılması gerekir;

(I) nolu çevreden;

$$R_E (i_b + i_c) + (R_C // R_L) \cdot i_c + \frac{1}{h_{oe}} \cdot i_c = \frac{h_{fe} \cdot i_b}{h_{oe}}$$

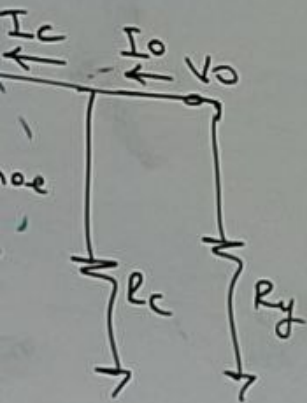
$$i_c = \frac{h_{fe} - R_E \cdot h_{oe}}{1 + h_{oe} (R_E + R_C // R_L)} \cdot i_b$$

A darsak

$$Z_B = \frac{h_{ie} i_b + R_E (i_b + A \cdot i_b)}{i_b}$$

$$Z_B = h_{ie} + R_E (1 + A)$$

$$Z_i = R_B // Z_B$$



Z_i bulmak için;

$$Z_B = \frac{V_i}{I_b} = \frac{h_{ie} I_b + R_E (I_b + I_c)}{I_b}$$

I_c 'nin I_b cinsinden yazılması gerekir;

(I) nolu devreden;

$$R_E (I_b + I_c) + (R_C // R_L) I_c + \frac{1}{h_{oe}} I_c = \frac{h_{fe} I_b}{h_{oe}}$$

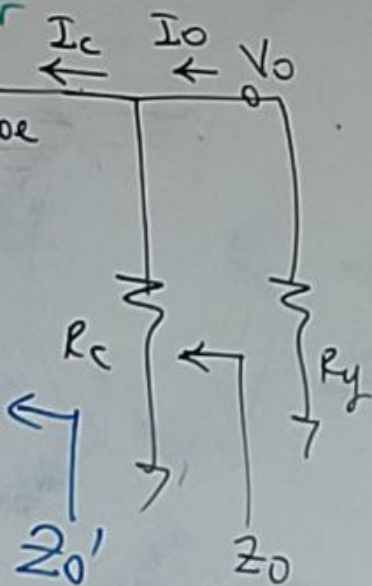
$$I_c = \frac{h_{fe} - R_E h_{oe}}{1 + h_{oe} (R_E + R_C // R_L)} I_b$$

A dersek

$$Z_B = \frac{h_{ie} I_b + R_E (I_b + A I_b)}{I_b}$$

$$Z_B = h_{ie} + R_E (1 + A)$$

$$Z_i = R_{BB} // Z_B$$



A_v 'yi bulmak için;

$$V_o = -I_c (R_C // R_L) = -A I_b (R_C // R_L)$$

$$V_i = I_b Z_B$$

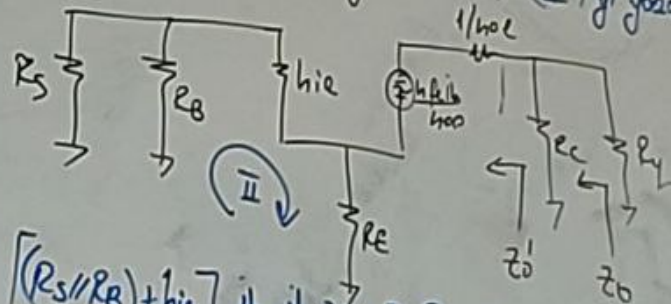
$$A_v = \frac{V_o}{V_i} = -\frac{A (R_C // R_L)}{Z_B}$$

Z_o 'ı bulmak için;

$$Z_o = R_C // Z_o'$$

$$Z_o' = \frac{V_o}{I_c} = \frac{I_c \cdot 1/h_{oe} - \frac{h_{fe} I_b}{h_{oe}} + R_E (I_b + I_c)}{I_c}$$

I_b 'leri I_c türünden yazmak için; (II)'yi yazalım



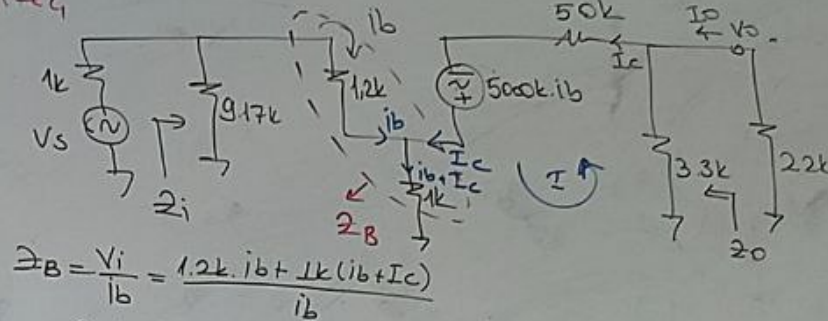
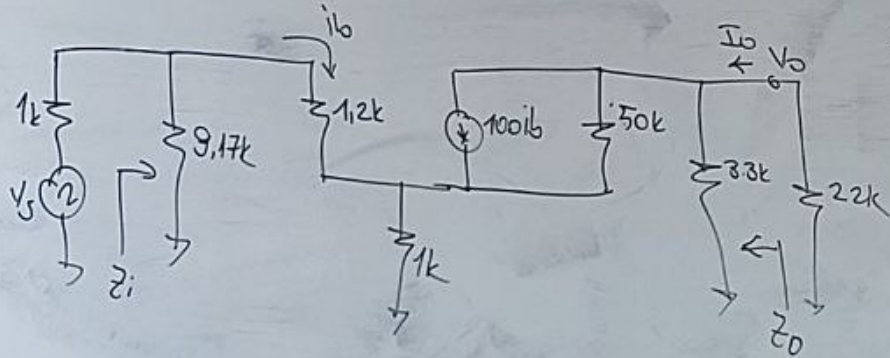
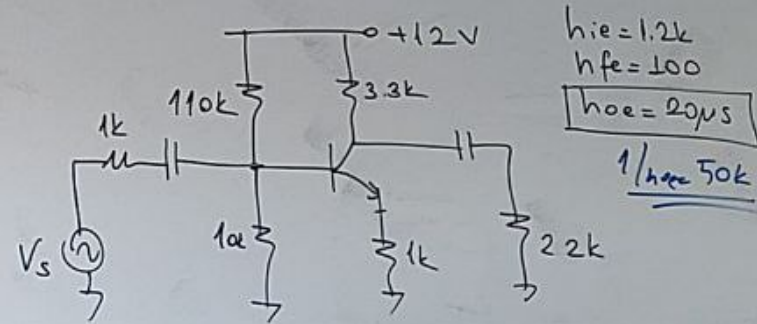
$$[(R_S // R_B) + h_{ie}] I_b + I_b R_E + I_c R_E = 0$$

$$I_b = -\frac{R_E}{R_E [h_{ie} + (R_S // R_B)]} I_c$$

B dersek

$$I_b = -B I_c$$

3. Ematör Köprülenmemiş Yükseltici (hoe VAR)



(I) den; $1k(I_b + I_c) + (3.3k // 22k) I_c + 50k I_c = 5000 I_b$

$I_c = 95.54 I_b$

$Z_B = \frac{1.2k I_b + 1k(95.54 I_b / I_b)}{I_b}$

$Z_B = 97.74k$

$Z_i = 9.17k // 97.74k$

$Z_i = 8.38k$

$V_o = -I_c (3.3k // 22k)$

$V_o = -95.54 I_b (1.32k)$

$V_i = I_b \cdot Z_B = 97.74k \cdot I_b$

$A_v = \frac{V_o}{V_i} = \frac{-95.54 I_b \cdot (1.32k)}{97.74k \cdot I_b}$

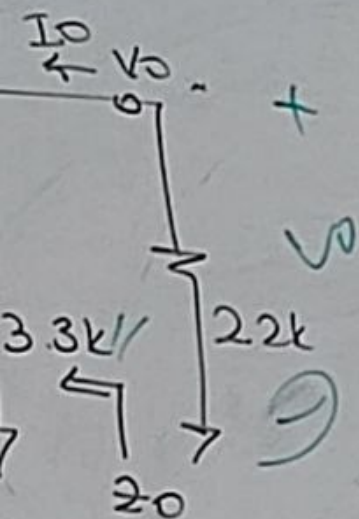
$A_v = -1.29$

$A_{v_s} = A_v \cdot \frac{Z_i}{Z_i + R_s} = (-1.29) \cdot \frac{8.38k}{9.38k}$

$A_{v_s} = -1.15$

$A_i = -A_v \cdot \frac{Z_i}{R_y} = -(-1.29) \cdot \frac{8.38k}{2.2k}$

$A_i = 4.91$



$$V_o = -I_c (3.3k \parallel 2.2k)$$

$$V_o = -95.54 \mu A (1.32k)$$

$$V_i = I_b \cdot Z_B = 97.74 \mu A \cdot I_b$$

$$A_v = \frac{V_o}{V_i} = -\frac{95.54 \mu A \cdot (1.32k)}{97.74 \mu A}$$

$$A_v = -1.29$$

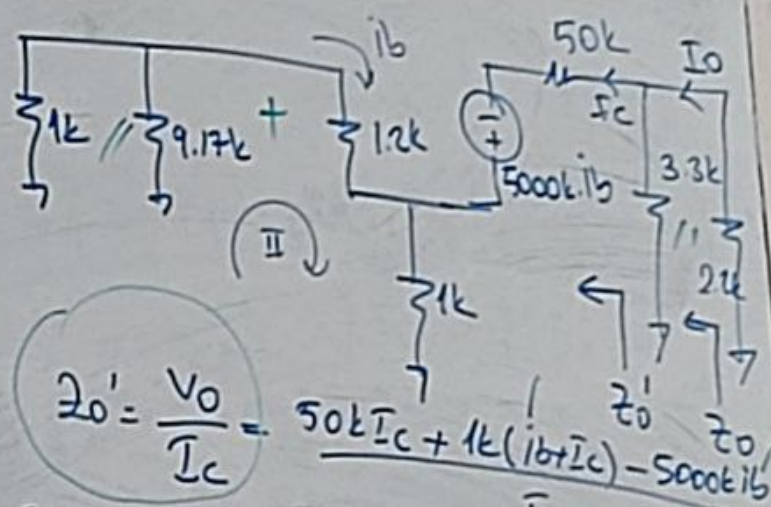
$$A_{vs} = A_v \cdot \frac{Z_i}{Z_i + R_s} = (-1.29) \cdot \frac{8.38k}{9.38k}$$

$$A_{vs} = -1.15$$

$$A_i = -A_v \cdot \frac{Z_i}{R_y} = -(-1.29) \cdot \frac{8.38k}{2.2k}$$

$$A_i = 4.91$$

ÖDEV: Z_o 'i bulunuz



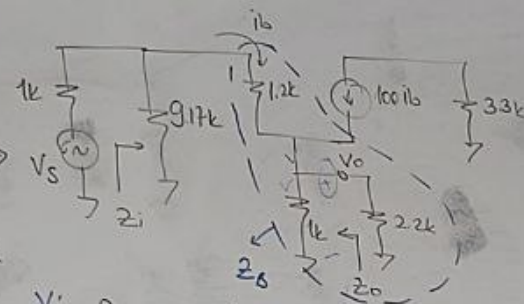
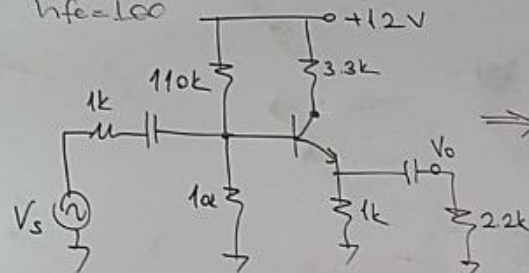
$$Z_o' = \frac{V_o}{I_c} = \frac{50k I_c + 1k(I_b + I_c) - 500k I_b}{I_c}$$

(II) 'nolu gerilimden $I_b = \dots I_c$ 'yi bul.

$$Z_o = Z_o' \parallel 3.3k \text{ yerine koy.}$$

4. Emetör Çıkışlı Yükseltici (hoe yok)

$h_{ie} = 1.2k$
 $h_{fe} = 100$



$$\frac{V_i}{i_b} = Z_B = 1.2k + (1 + h_{fe})(R_E || R_Y)$$

$$Z_B = 1.2k + (101)(0.68k)$$

$$Z_B = 70.63k$$

$$Z_i = 9.17k || 70.63k \Rightarrow Z_i = 8.12k$$

$$V_o = (1 + h_{fe})i_b(R_E || R_Y)$$

$$V_o = 101 \cdot i_b \cdot (0.68k)$$

$$V_i = i_b \cdot Z_B = 70.63k \cdot i_b$$

$$A_v = \frac{V_o}{V_i} = \frac{68.68k \cdot i_b}{70.63k \cdot i_b}$$

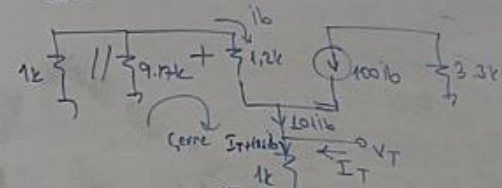
$$A_v = 0.97$$

$$A_{vs} = 0.97 \cdot \frac{8.12k}{9.12k} \Rightarrow A_{vs} = 0.86$$

$$A_i = -A_v \cdot \frac{Z_i}{R_Y} = -(0.97) \cdot \frac{8.12k}{2.2k}$$

$$A_i = -3.58$$

Z_o için



$$V_T = 1k \cdot (101i_b + I_T)$$

Girişte gerçe yazılırsa;

$$[(1k/9.17) + 1.2k] \cdot i_b + 1k(101i_b + I_T) = 0$$

$$i_b = -\frac{I_T}{103.1}$$

$$Z_o = \frac{V_T}{I_T} = \frac{1k \cdot [101 \cdot (-\frac{I_T}{103.1}) + I_T]}{I_T}$$

$$Z_o = 20.36\Omega$$

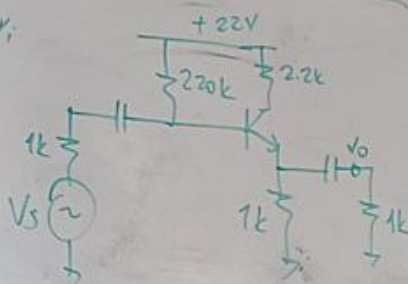
ÖDER:

$h_{ie} = 1.2k$
 $h_{fe} = 80$

Z_i, Z_o

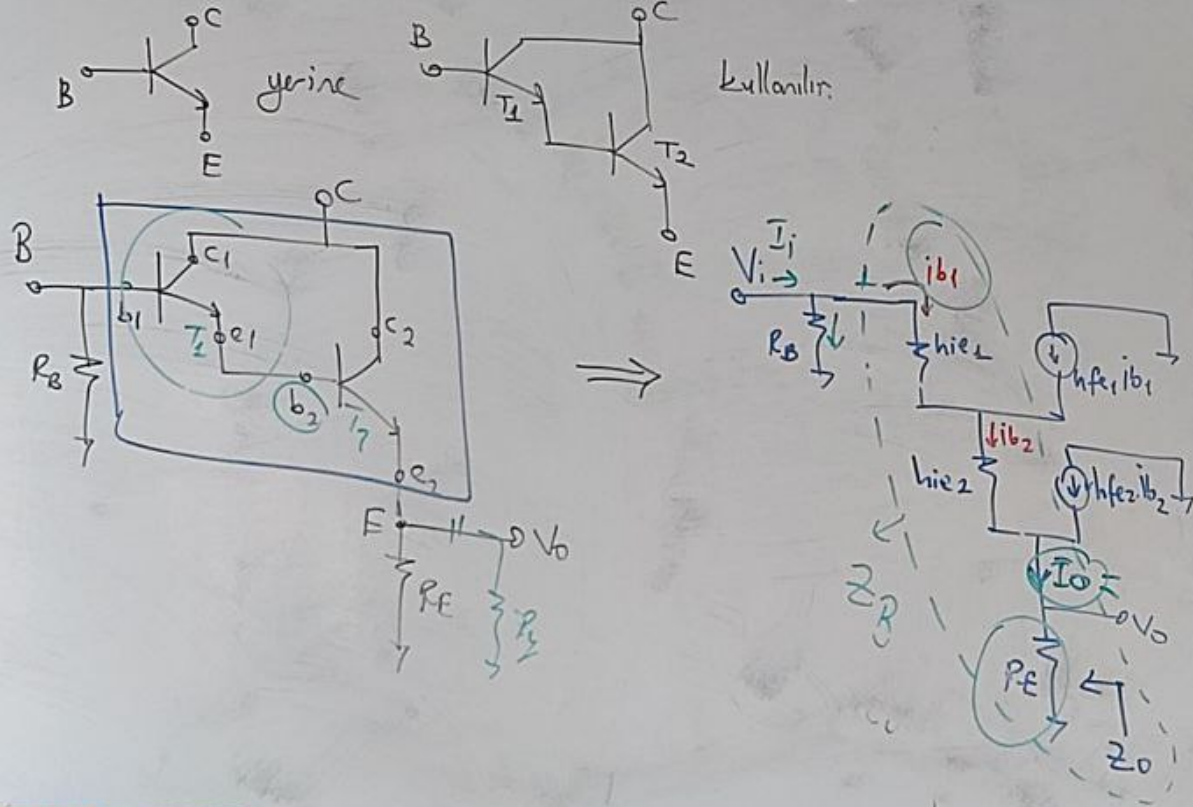
A_v, A_{vs}

$A_i = ?$



5. Darlington Çifti

Yüksek giriş direnci, yüksek akım kazancı, düşük çıkış direnci sağlamak için kullanılır. Yük Ematöre bağlanır.



$$\begin{aligned} I_{b2} &= (1+h_{fe1}) I_{b1} \\ I_o &= (1+h_{fe2}) I_{b2} \\ I_o &= (1+h_{fe2})(1+h_{fe1}) I_{b1} \end{aligned}$$

$$Z_B = \frac{V_i}{I_{b1}} \quad V_i = h_{ie1} I_{b1} + h_{ie2} I_{b2} + R_E I_o$$

$$V_i = h_{ie1} I_{b1} + h_{ie2} (1+h_{fe1}) I_{b1} + R_E (1+h_{fe2})(1+h_{fe1}) I_{b1}$$

$$Z_B = \frac{V_i}{I_{b1}} \Rightarrow Z_B = h_{ie1} + h_{ie2} (1+h_{fe1}) + R_E (1+h_{fe2})(1+h_{fe1})$$

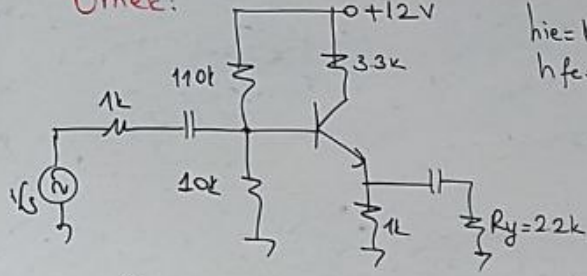
$$Z_i = R_B \parallel Z_B$$

$$A_i = \frac{I_o}{I_i} = \frac{(1+h_{fe1})(1+h_{fe2}) I_{b1}}{I_{b1} \left(1 + \frac{V_i}{R_B}\right)}$$

$$A_v = \frac{V_o}{V_i} = \frac{R_E I_o}{Z_B I_{b1}}$$

$$A_v = \frac{(1+h_{fe1})(1+h_{fe2}) R_E}{Z_B}$$

Örnek:



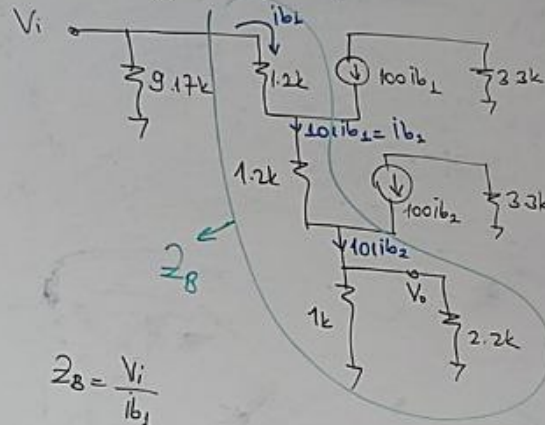
$h_{ie} = 1.2k$
 $h_{fe} = 100$

Data önce çözülerek;

$Z_i = 8,12k$
 $Z_o = 20,36\Omega$
 $A_v = 0,97$
 $A_i = -3,58$

bulunmuştur.

T₁ = T₂ olarak şekilde
darlington bağlanırsa;



$Z_B = \frac{V_i}{I_{B1}}$

$V_i = 1,2k I_{B1} + 1,2k (101 I_{B1}) + (1k // 2,2k) \cdot 101 I_{B2}$
 $I_{B2} = 101 I_{B1}$ old. göre,

$V_i = 7058,4k I_{B1}$

$Z_B = \frac{V_i}{I_{B1}} \Rightarrow Z_B \approx 7,06M$

$Z_i = 9,17k // 7,06M$

$Z_i = 9,15k$

$A_v = \frac{V_o}{V_i}$

$V_o = 101 I_{B2} \cdot (1k // 2,2k)$

$V_o = 6936,68k \cdot I_{B1}$

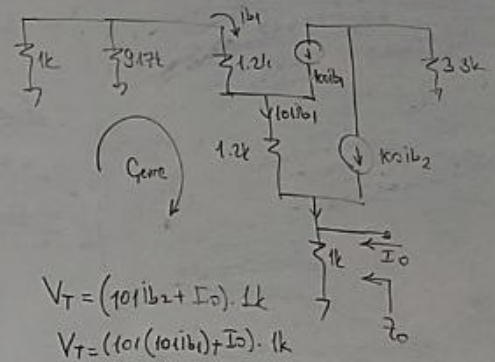
$V_i = 7058,4k \cdot I_{B1}$

$A_v = \frac{V_o}{V_i} = 0,98$

$A_i = -A_v \cdot \frac{Z_i}{R_L} = -0,98 \cdot \frac{9,15k}{2,2k}$

$A_i = -4,07$

Z_o bulmak için;



$V_T = (101 I_{B2} + I_o) \cdot 1k$

$V_T = (101 (101 I_{B1}) + I_o) \cdot 1k$

$[(1k // 9,17k) + 1,2k] I_{B1} + 1,2k (101 I_{B1}) + 1k (101001 I_{B1} + I_o)$
 $I_{B1} = -\frac{1}{10223,3} I_o$

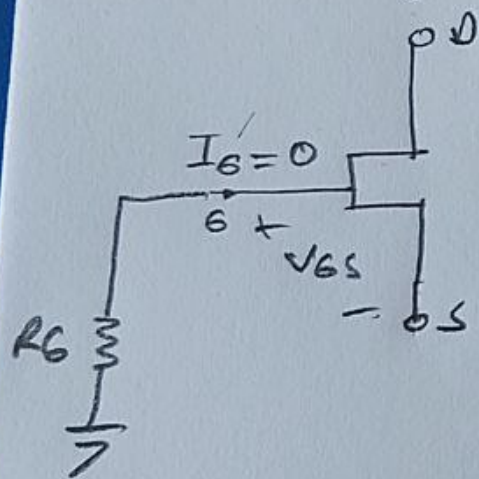
$V_T = 12,10 I_o$

$Z_o = \frac{V_T}{I_o} \Rightarrow Z_o = 12,1\Omega$

Collector çıkış emir göre emitor kopulduğunu ve
hoc varsa Z_o kullanılır.

Transistörler (BJT) akım kontrollü akım elemanıdır.

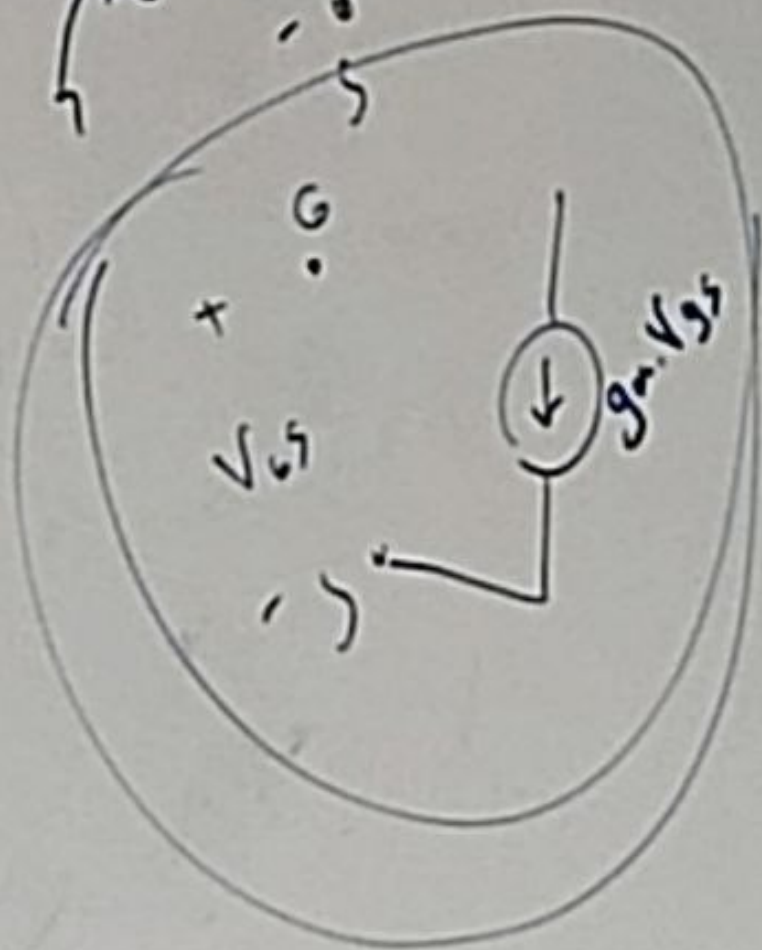
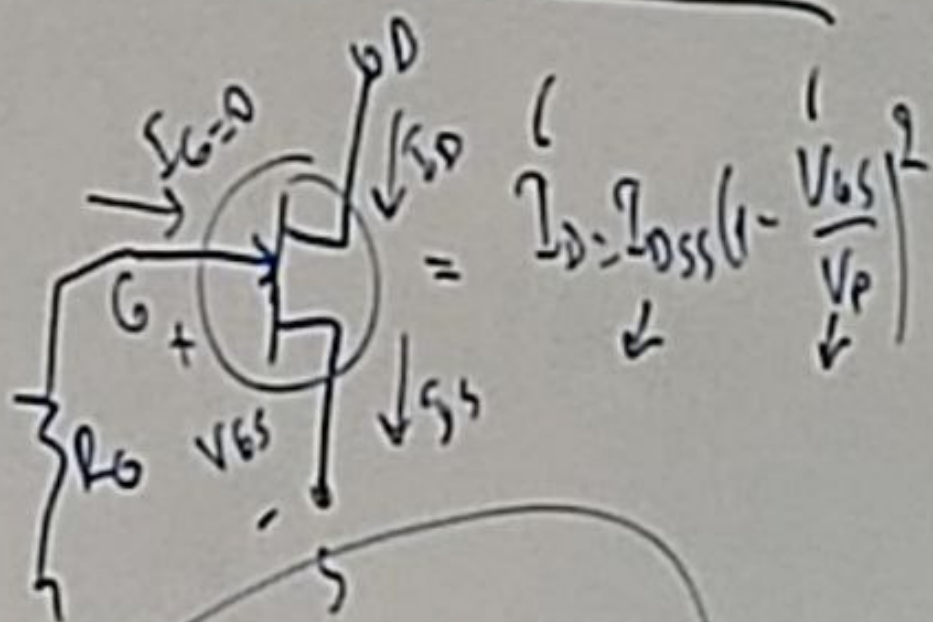
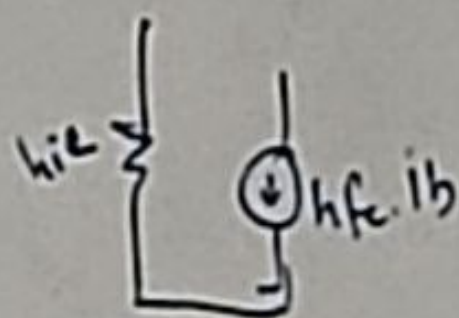
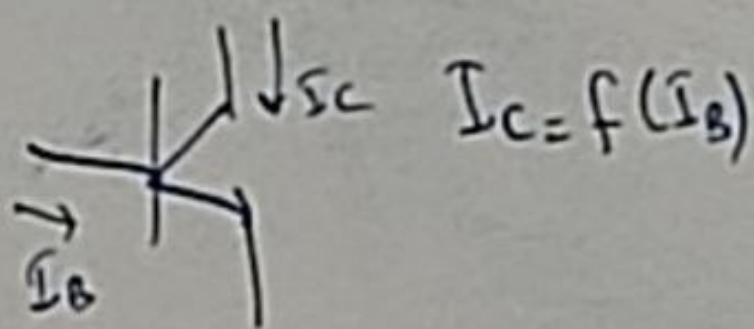
FET'ler gerilim kontrollü gerilim elemanıdır.



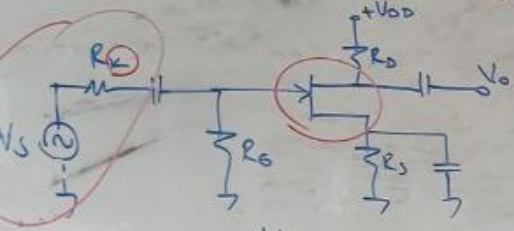
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$I_D = \text{Değişken}$ $V_{GS} = \text{Değişken}$
 $I_{DSS} = \text{sabit}$ $V_P = \text{sabit}$

FETler gerilim kontrollü ($V_{GS} = \text{Değişken}$)
akım elemanı veya kaynağıdır. ($I_D = \text{Değişken}$)



FET Küçük Sinyal Analizi



AC eşdeğeri



JFET ————— } Yüksek frekanslarda
 MOSFET — { Kanal Açılabilir }
 — { Kanal Kilitlenir } } Dijital devrelerde
 CMOS ————— }

JFET ve Kanal Açılabilir MOSFET için
 AC Model



$r_d = 1/y_{os}$ y_{os} : Çıkış iletkenliği

g_m : FET geçiş iletkenliği //

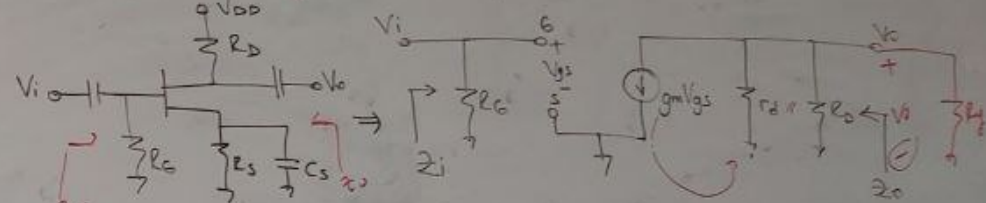
$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

$$g_m = \frac{2 I_{DSS}}{|V_P|} \left[1 - \frac{V_{GS}}{V_P} \right]$$

$$g_{mo} = \frac{2 I_{DSS}}{|V_P|} \Rightarrow$$

$$g_m = g_{mo} \left(1 - \frac{V_{GS}}{V_P} \right)$$

1. Kendiliğinden Kutuplanalı Devre

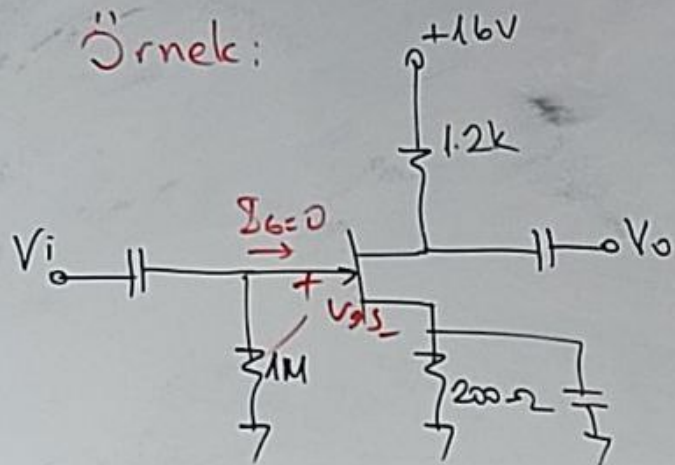


① DC analiz ile g_m bulunur.

$$\textcircled{2} \quad Z_i = R_G \quad Z_o = r_d \parallel R_D$$

$$\textcircled{3} \quad \left. \begin{array}{l} V_o = -g_m V_{gs} (r_d \parallel R_D) \\ V_i = V_{gs} \end{array} \right\} A_v = \frac{V_o}{V_i} = -g_m (r_d \parallel R_D)$$

Örnek:



$$I_{DSS} = 8\text{mA}, V_p = -4\text{V} \quad Z_i, Z_o, A_v = ?$$

gm verilmediği için DC analiz yapılır;

$$V_{GS} = -R_S \cdot I_D = -(0.2\text{k}) \cdot I_D \quad (1)$$

$$(2) \quad I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2 = 8\text{m} \left(1 - \frac{(-0.2\text{k}) \cdot I_D}{(-4)}\right)^2$$

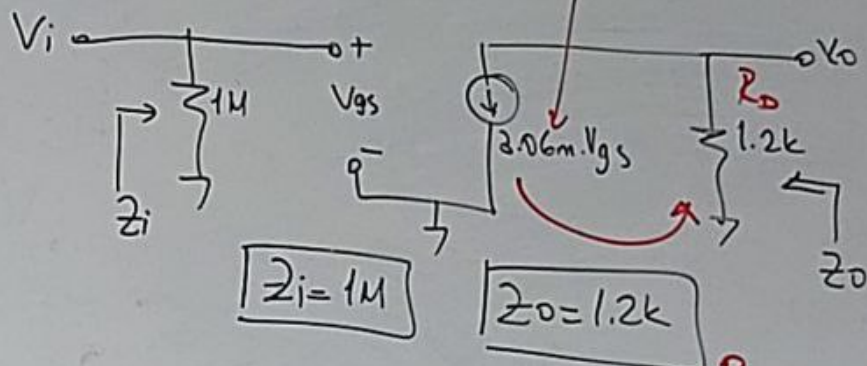
denklemler çözülürse;

$$I_{DQ} = 4.7\text{mA} \quad V_{GSQ} = -0.94\text{V}$$

$$g_m = \frac{2 I_{DSS}}{|V_p|} \left(1 - \frac{V_{GS}}{V_p}\right)$$

$$g_m = \frac{2 \cdot 8\text{m}}{4} \left(1 - \frac{(-0.94)}{(-4)}\right)$$

$$g_m = 3.06 \text{ m} \frac{\text{A}}{\text{V}} = 3.06 \text{ mS}$$



$$A_v = \frac{V_o}{V_i}$$

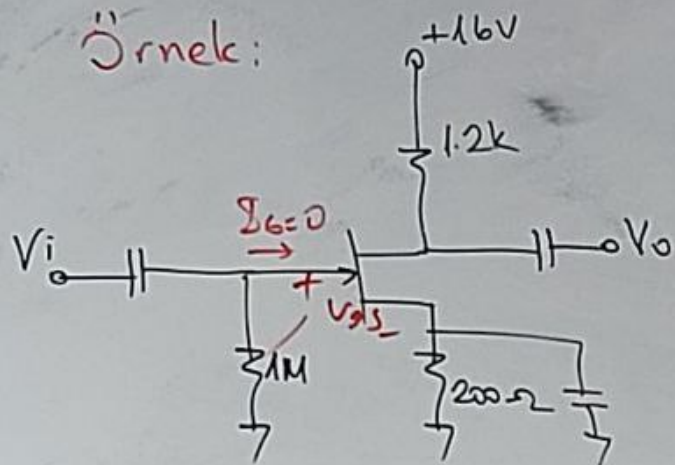
$$V_o = -g_m V_{GS} (1.2\text{k})$$

$$\frac{V_o}{V_i} = \frac{-3.06\text{m} \cdot V_{GS} (1.2\text{k})}{V_{GS}}$$

$$A_v = -3.06\text{m} (1.2\text{k}) = -g_m R_D$$

$$A_v = -3.67$$

Örnek:



$$I_{DSS} = 8\text{mA}, V_P = -4\text{V} \quad Z_i, Z_o, A_v = ?$$

gm verilmediği için DC analiz yapılır;

$$V_{GS} = -R_S \cdot I_D = -(0.2\text{k}) \cdot I_D \quad (1)$$

$$(2) \quad I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 = 8\text{m} \left(1 - \frac{(-0.2\text{k}) \cdot I_D}{(-4)}\right)^2$$

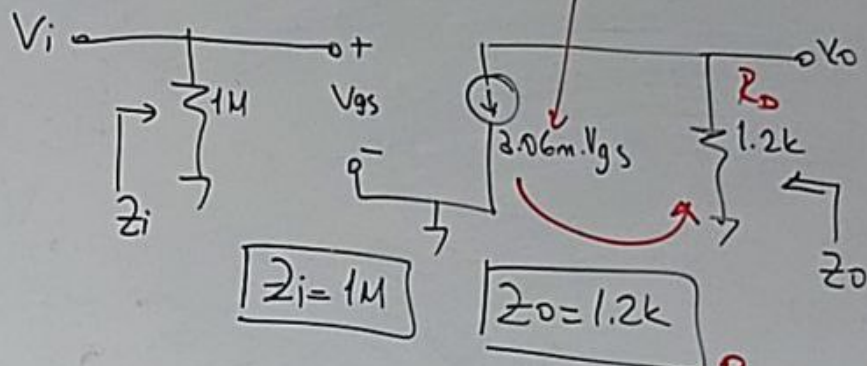
denklemler çözülürse;

$$I_{DQ} = 4.7\text{mA} \quad V_{GSQ} = -0.94\text{V}$$

$$g_m = \frac{2 I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right)$$

$$g_m = \frac{2 \cdot 8\text{m}}{4} \left(1 - \frac{(-0.94)}{(-4)}\right)$$

$$g_m = 3.06 \text{ m} \frac{\text{A}}{\text{V}} = 3.06 \text{ mS}$$



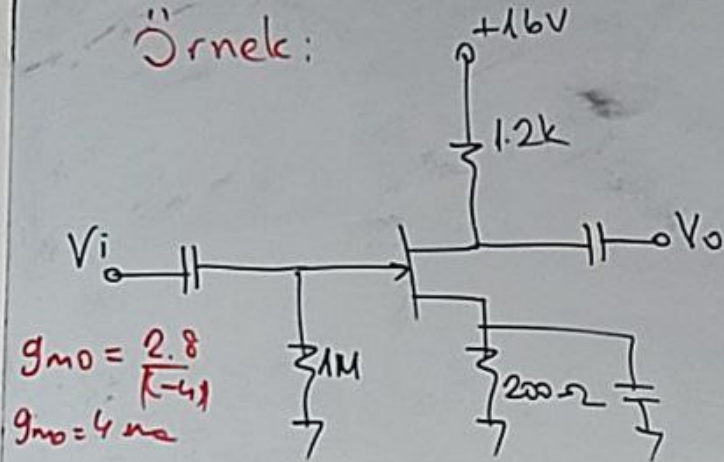
$$A_v = \frac{V_o}{V_i}$$

$$V_o = -g_m V_{GS} (1.2\text{k})$$

$$\frac{V_o}{V_i} = \frac{-3.06\text{m} \cdot V_{GS} (1.2\text{k})}{V_{GS}}$$

$$A_v = -3.06\text{m} (1.2\text{k}) = -3.67$$

Örnek:



$$g_{m0} = \frac{2.8}{(-4)}$$

$$g_{m0} = 4 \text{ mA/V}$$

$$I_{DSS} = 8 \text{ mA}, V_p = -4 \text{ V} \quad z_i, z_o, A_v = ?$$

g_m verilmediği için DC analiz yapılır;

$$V_{GS} = -R_s \cdot I_D = -(0.2 \text{ k}) \cdot I_D$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2 = 8 \text{ m} \left(1 - \frac{(-0.2 \text{ k}) \cdot I_D}{(-4)}\right)^2$$

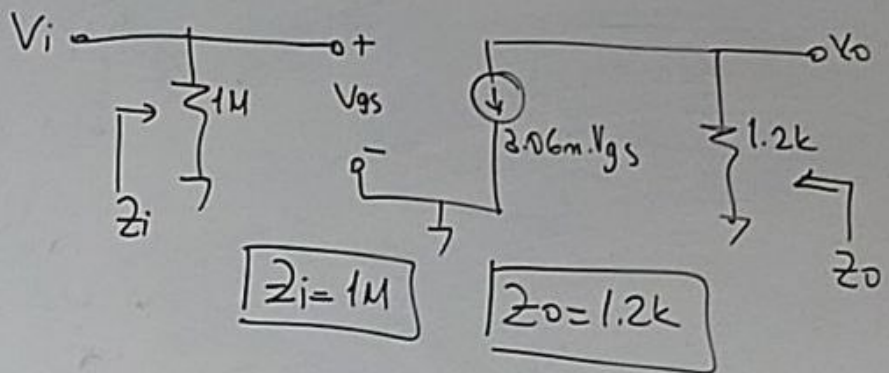
denklemler çözülürse;

$$I_{DQ} = 4.7 \text{ mA} \quad V_{GSQ} = -0.94 \text{ V}$$

$$g_m = \frac{2 I_{DSS}}{|V_p|} \left(1 - \frac{V_{GS}}{V_p}\right)$$

$$g_m = \frac{2 \cdot 8 \text{ m}}{4} \left(1 - \frac{(-0.94)}{(-4)}\right)$$

$$g_m = 3.06 \text{ m A/V} = 3.06 \text{ mS}$$



$$z_i = 1 \text{ M}$$

$$z_o = 1.2 \text{ k}$$

$$A_v = \frac{V_o}{V_i}$$

$$V_o = -g_m V_{GS} (1.2 \text{ k})$$

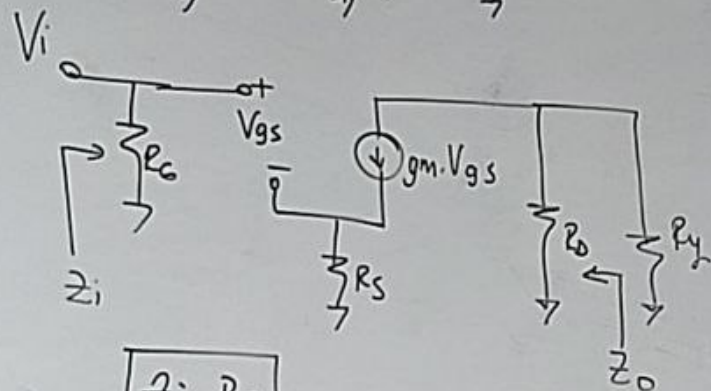
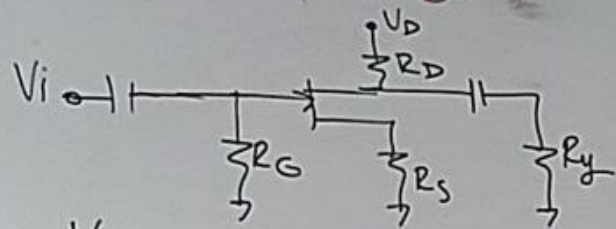
$$V_o = -3.06 \text{ m} \cdot V_{GS} (1.2 \text{ k})$$

$$V_i = V_{GS}$$

$$A_v = -3.06 \text{ m} (1.2 \text{ k})$$

$$A_v = -3.67$$

2. Source Dirençli Yükseltici (r_d yok)



$$Z_i = R_G$$

Z_o için; V_i kısa devre olursa;

$$V_{gs} + g_m V_{gs} R_s = 0$$

$$V_{gs} (1 + g_m R_s) = 0$$

$V_{gs} = 0$ dir. Akım kaynağı açık devre

$$Z_o = R_D$$

$$A_v = \frac{V_o}{V_i}$$

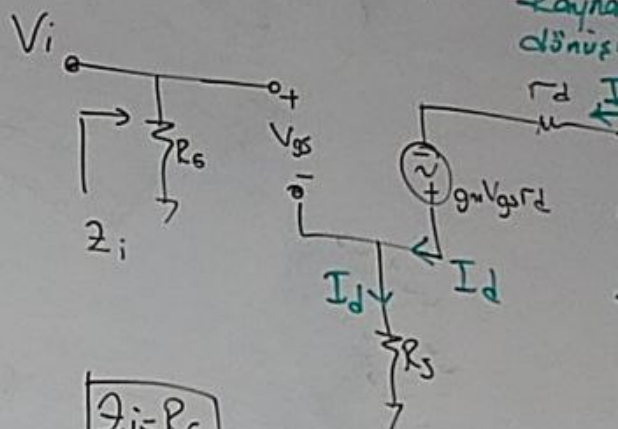
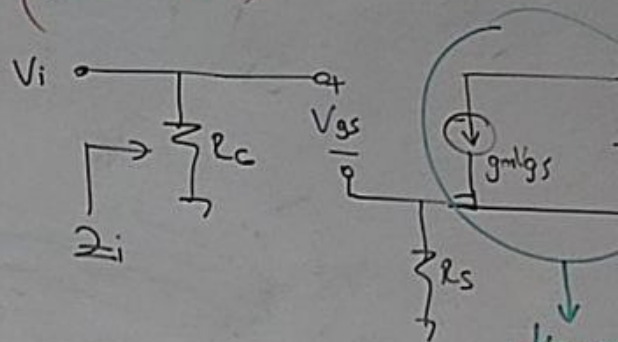
$$V_o = -g_m V_{gs} (R_D \parallel R_L)$$

$$V_i = V_{gs} + g_m V_{gs} R_s$$

$$A_v = - \frac{V_{gs} (g_m) (R_D \parallel R_L)}{V_{gs} (1 + g_m R_s)}$$

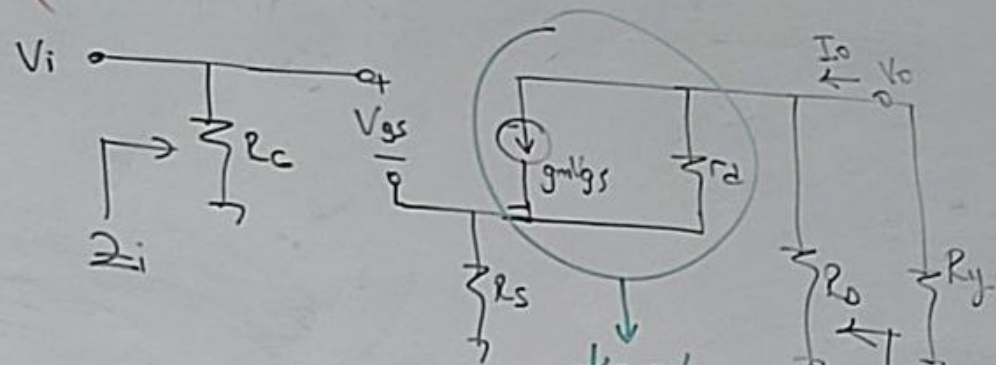
$$A_v = - \frac{g_m (R_D \parallel R_L)}{1 + g_m R_s}$$

(r_d var)

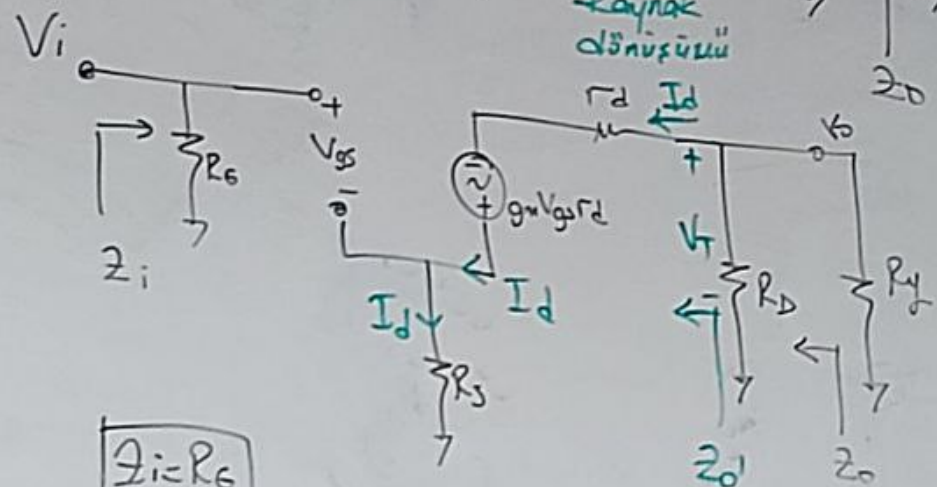


$$Z_i = R_G$$

(r_d Var)



Kaynak
dönüşümü



$$Z_i = R_c$$

$$\begin{aligned} V_o &= -I_d (R_o \parallel R_L) \\ V_i &= V_{gs} + I_d R_s \end{aligned}$$

$I_d = \dots V_{gs}$ türünden ifade etmek için
Ağırdaki çevre yazılır.

$$A_v = - \frac{g_m r_d (R_o \parallel R_L)}{g_m r_d R_s + r_d + R_s + (R_o \parallel R_L)}$$

Z_o 'i bulmak için önce Z_o' bulunur.

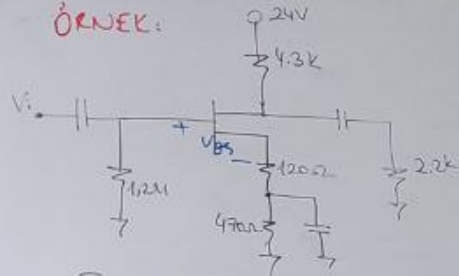
$$Z_o' = \frac{V_T}{I_d}$$

$V_{gs} = \dots I_d$ bulunmalı.

$V_i = 0$ iken girişten çevre yazılır.

$$Z_o = r_d + R_s + g_m r_d R_s$$

ÖRNEK:

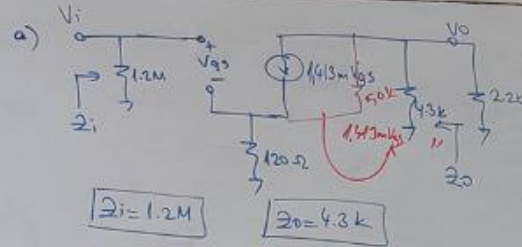


$$I_{DSS} = 6\text{mA}, V_p = -6\text{V}$$

- a) $Z_i, Z_o, A_v, A_i = ?$ b) $I_D = 50\text{k}\Omega$ için Q noktası bulun.

$$V_{GS} + I_D(570\Omega) = 0$$

DC analizden $g_m = 1,413\text{mS}$ çıkar.



$$Z_i = 1.2\text{M}$$

$$Z_o = 4.3\text{k}$$

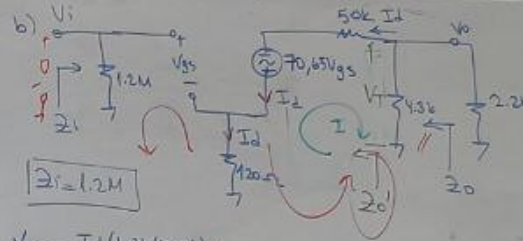
$$A_v = \frac{-g_m(R_D \parallel R_L)}{1 + g_m R_S}$$

$$A_v = \frac{-1,413\text{m}(4.3\text{k} \parallel 2.2\text{k})}{1 + 1,413\text{m}(0,12\text{k})}$$

$$A_v = -1,75$$

$$A_i = -A_v \frac{Z_i}{R_L} = (1,75) \frac{1,2\text{M}}{2,2\text{k}}$$

$$A_i = 954$$



$$V_o = -I_D(4.3\text{k} \parallel 2.2\text{k})$$

$$V_i = V_{GS} + I_D(0,12\text{k})$$

I notu g_{m1} üzerinden;

$$0,12\text{k} \cdot I_D + (4.3\text{k} \parallel 2.2\text{k}) \cdot I_D + 50\text{k} I_D = 70,65 V_{GS}$$

$$V_{GS} = 730 I_D$$

$$V_i = V_{GS} + I_D(0,12\text{k}) \Rightarrow V_i = 850 I_D$$

$$A_v = \frac{V_o}{V_i} = \frac{-I_D(4,45\text{k})}{850 I_D}$$

$$A_v = -1,711$$

$$A_i = (1,711) \frac{1,2\text{M}}{2,2\text{k}}$$

$$A_i = 933,7$$

Z_o' bulmak için;

$$Z_o' = \frac{V_T}{I_D}$$

$$V_T = 50\text{k} I_D + 0,12\text{k} I_D + 70,65 V_{GS}$$

$$V_i = 0 \Rightarrow V_{GS} + 120 I_D = 0 \Rightarrow V_{GS} = -120 I_D$$

$$V_T = 50\text{k} I_D + 0,12\text{k} I_D + 8,478\text{k} I_D$$

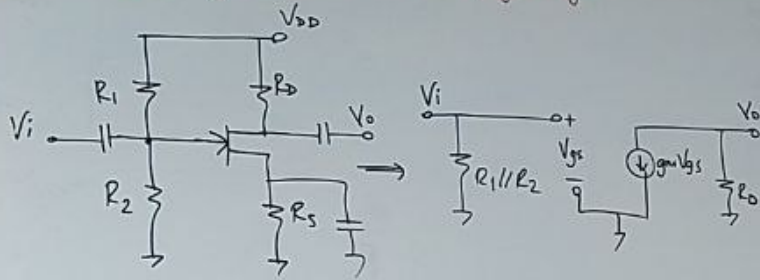
$$Z_o' = \frac{V_T}{I_D} \Rightarrow Z_o' = 58,6\text{k}$$

$$Z_o = Z_o' \parallel 4,3\text{k} \Rightarrow 58,6\text{k} \parallel 4,3\text{k} = Z_o$$

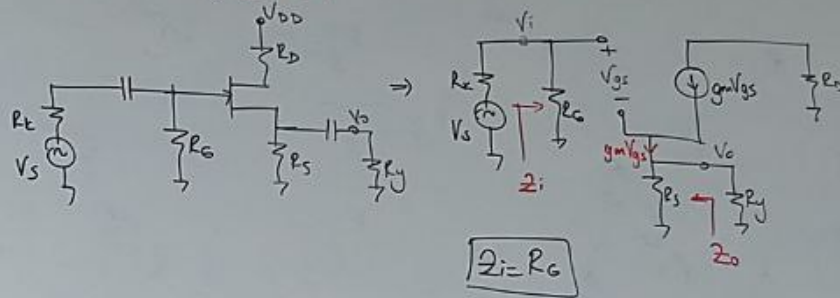
$$Z_o = 4\text{k}$$

Ödev: R_S koparılmamış bir JFET devresini, $I_D = 50\text{k}\Omega$ vaktin ve yolları analiz ediniz.
NOT: Devrede V_{GS} ve $R_L = 1\text{k}$ yer almaktadır.

Voltaj Bölücü Konfigurasyon



Source İzleyicili (Ortak Drain) Yükseltici ($r_d \rightarrow \infty$)

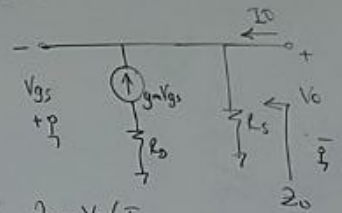


$$V_o = g_m V_{gs} (R_s \parallel R_L)$$

$$V_i = V_{gs} + g_m V_{gs} (R_s \parallel R_L)$$

$$A_v = \frac{V_o}{V_i} = \frac{g_m (R_s \parallel R_L)}{1 + g_m (R_s \parallel R_L)}$$

Z_o için (V_o ve V_i kısa devre)



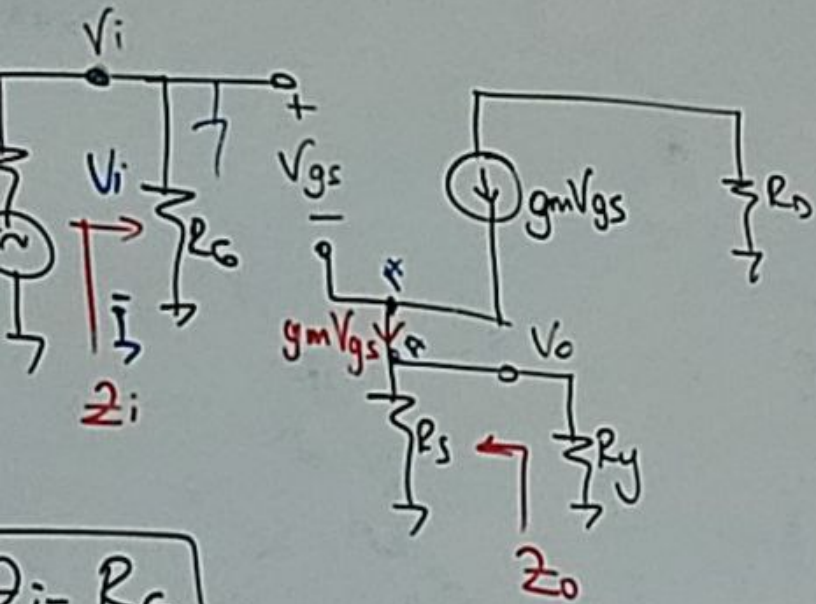
$$Z_o = V_o / I_o$$

$$V_o = -V_{gs}$$

$$I_o = \frac{V_o}{R_s} - g_m V_{gs} = \frac{-V_{gs}}{R_s} - g_m V_{gs}$$

$$Z_o = \frac{V_{gs}}{\frac{V_{gs}}{R_s} + g_m V_{gs}} \Rightarrow \boxed{Z_o = \frac{R_s}{1 + g_m R_s}}$$

eltes

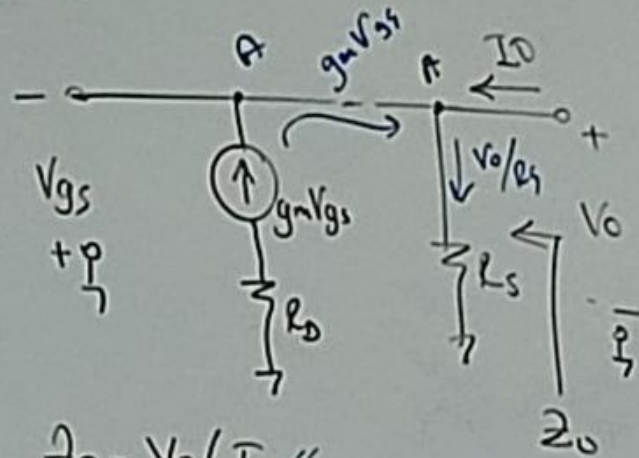


$$Z_i = R_G$$

$$g_m V_{gs} (R_S \parallel R_y)$$

$$-V_{gs} + g_m V_{gs} (R_S \parallel R_y)$$

Z_o için (V_i ve V_i kısa devre)



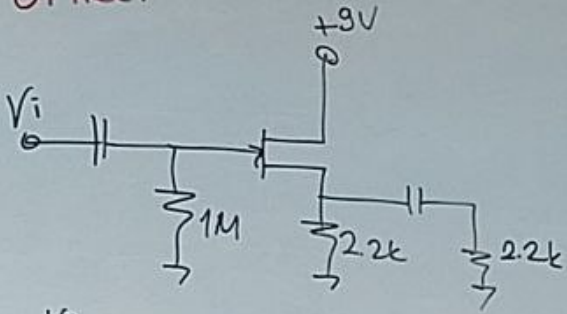
$$Z_o = V_o / I_o$$

$$V_o = -V_{gs}$$

$$I_o = \frac{V_o}{R_S} - g_m V_{gs} = -\frac{V_{gs}}{R_S} - g_m V_{gs}$$

$$Z_o = \frac{-V_{gs}}{\frac{V_{gs}}{R_S} + g_m V_{gs}} \Rightarrow \boxed{Z_o = \frac{R_S}{1 + g_m R_S}}$$

Örnek:



$g_m = 2.28 \text{ mS}$
 Z_i, Z_o, A_v, A_i ?

$$A_v = \frac{g_m (R_s \parallel R_d)}{1 + g_m (R_s \parallel R_d)}$$

$$A_v = \frac{2.28 \text{ m} (2.2 \text{ k} \parallel 2.2 \text{ k})}{1 + 2.28 \text{ m} (2.2 \text{ k} \parallel 2.2 \text{ k})}$$

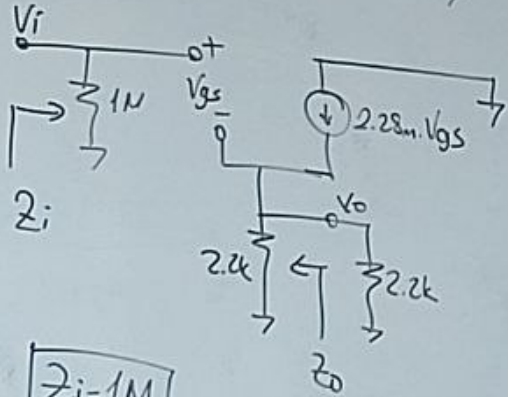
$$A_v = 0.714$$

$$A_i = -0.714 \cdot \frac{1 \text{ M}}{2.2 \text{ k}}$$

$$A_i = -325$$

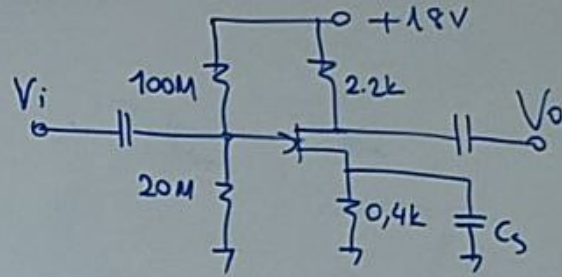
$$Z_o = \frac{R_s}{1 + g_m R_s} = \frac{2.2 \text{ k}}{1 + (2.28 \text{ mS})(2.2 \text{ k})}$$

$$Z_o = 365.7 \Omega$$



$$Z_i = 1 \text{ M}$$

ÖDEV:

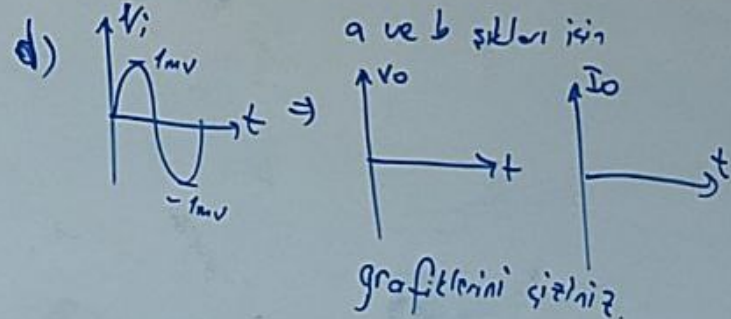


$I_{DSS} = 6 \text{ mA}$ $V_p = -3 \text{ V}$

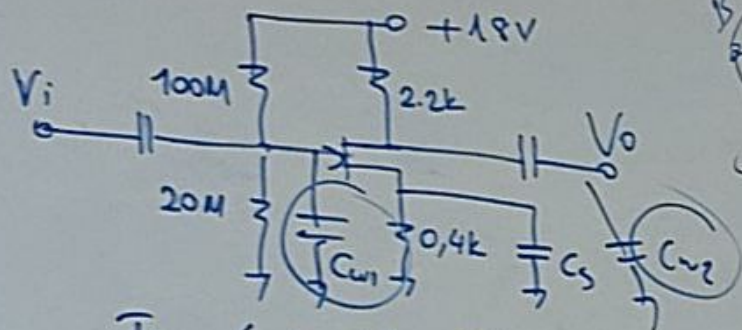
a) Z_i, Z_o, A_v, A_i = ?

c) 'Hükümet, Source çıkışı olarak tasarlanırsa ($R_g = 1 \text{ k}$), Z_o, Z_i, A_v, A_i ?

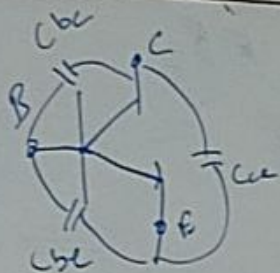
b) C_s yokken a) çıktıyı tekrar gözünüz.



ÖDEV:



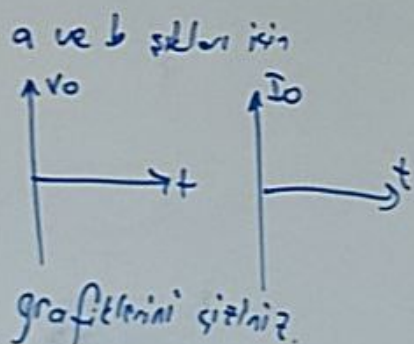
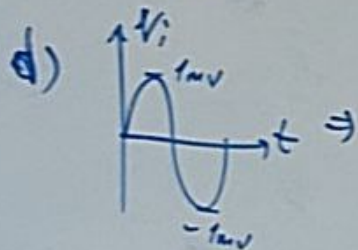
$I_{DSS} = 6mA$ $V_p = -3V$



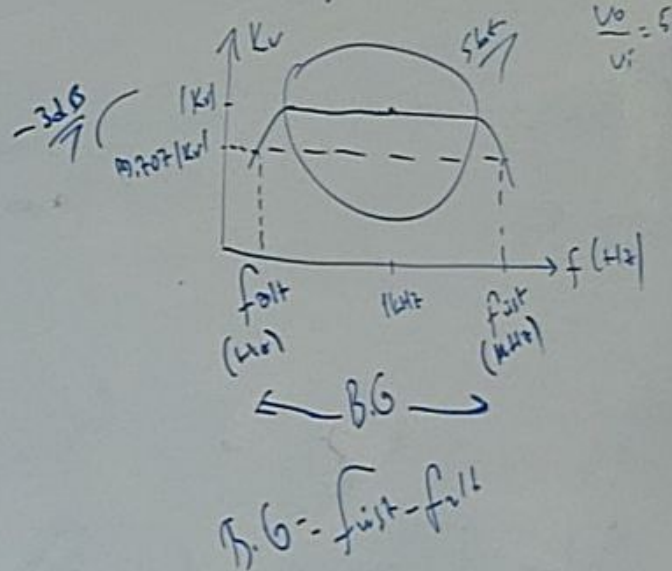
a) $Z_i, Z_o, A_v, A_i = ?$

b) C_s yokken
a şıkkını tekrar çözünüz.

c) Yükleli, Source çıkış olarak tasarlıyorsa
($R_L = 1k$), Z_o, Z_i, A_v, A_i ?



Orta Freq. Bölgesi: $f = 1kHz$



BJT ve JFET Frekans Cevabı

Kapasite - frekans ilişkisi

$$X_c = \frac{1}{\omega c} = \frac{1}{2\pi f c}$$

$C = 2 \text{ pF}$ olsun;

$$f = 1 \text{ Hz} \Rightarrow X_c = \frac{1}{2\pi \cdot 1 \cdot 2 \cdot 10^{-12}} = \frac{10^{12}}{4\pi} \approx \infty \text{ (Açık devre)}$$

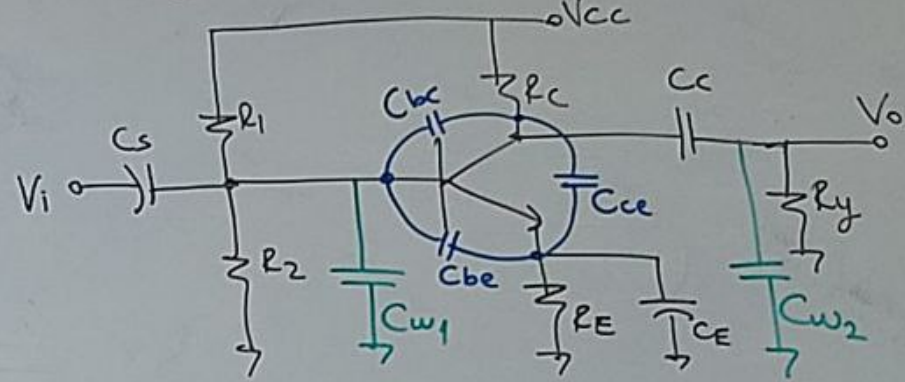
$$f = 10 \text{ MHz} \Rightarrow X_c = \frac{1}{2\pi \cdot 10 \cdot 10^6 \cdot 2 \cdot 10^{-12}} \approx 7,96 \text{ k} \text{ (ihmal edilemez)}$$

$C = 100 \text{ }\mu\text{F}$ olsun;

$$f = 1 \text{ Hz} \Rightarrow X_c = \frac{1}{2\pi \cdot 1 \cdot 100 \cdot 10^{-6}} = 1,6 \text{ k} \text{ (ihmal edilemez)}$$

$$f = 10 \text{ MHz} \Rightarrow X_c = \frac{1}{2\pi \cdot 10 \cdot 10^6 \cdot 100 \cdot 10^{-6}} = \frac{1}{2000\pi} \approx 0 \text{ (kısa devre)}$$

1. BJT Frekans Etkisi



C_s, C_c, C_E : Kutuplama ve köprüleme kond.

μF mertebesinde

Orta ve yüksek frk'da KISA DEVRE

Düşük frk'da İHMAL EDİLEMEZ

$C_{bc}, C_{ce}, C_{be}, C_{w1}, C_{w2}$

Transistörün iç yapısından kaynaklanır

Kabloların kapasiteleri

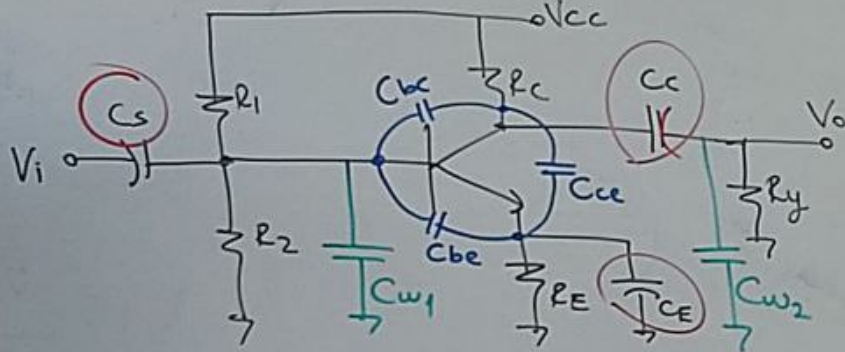
pF mertebesinde

Düşük ve orta frk. AÇIK DEVRE

Yüksek frk. İHMAL EDİLEMEZ

ab)

1. BJT Frekans Etkisi



c7)

C_S, C_C, C_E : Kutuplama ve köprülere kond.
MF mertebesinde

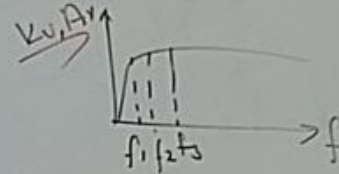
Orta ve yüksek frk'da KISA DEVRE
Düşük frk'da İHMAL EDİLEMEZ

$C_{bc}, C_{ce}, C_{be}, C_{w1}, C_{w2}$:
Tranzistörün iç yapısından kaynaklanır
Kabloların kapasiteleri

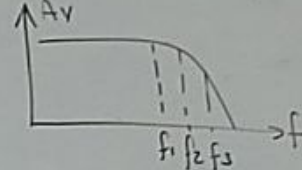
pF mertebesinde
Düşük ve orta frt. AÇIK DEVRE
Yüksek frt. İHMAL EDİLEMEZ

A_v, k_v

* C_S, C_C, C_E düşük frk'da Yüksek Geçiren Filtre

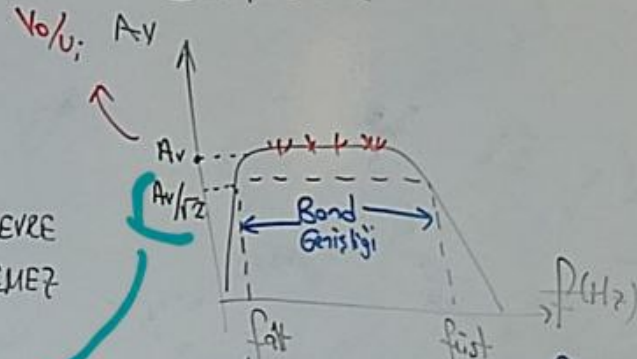


* $C_{bc}, C_{be}, C_{ce}, C_{w1}, C_{w2}$ yüksek frk'larda AÇIK Geçiren Filtre



NOT: Ders notlarından
filtre dereceleri
bölümüne göz at!

Sonuç Olarak



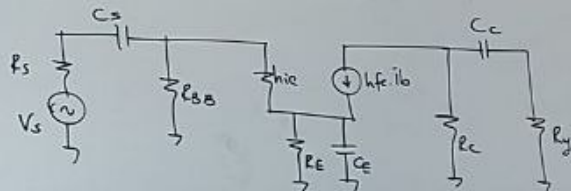
C_S, C_C, C_E etkili
B.G. = $f_2 - f_1$
 $C_{bc}, C_{be}, C_{ce}, C_{w1}, C_{w2}$ etkili

Yarı Güç Noktası

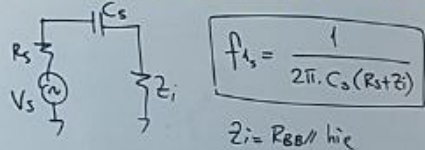
devre)

BJT'nin Frekans Analizi

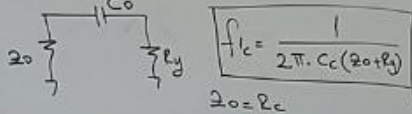
a) Alçak Frekans Analizi ($C_{bc}, C_{ce}, C_{be}, C_{w1}, C_{w2}$: açık devre)



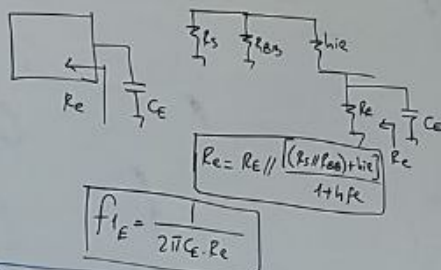
* C_s 'nin Etkisi (C_E ve C_C kısa devre)



* C_c 'nin Etkisi (C_s, C_E : kısa devre)

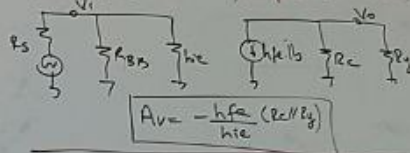


* C_E 'nin Etkisi (C_s, C_C : kısa devre)

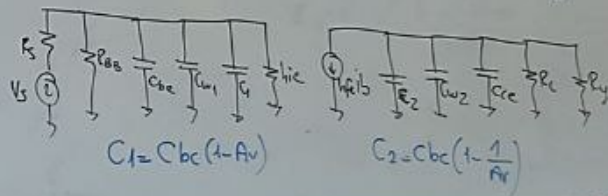
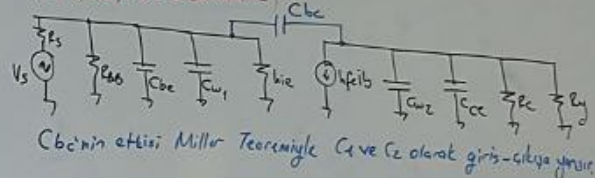


$$f_{alt} = \max(f_{L_s}, f_{L_c}, f_{L_E})$$

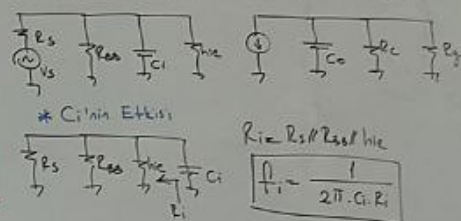
* Orta Frekans Analizi (C_s, C_E, C_C : kısa devre; diğerleri: açık devre)



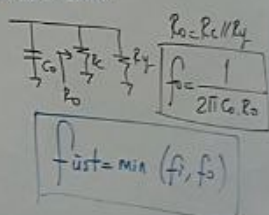
* Yüksek Frekans Analizi (C_c, C_E, C_s : kısa devre)



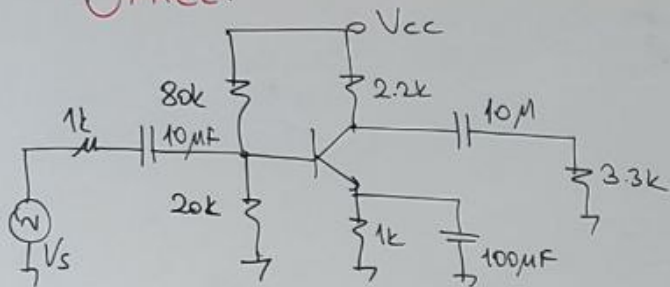
* C_{bc} 'nin Etkisi



* C_o 'nin Etkisi



Örnek:



$h_{ie} = 1.8k$, $h_{fe} = 150$, $C_{bc} = 3pF$, $C_{ce} = 4pF$, $C_{be} = 8pF$
 $C_{w1} = C_{w2} = 10pF$

a) $A_v = ?$

b) $f_{alt} = ?$

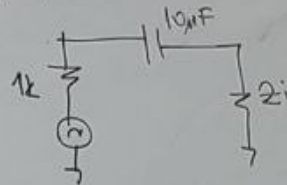
c) $f_{üst} = ?$

a) $A_v = -\frac{h_{ie}(R_c \parallel R_L)}{h_{fe}}$

$A_v = -\frac{1.8k(2.2k \parallel 3.3k)}{150}$

$A_v = -10$

b) * C_s 'nin etkisi:



$Z_i = (80k \parallel 20k) \parallel 1.8k$

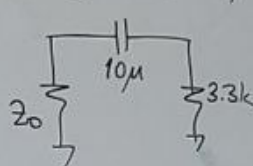
$Z_i = 1.62k$

$f_s = \frac{1}{2\pi \cdot C_s \cdot (1k + Z_i)}$

$f_s = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} (1k + 1.62k)}$

$f_s = 6Hz$

* C_c 'nin etkisi:



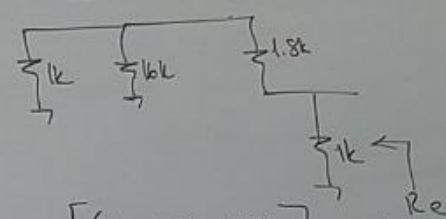
$Z_o = R_c = 2.2k$

$f_c = \frac{1}{2\pi \cdot C_c \cdot (Z_o + 3.3k)}$

$f_c = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} (2.2k + 3.3k)}$

$f_c = 2.9Hz$

* C_E 'nin Etkisi



$R_E = \left[\frac{(1k \parallel 16k) + 1.8k}{100 + 1} \right] \parallel 1k$

$R_E = 17.82\Omega$

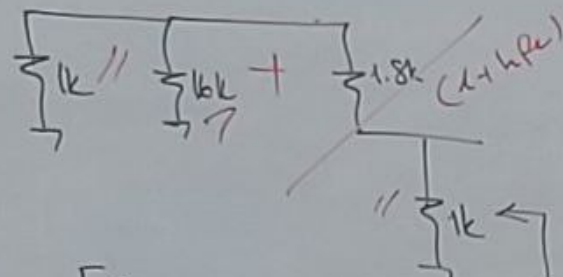
$f_e = \frac{1}{2\pi \cdot C_E \cdot R_E}$

$f_e = \frac{1}{2\pi \cdot 100 \cdot 10^{-6} (17.82)} \Rightarrow f_e = 89.35Hz$

$f_{alt} = \max(6Hz, 2.9Hz, 89.35Hz)$

$f_{alt} = 89.35Hz$

* CE'nin Etkisi



$$R_e = \left[\frac{(1k // 16k) + 1.8k}{150 + 1} \right] // 1k$$

$$R_e = 17.82 \Omega$$

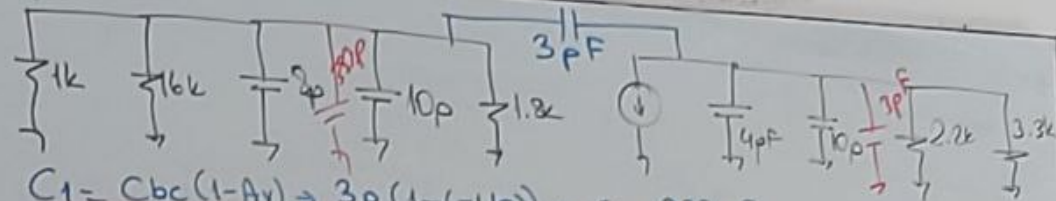
$$f_e = \frac{1}{2\pi \cdot C_E \cdot R_e}$$

$$f_e = \frac{1}{2\pi \cdot 100 \cdot 10^{-6} \cdot (17.82)} \Rightarrow \boxed{f_e = 89.35 \text{ Hz}}$$

$$f_{aH} = \max(6 \text{ Hz}; 2.9 \text{ Hz}; 89.35 \text{ Hz})$$

$$\boxed{f_{aH} = 89.35 \text{ Hz}}$$

c)



$$C_1 = C_{bc}(1 - A_v) \Rightarrow 3p(1 - (-110)) \Rightarrow C_1 = 333pF$$

$$C_2 = C_{bc}(1 - \frac{1}{A_v}) \Rightarrow 3p(1 - \frac{1}{(-110)}) \Rightarrow C_2 \approx 3pF$$

$$C_i = 8p + 10p + 333pF \Rightarrow C_i = 351pF$$

$$C_o = 4p + 10p + 3p = 17p$$

Giriş için;

$$f_i = \frac{1}{2\pi \cdot C_i \cdot R_i}$$

$$R_i = 1k // 16k // 1.8k$$

$$R_i = 0.617k$$

$$f_i = \frac{1}{2\pi \cdot (351 \cdot 10^{-12}) \cdot (0.617 \cdot 10^3)} = 735.2 \text{ kHz}$$

Çıkış için

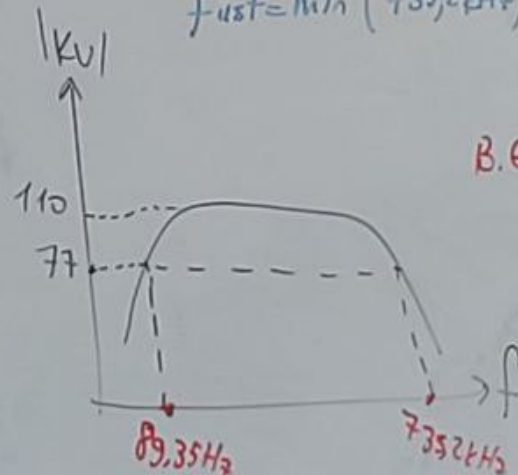
$$f_o = \frac{1}{2\pi \cdot C_o \cdot R_o}$$

$$R_o = 2.2k // 3.3k$$

$$R_o = 1.32k$$

$$f_o = \frac{1}{2\pi \cdot (17 \cdot 10^{-12}) \cdot (1.32 \cdot 10^3)} = 7.09 \text{ MHz}$$

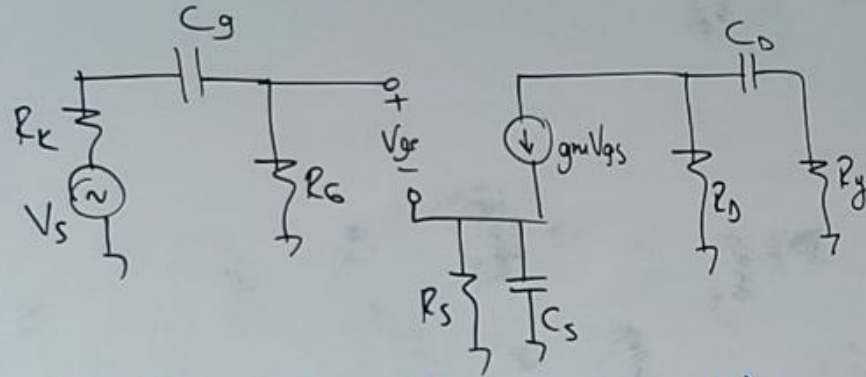
$$f_{üst} = \min(735.2 \text{ kHz}; 7.09 \text{ MHz}) \Rightarrow \boxed{f_{üst} = 735.2 \text{ kHz}}$$



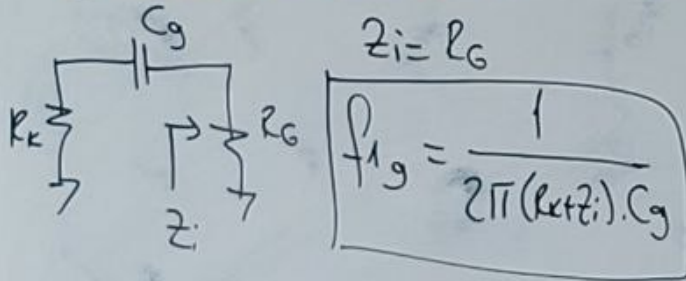
$$B.G = 735.2 \text{ kHz} - 89.35 \text{ Hz}$$

FET'in Frekans Analizi

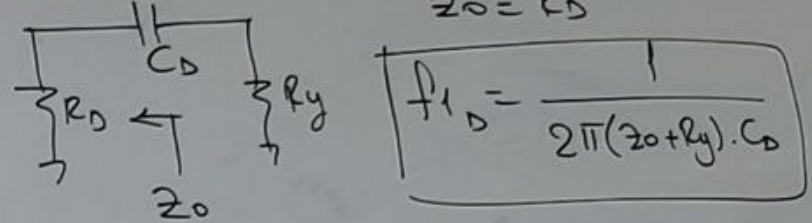
1. Dc Frekans Analizi ($C_{gs}, C_{ds}, C_{gd}, C_{w1}, C_{w2}$ Açık devre)



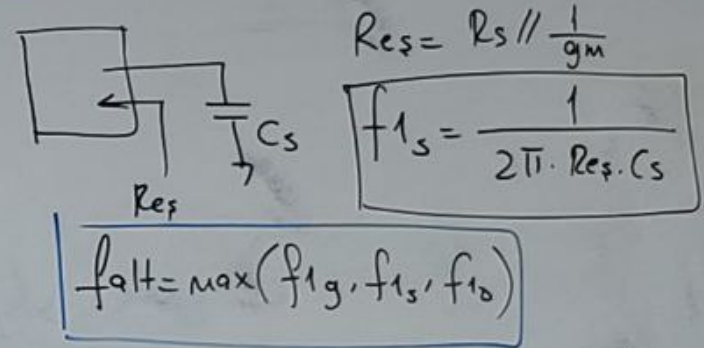
* C_g 'nin Etkisi (C_s ve C_D : kısa dev.)



* C_D 'nin Etkisi (C_s ve C_g : kısa dev.)



* C_s 'in Etkisi (C_g ve C_D : kısa dev.)



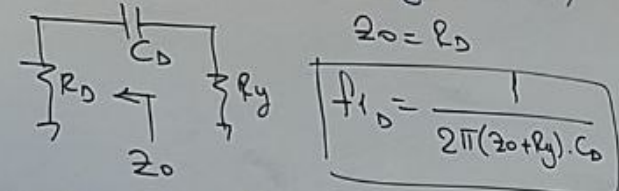
2. Orta Frekans Analizi (C_g, C_s, C_D : kısa dev.) diğerleri: açık dev.)

$$A_v = -g_m (R_D \parallel R_L)$$

li zi

C_{gd}, C_{w1}, C_{w2}
ile devre

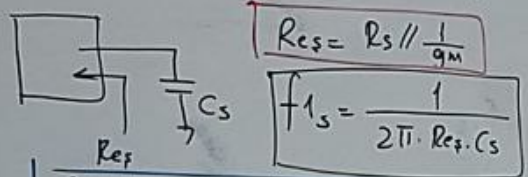
* C_D 'nin Etkisi (C_{gs} ve C_D : kısa dev.)



$$Z_o = R_D$$

$$f_{1D} = \frac{1}{2\pi(Z_o + R_D) \cdot C_D}$$

* C_S 'in Etkisi (C_{gs} ve C_D : kısa dev.)



$$R_{eq} = R_S \parallel \frac{1}{g_m}$$

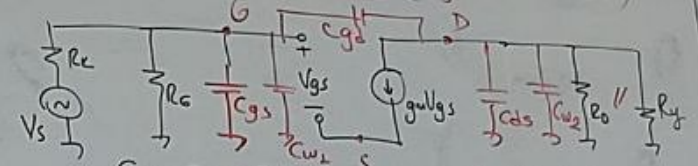
$$f_{1S} = \frac{1}{2\pi \cdot R_{eq} \cdot C_S}$$

$$f_{alt} = \max(f_{1g}, f_{1s}, f_{1D})$$

2 Orta Frekans Analizi (C_{gs}, C_{gs}, C_D : kısa dev.)
diğerleri: açıl dev.

$$A_v = -g_m (R_D \parallel R_L)$$

3. Yüksek Frekans Analizi (C_{gs}, C_{gs}, C_D : kısa dev.)



Miller Teo $\Rightarrow$

$$C_i = C_{gs} + C_{w1} + C_1$$

$$C_1 = C_{gd} (1 - A_v)$$

$$R_i = R_k \parallel R_g$$

$$f_i = \frac{1}{2\pi \cdot R_i \cdot C_i}$$

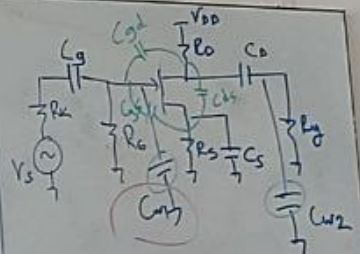
$$C_o = C_{ds} + C_{w2} + C_2$$

$$C_2 = C_{gd} (1 - \frac{1}{A_v})$$

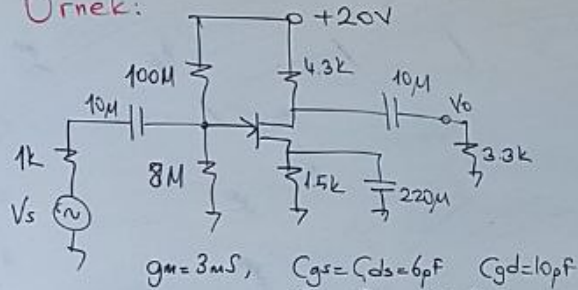
$$R_o = R_D \parallel R_L$$

$$f_o = \frac{1}{2\pi \cdot R_o \cdot C_o}$$

$$f_{üst} = \min(f_i, f_o)$$



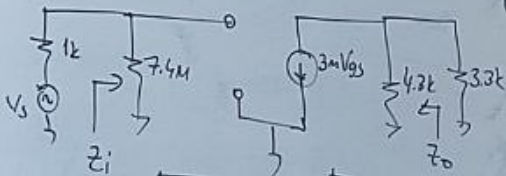
Örnek:



$A_v, f_{alt}, f_{ist} = ?$

$$A_v = -g_m(R_o || R_L)$$

$$A_v = -3m(4.3k || 3.3k)$$

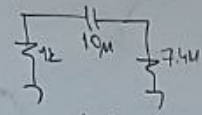


$$Z_i = 7.4M$$

$$Z_o = 4.3k$$

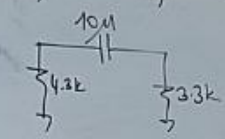
$$A_v = -5.6$$

f_{alt} için;



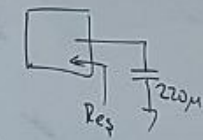
$$f_{1g} = \frac{1}{2\pi(1k + 7.4M) \cdot 10 \cdot 10^{-6}}$$

$$f_{1g} = 0.002 \text{ Hz}$$



$$f_{1b} = \frac{1}{2\pi(4.2k + 3.3k) \cdot 10 \cdot 10^{-6}}$$

$$f_{1b} = 2.08 \text{ Hz}$$



$$R_{eq} = R_s || \frac{1}{g_m} = 15k || \frac{1}{3mS}$$

$$R_{eq} = 0.27k$$

$$f_{1s} = \frac{1}{2\pi(0.27k)(220 \cdot 10^{-6})}$$

$$f_{1s} = 2.68 \text{ Hz}$$

$$f_{alt} = \max(0.002 \text{ Hz}; 2.08 \text{ Hz}; 2.68 \text{ Hz})$$

$$f_{alt} = 2.68 \text{ Hz}$$

f_{ist} için;

$$C_i = C_{gs} + C_{w1} + C_1$$

$$C_1 = C_{gd}(1 - A_v)$$

$$C_1 = 10p(1 - (-5.6)) = 66p$$

$$C_i = 6p + 20p + 66p$$

$$C_i = 92pF$$

$$R_i = 1k || 7.4M \approx 1k$$

$$f_i = \frac{1}{2\pi \cdot 92 \cdot 10^{-12} \cdot 1 \cdot 10^3}$$

$$f_i = 1.73 \text{ MHz}$$

$$C_o = C_{ds} + C_{w2} + C_2$$

$$C_2 = C_{gd}(1 - \frac{1}{A_v})$$

$$C_2 = 10p(1 - \frac{1}{-5.6}) = 11.78p$$

$$C_o = 6p + 20p + 11.78p$$

$$C_o = 37.78pF$$

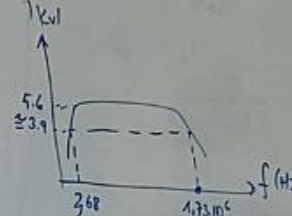
$$R_o = 4.3k || 3.3k = 1.86k$$

$$f_o = \frac{1}{2\pi \cdot (1.86 \cdot 10^3) \cdot (37.78 \cdot 10^{-12})}$$

$$f_o = 2.26 \text{ MHz}$$

$$f_{ist} = \min(1.73 \text{ MHz}; 2.26 \text{ MHz})$$

$$f_{ist} = 1.73 \text{ MHz}$$

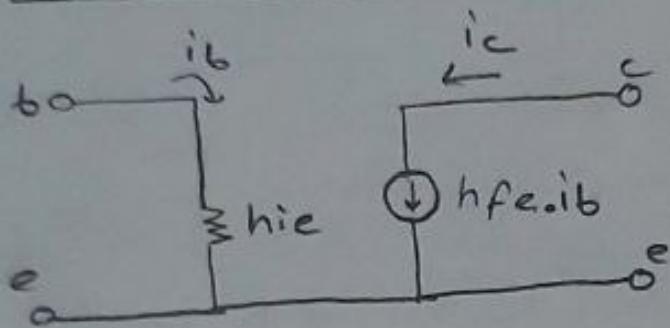


(1)

Transistörlerin AC eşdeğer devre modelleri

1) BJT & ScwK Sinyal modelleri

Yaklaşık hibrid Model



$$\beta = h_{fe} \rightarrow i_c = h_{fe} \cdot i_b$$

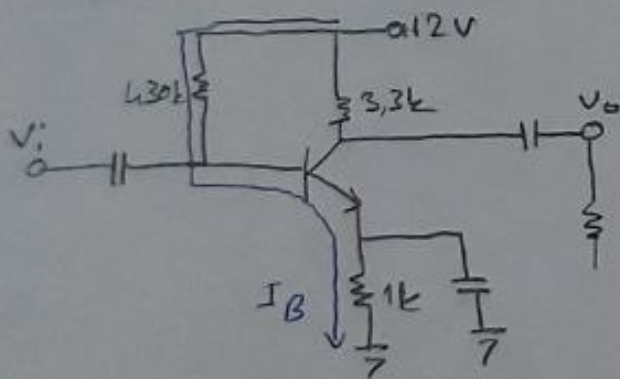
$$h_{ie} = h_{fe} \cdot r_e$$

★ Eğer soruda h_{ie} verilmezse r_e bul!

$$r_e = \frac{26 \text{ mV}}{i_e}$$

i_e , Dc emetör akımıdır.
Dc emetör akımı hesapla.

$$h_{ie} = \beta \cdot r_e$$



h_{ie} verilmemişse Dc Akımlar

$$12 \text{ V} = 430 \text{ k} \cdot I_B + 0.7 + 101 \cdot I_B \cdot 1 \text{ k}$$

$$I_B = 21,28 \mu\text{A}$$

$$I_E = I_B + I_C$$

$$I_E = (\beta + 1) I_B$$

$$I_E = 2,15 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{2,15 \text{ mA}} = 12,1 \Omega$$

$$r_e = 12,1 \Omega$$

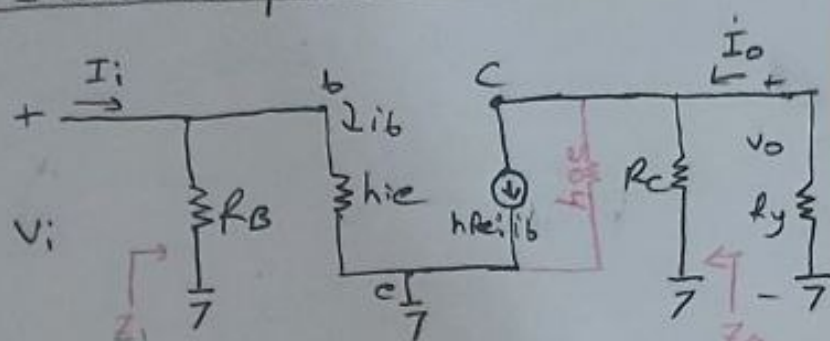
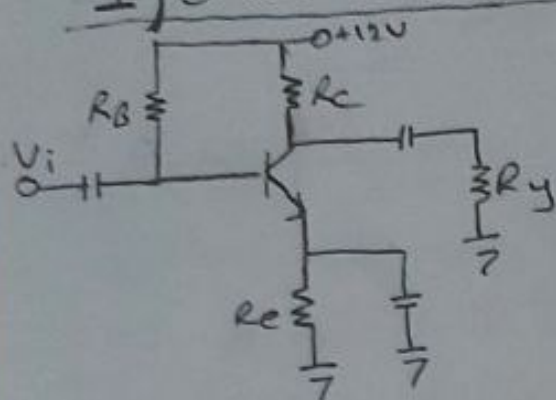
$$h_{ie} = h_{fe} \cdot r_e = 100 \cdot (12,1 \Omega)$$

$$h_{ie} = 1,2 \text{ k}\Omega$$

(2)

2) BJT'li Yükseltgeç Devrelerinin Analizi

1) Ortak Emitör'lü Sabit Kutuplamalı Yükseltgeç



hoe Yokken

$$Z_i = R_B \parallel h_{ie} \quad Z_o = R_C$$

$$V_i = i_b \cdot h_{ie} \quad V_o = -h_{fe} \cdot i_b (R_C \parallel R_L)$$

Ai = Standart

Av = Standart

$$A_i = -A_v \frac{Z_i}{R_L}$$

Av, Ai = Standart

hoe Yokken $\rightarrow R_o = \frac{1}{h_{oe}} = 50k$

$$Z_i = R_B \parallel h_{ie}$$

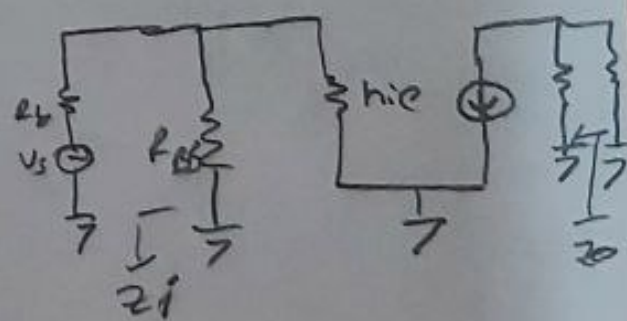
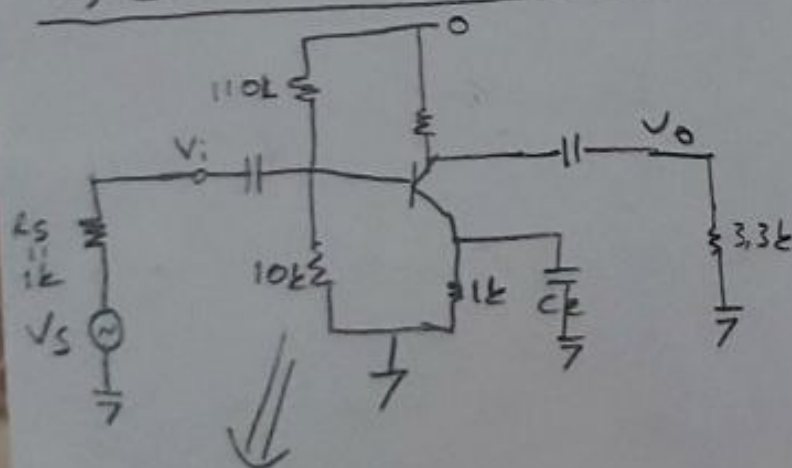
$$V_o = -h_{fe} \cdot i_b (R_C \parallel R_L \parallel \frac{1}{h_{oe}})$$

$$Z_o = R_C \parallel \frac{1}{h_{oe}}$$

$$V_i = h_{ie} \cdot i_b$$

$$A_v = \frac{V_o}{V_i} \quad A_i = -A_v \cdot \frac{Z_i}{R_L} \quad A_{vs} = A_v \cdot \frac{Z_i}{Z_i + R_s}$$

2) Gerilim Bölücü'lü Yükseltgeç

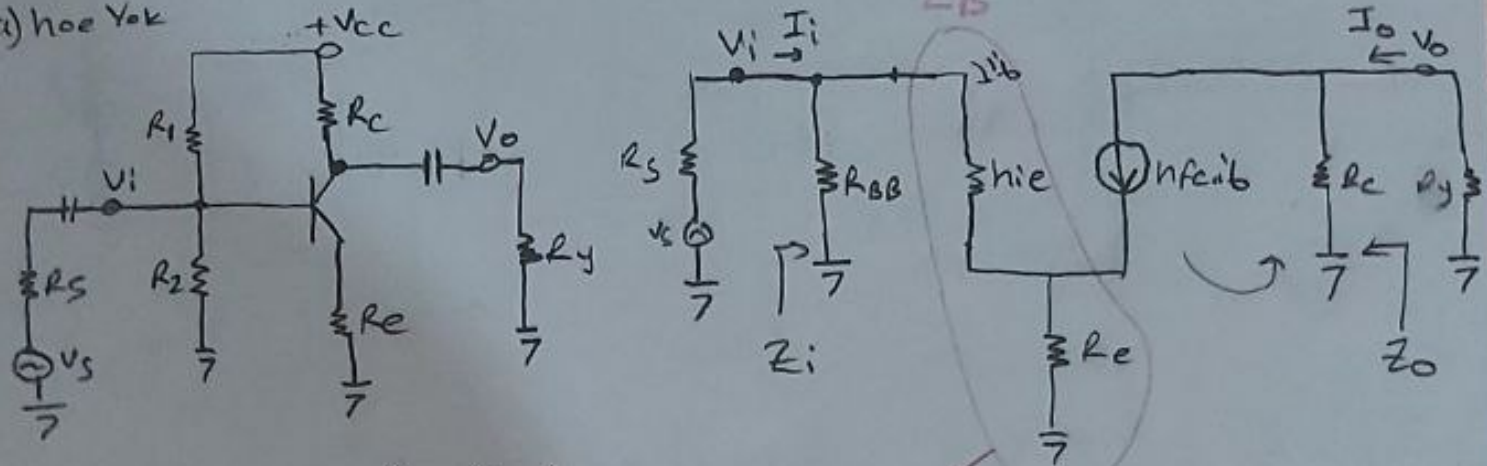


$$110k \parallel 10k \rightarrow R_{BB}$$

→ Tüm hesaplamalar yukarıdaki
ile aynı
Sadece RBB bulunmalı

3. Emeter Köproblemini Yönetmek (hoe Yok)

a) hoe Yok



$A_v, A_i, A_v \rightarrow$ standart

A_v

$V_i = i_b \cdot Z_B$

$V_o = -h_{fe} \cdot i_b (R_C \parallel R_L)$

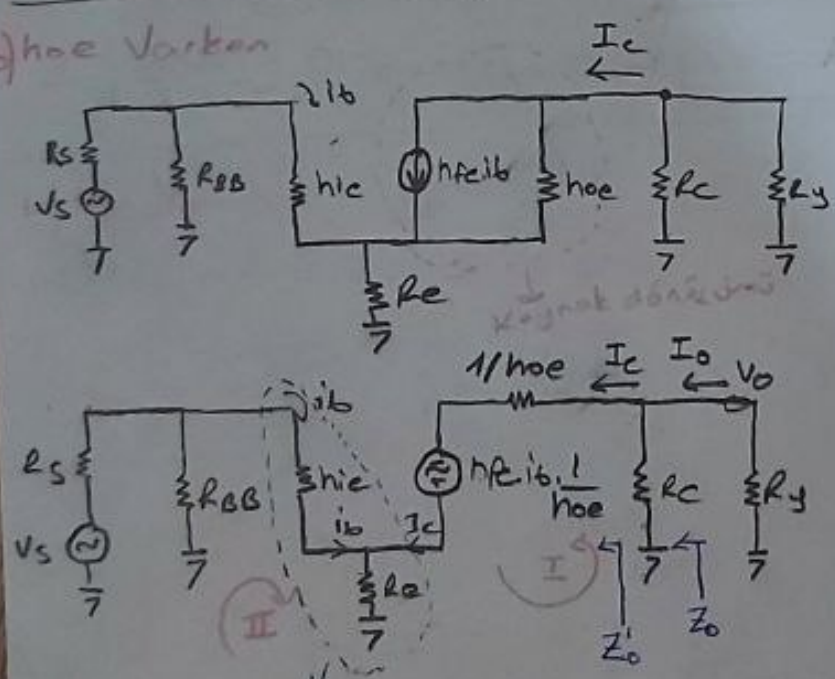
$V_s \rightarrow 0 \Rightarrow i_b = 0$

$Z_o = R_C$

$Z_B = \frac{V_i}{i_b} = \frac{h_{ie} \cdot i_b + R_E(h_{fe}+1)i_b}{i_b}$

$Z_i = R_{BB} \parallel Z_B$

b) hoe Varken



① Z_i bulmak için

$Z_B = \frac{V_i}{i_b} = \frac{h_{ie} \cdot i_b + R_E(I_b + I_c)}{i_b}$

I_c 'yi I_b cinsinden bulmak için

① numaralı devreyi kullan

$R_E(i_b + I_c) + (R_C \parallel R_L) \cdot I_c + \frac{1}{h_{oe}} \cdot I_c = \frac{h_{fe} \cdot i_b}{h_{oe}}$

$I_c = A \cdot i_b \Rightarrow Z_B = \frac{h_{ie} \cdot i_b + R_E(i_b + A \cdot i_b)}{i_b} \Rightarrow Z_B = h_{ie} + R_E[1+A] \Rightarrow Z_i = Z_B \parallel R_{BB}$

② A_v 'yi bulmak için

$I_c = A \cdot i_b \Rightarrow V_o = -I_c(R_C \parallel R_L) = -A \cdot i_b(R_C \parallel R_L)$

$V_i = i_b \cdot Z_B \Rightarrow A_v = \frac{V_o}{V_i}$

③ $Z_o = R_C \parallel Z_o'$

$Z_o' = \frac{V_o}{I_c} = \frac{I_c \cdot \frac{1}{h_{oe}} - \frac{h_{fe} \cdot i_b}{h_{oe}} + R_E(i_b + I_c)}{I_c}$

i_b 'leri I_c cinsinden yazmak için

② numaralı devreyi yaz

$[(R_S \parallel R_B) + h_{ie}] \cdot i_b + i_b \cdot R_E + I_c \cdot R_E = 0$

$i_b = -\beta \cdot I_c$

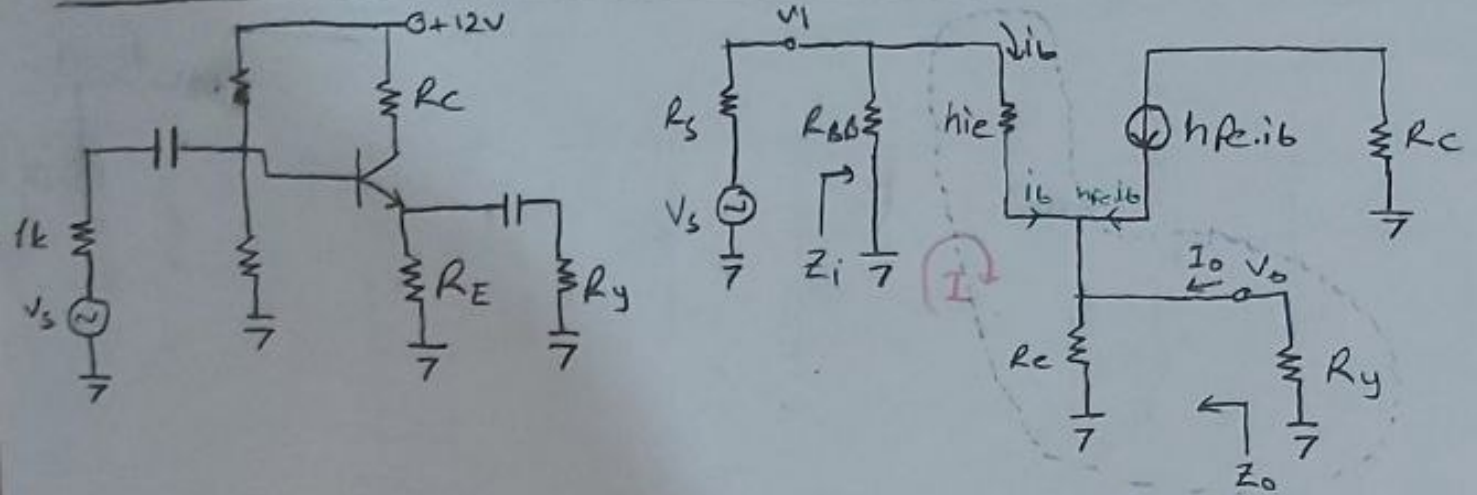
$Z_o' = \frac{I_c \cdot \frac{1}{h_{oe}} - \frac{h_{fe} \cdot \beta \cdot I_c}{h_{oe}} + R_E \cdot I_c(1+\beta)}{I_c}$

$Z_o' = \dots$

$Z_o = Z_o' \parallel R_C$

4) Emetör Gıkışlı Vgkseteq (hoe Yok)

④



$$Z_B = \frac{V_i}{i_b} = \frac{h_{ie} \cdot i_b + (R_E \parallel R_Y)(1+h_{fe}) \cdot i_b}{i_b}$$

Z_B

$$Z_B = h_{ie} + (R_E \parallel R_Y)(1+h_{fe})$$

$$Z_i = R_B \parallel Z_B$$

V_o için

$$V_o = (1+h_{fe}) \cdot i_b (R_E \parallel R_Y)$$

$$V_i = i_b \cdot Z_B$$

$$A_v = \frac{V_o}{V_i}$$

$$Z_o = \frac{V_o}{I_o}$$

$$V_o = R_E (I_o + i_b(1+h_{fe}))$$

i_b 'yi I_o cinsinden yazmak için $\textcircled{I}$ node deneyi kullan.

$$\textcircled{I} [(R_S \parallel R_B) + h_{ie}] \cdot i_b + R_E [I_o + i_b(1+h_{fe})] = 0$$

$$i_b [A] = -R_E \cdot I_o$$

$$i_b = -\frac{R_E}{A} \cdot I_o$$

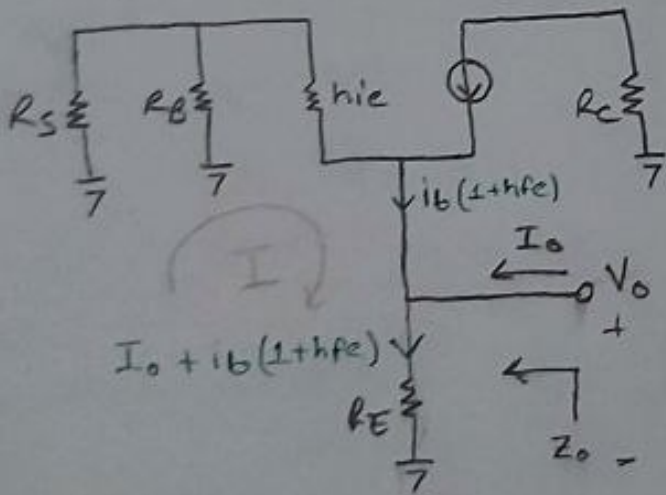
$$V_o = R_E (I_o - \frac{R_E}{A} \cdot I_o (1+h_{fe}))$$

$$V_o = I_o (---)$$

$$Z_o = \frac{V_o}{I_o} = ---$$

Z_o 'ı bulmak için

V_S uçları ile kısa devre

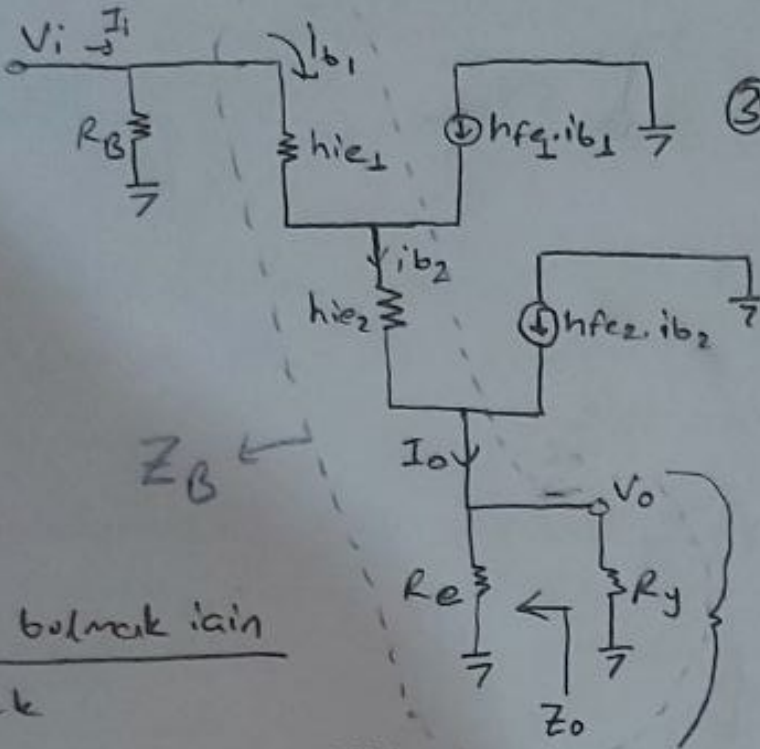


Z_o formülü

$$\frac{(R_S \parallel R_B) + h_{ie}}{h_{fe} + 1} \parallel R_E$$

Emetör çıkışı

5.) Darlington Cıftı



$$② Z_B = \frac{V_i}{i_{b1}}$$

$$③ V_i = h_{ie1} \cdot i_{b1} + h_{ie2} \cdot i_{b2} + (R_E \parallel R_L) I_O$$

$$Z_i = R_B \parallel Z_B$$

$$A_v = \frac{V_o}{V_i} = \frac{I_O (R_E \parallel R_L)}{Z_B \cdot i_{b1}}$$

$$A_i = -A_v \left(\frac{Z_i}{R_L} \right)$$

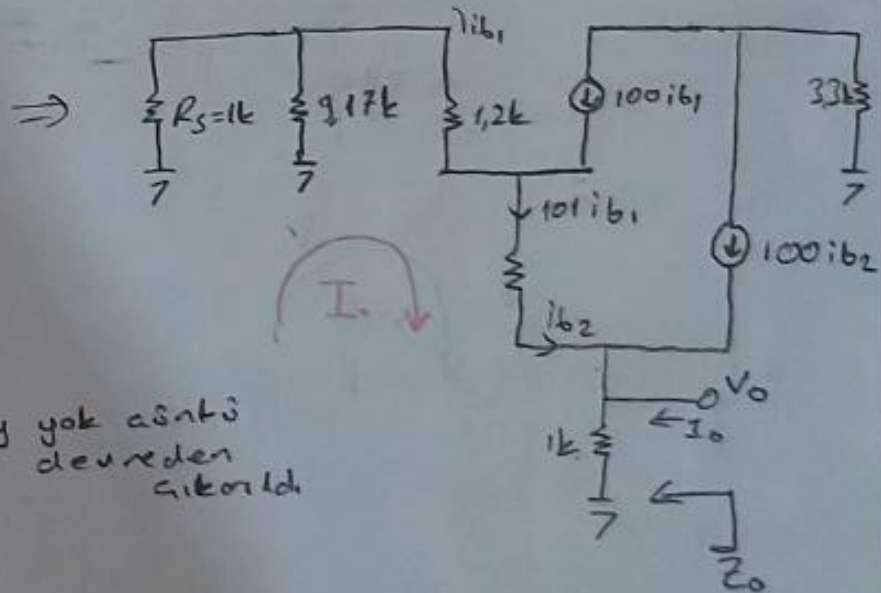
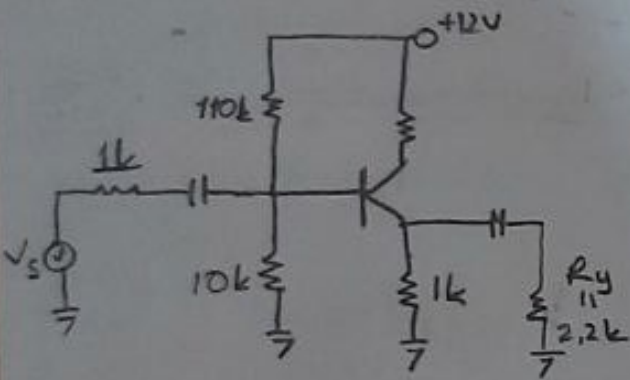
$$① i_{b2} = (1 + h_{fe1}) i_{b1} \quad ⑤$$

$$I_O = (1 + h_{fe2}) i_{b2}$$

$$I_O = (1 + h_{fe2})(1 + h_{fe1}) i_{b1}$$

Z_o bulmak için

Örnek



$$① V_o = (101 i_{b2} + I_O) \cdot 1k$$

R_L yok aşıtı
devreden
çıkarıldı

$$② V_o = (101(101 i_{b1}) + I_O) \cdot 1k$$

$$③ I_{devreden} i_{b1} = \dots I_O \text{ şeklinde bir bağlantı elde ederiz}$$

$$[(1k \parallel 9.17k) + 1.2k] i_{b1} + 1.2k(101 i_{b1}) + 1k(101000 i_{b1} + I_O)$$

$$i_{b1} = - \frac{1}{1023.2} \cdot I_O$$

$$① \rightarrow V_o = (101 \cdot 101 i_{b1} + I_O) \cdot 1k$$

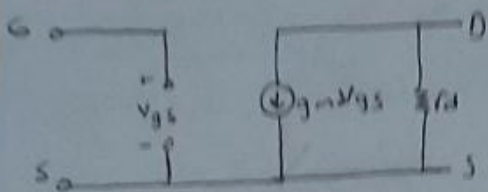
$$\frac{V_o}{I_O} = 12,10 \Omega$$

Önemli Not

4) Emeter çıkışı ve 5)
Darlington çifti aygıtın
Darlington çiftinde
sadece V_i , V_o , Z_o bulur
ken tsm akımları i_{b1}
cişinden yazmalısın

Bunun için
en yukarıdaki
① nolu eşitliği kullan

JFET ve Kanallı Akışkanlı MOSFET için

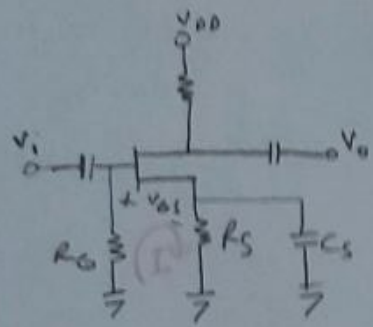


g_m = Fet genlik Hassasiyeti

$$g_m = \frac{2 I_{DSS}}{|V_p|} \left[1 - \frac{V_{GS}}{V_p} \right]$$

$$g_{mo} = \frac{2 I_{DSS}}{|V_p|}$$

I_{DSS} ve V_p soruda verilir. Eğer soruda g_m verilmezse DC analiz yapılarak V_{GS} bulunur



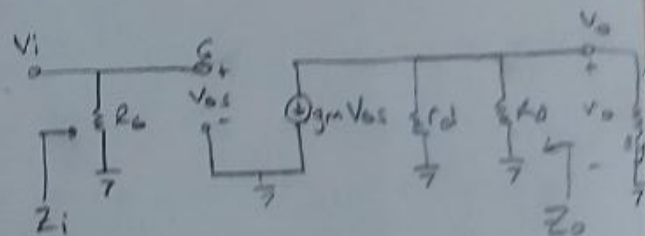
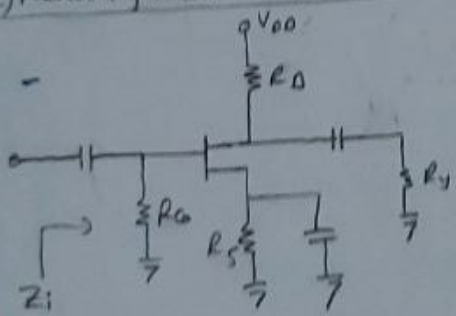
Önemli Not: V_{GS} de analizle bulunurken I_D değeriye gerekiz. Ancak yazarken R_G üzerine düşen gerilimi hesaplamamız

$$V_{GS} + I_D \cdot R_S = 0 \Rightarrow I_D = -\frac{V_{GS}}{R_S}$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2 \rightarrow \text{Buradan } V_{GS} \text{ bulunur}$$

g_m bu şekilde $g_m = \frac{2 I_{DSS}}{|V_p|} \left[1 - \frac{V_{GS}}{V_p} \right]$ bulunur

1) Kendiliğinden Kutuplanmalı Devre



① DC analizde g_m bulunur.

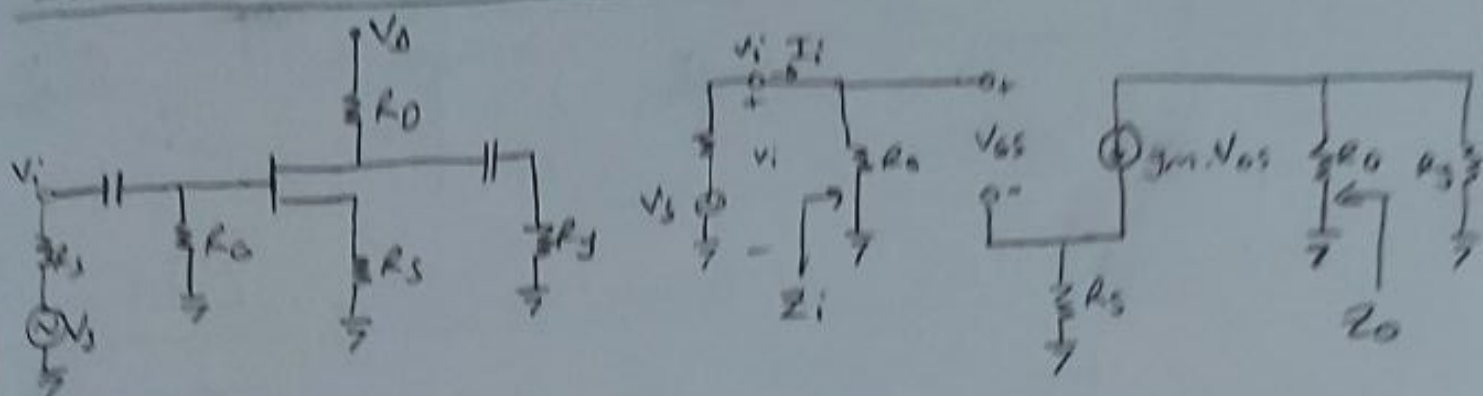
② $Z_i = R_G$ $Z_o = r_d \parallel R_D$

$$\left. \begin{aligned} \text{③ } V_o &= -g_m \cdot V_{gs} (r_d \parallel R_D) \\ V_i &= V_{gs} \end{aligned} \right\} A_v = \frac{V_o}{V_i} = -g_m (R_D \parallel r_d)$$

$$A_i = -A_v \cdot \frac{Z_i}{R_y}$$

2. Source Dirençli Yükseltici (Full-yük)

(7)

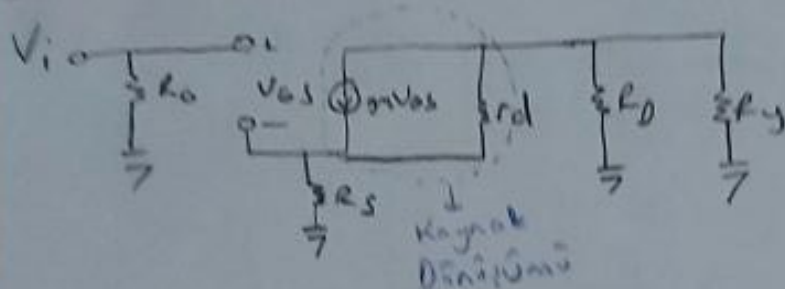


$Z_i = R_g$ Z_o için V_s ve V_i kısa devre alırsa $V_{gs} = 0$ olur. Akım kaynağına kısa devre $\rightarrow Z_o = R_D$

$V_o = -g_m V_{gs} (R_D \parallel R_L)$ $A_v = \frac{V_o}{V_i} = \frac{-g_m (R_D \parallel R_L)}{1 + g_m R_s}$ $A_i = -A_v \cdot \frac{Z_i}{R_L}$

$V_i = V_{gs} + g_m \cdot V_{gs} \cdot R_s$

(1) Yükten ($r_d = \frac{1}{Y_{os}}$)



③ Z_o bulmak için önce Z_o' bulunur.

$Z_o' = \frac{V_T}{I_d}$ $V_{gs} = \dots I_d$ bulunmalı.

④ $V_T = I_d \cdot r_d + R_s \cdot I_d - g_m V_{gs} \cdot r_d$

Z_o için I_d ile V_{gs} arasındaki ilişkiyi $V_i = 0 \rightarrow V_{gs} + R_s \cdot I_d = 0$ $V_{gs} = -R_s \cdot I_d$

⑤ $V_T = \dots I_d + \dots I_d + \dots I_d$

⑥ $Z_o' = \frac{V_T}{I_d} = \dots$

$Z_o = Z_o' \parallel R_D$

① $Z_i = R_g$

② $V_o = -I_d (R_D \parallel R_L)$

$V_i = V_{gs} + I_d R_s$

$I_d = \dots V_{gs}$ + r_{ds} üzerinden ifade etmek için (I) 'naly. $I_d = \dots V_{gs}$ g_m ile V_{gs} arasındaki ilişkiyi $V_i = 0$ ve V_{gs} ile I_d arasında yazılır. Sonra V_o ve V_i ile I_d yerine V_o ve V_i yazılır. V_o ve V_i I_d yerine V_o ve V_i yazılır. V_o ve V_i I_d yerine V_o ve V_i yazılır.

$I_d \cdot R_s + I_d \cdot r_d + I_d (R_D \parallel R_L) = g_m V_{gs} r_d$

$I_d = \frac{g_m V_{gs} r_d}{R_s + r_d + (R_D \parallel R_L)}$

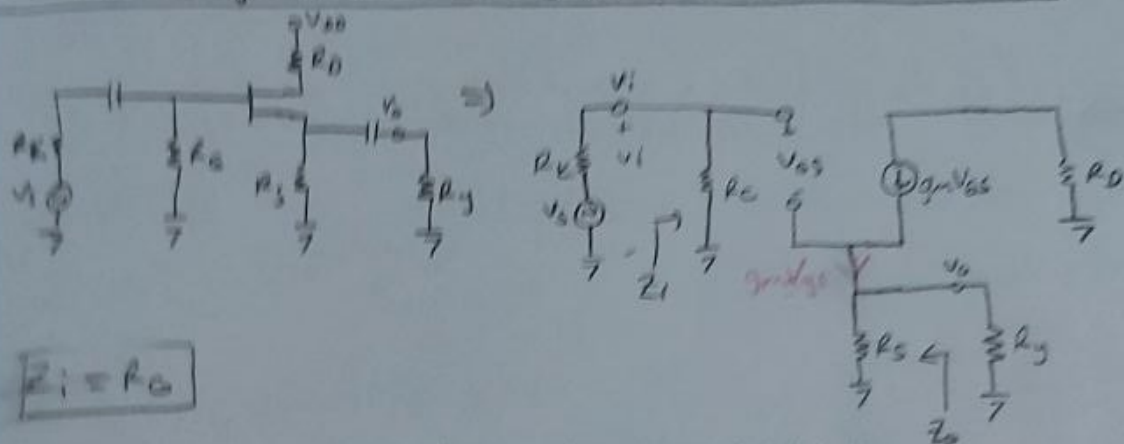
$\Rightarrow A_v = \frac{-g_m \cdot r_d (R_D \parallel R_L)}{g_m r_d R_s + r_d + R_s + (R_D \parallel R_L)}$

Volta) B513.012 Konfigürasyon

(1)



3) Source izleyicili (Ortak Drain) yapısalatın (Source çıkışlı)



$$Z_i = R_G$$

$V_o = g_m V_{gs} (R_S \parallel R_L) \rightarrow$ Akım yakardan indirdiği için
Güçte - yok.

$$V_i = V_{gs} + g_m V_{ds} (R_S \parallel R_L)$$

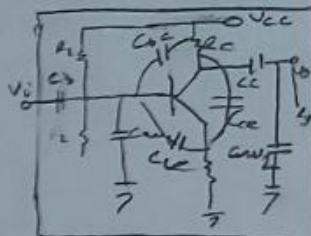
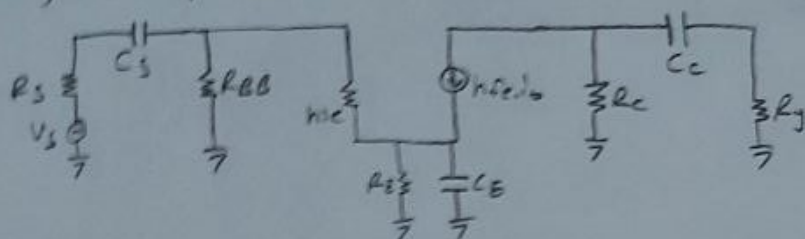
Source çıkışlı için $Z_o \Rightarrow$
$$Z_o = \frac{R_S}{1 + g_m R_S}$$

$\rightarrow$ Ezberlenicek

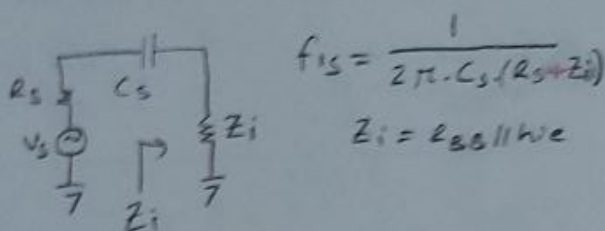
Ortams Frekans Analizi:

(C_s, C_c, C_e hesabi)

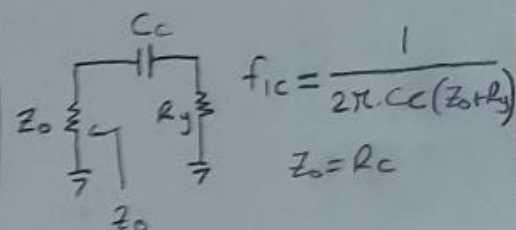
a) Aşağı frekans analizi ($C_{bc}, C_{be}, C_{ce}, C_{w1}, C_{w2}$ acık devre)



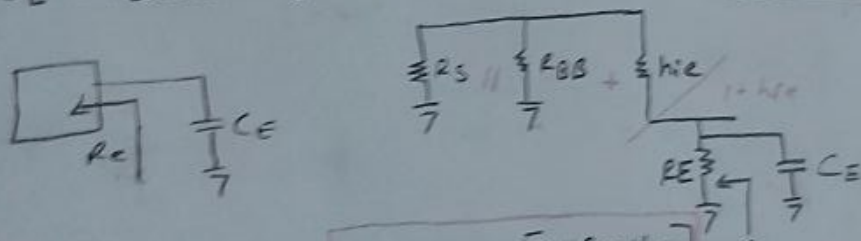
1) C_s 'nin etkisi (C_E u C_c kısa devre)



2) C_c 'nin etkisi (C_s, C_E kısa devre)



3) C_E 'nin Etkisi (C_s, C_c kısa devre)

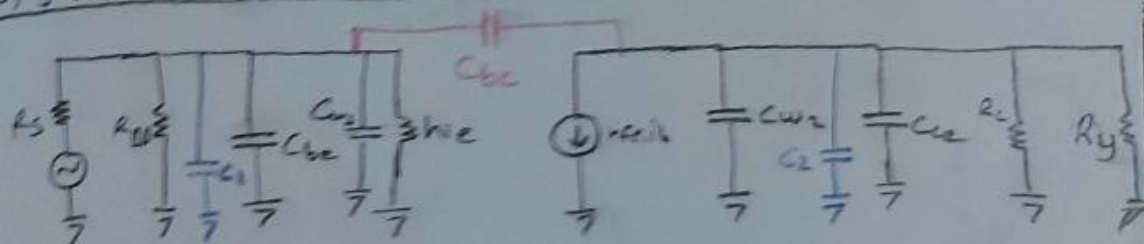


$$f_{IE} = \frac{1}{2\pi \cdot C_E R_E}$$

$$R_E = R_E \parallel \frac{R_s \parallel R_{BB} + h_{ie}}{1 + h_{fe}}$$

→ Etki en az

b) Yüksek Frekans Analizi (C_s, C_c, C_E = Kısa devre)

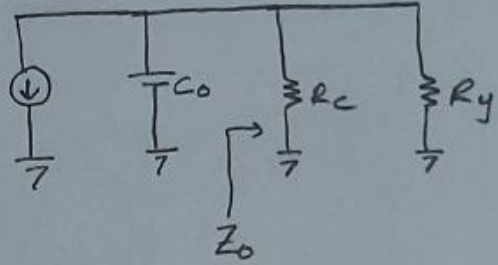
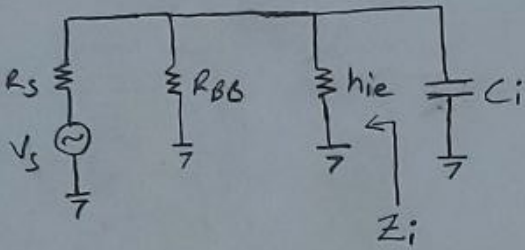


① C_{bc} → ② C_{bc} → C_1, C_2

$$C_1 = C_{bc}(1 - A_v) \quad C_2 = C_{bc}(1 - \frac{1}{A_v})$$

$$C_i = C_{be} + C_{w1} + C_1$$

$$C_o = C_{ce} + C_{w2} + C_2$$



C_i 'nin Etkisi

$$R_i = R_s \parallel R_B \parallel h_{ie}$$

$$f_i = \frac{1}{2\pi \cdot C_i \cdot R_i}$$

C_o 'nin Etkisi

$$R_o = R_c \parallel R_y$$

$$f_o = \frac{1}{2\pi \cdot C_o \cdot R_o}$$

$$f_{üst} = \min(f_i, f_o)$$

E_z berlenecek formüller

Emetör çıkışlı $R_o \rightarrow Z_o = \frac{(R_s \parallel R_B) + h_{ie}}{h_{fe} + 1} \parallel R_E$

Source çıkışlı $\rightarrow Z_o = \frac{R_s}{1 + g_m R_s}$

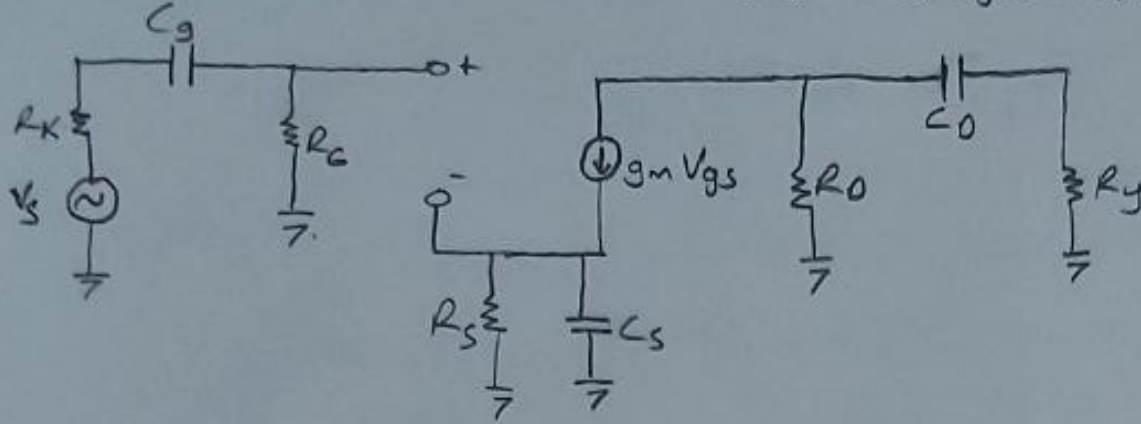
C_E hesabı $= \frac{(R_s \parallel R_B) + h_{ie}}{1 + h_{fe}} \parallel R_E = R_e$ $f_{iE} = \frac{1}{2\pi C_E R_e}$

C_s hesabı $= R_s \parallel \frac{1}{g_m} \rightarrow f_{is} = \frac{1}{2\pi C_s R_s}$

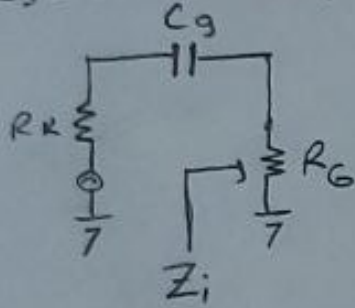
FET frekans Analizi

(11)

d) Aşağı Frekans Analizi (C_{gs} , C_{ds} , C_{gd} , C_{w1} , C_{w2} Aşağı Fre)



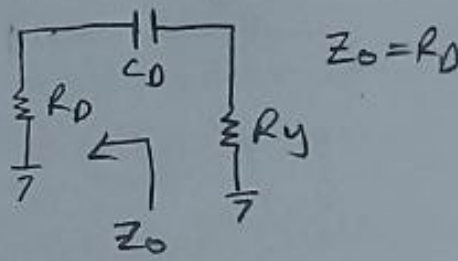
1) C_g 'nin etkisi
(C_S ve C_D 'yi kısa devre)



$$Z_i = R_G$$

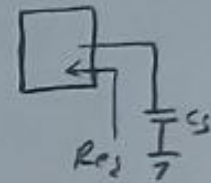
$$f_{ig} = \frac{1}{2\pi \cdot C_g (R_K + Z_i)}$$

2) C_D 'nin Etkisi
(C_S ve C_g 'yi kısa devre)



$$f_{io} = \frac{1}{2\pi \cdot C_D (Z_o + R_L)}$$

3) C_S 'nin Etkisi



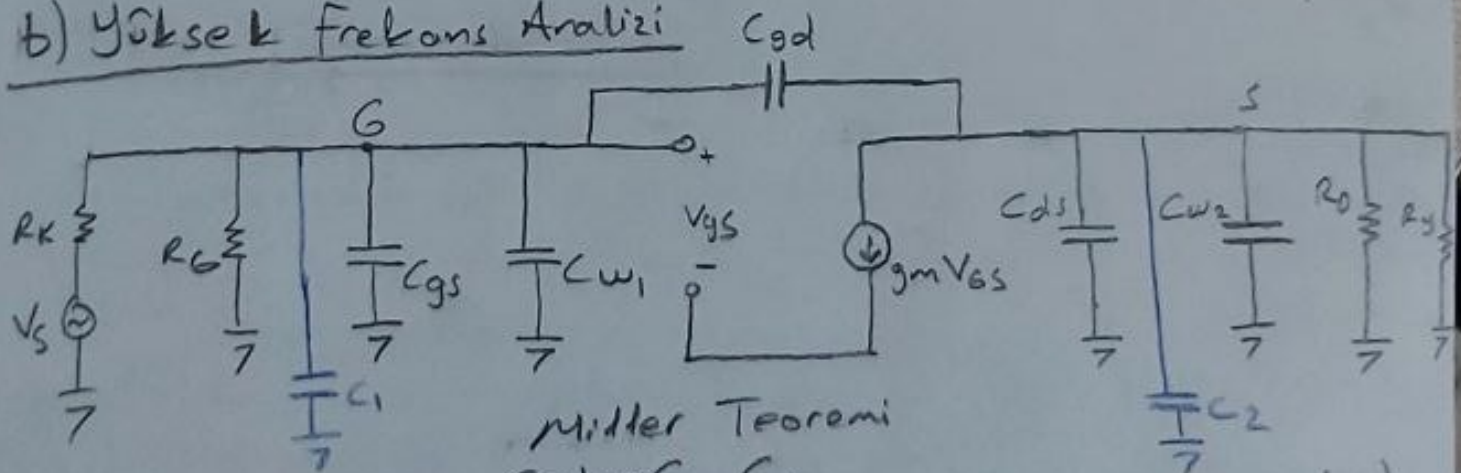
$$R_{es} = R_S \parallel \frac{1}{g_m}$$

Gezberlenecek

$$f_{is} = \frac{1}{2\pi \cdot R_{es} \cdot C_S}$$

$$f_{alt} = \max(f_{ig}, f_{is}, f_{io})$$

b) Yüksek Frekans Analizi



$$C_1 = C_{gd} (1 - A_v)$$

$$C_2 = C_{gd} (1 - \frac{1}{A_v})$$

$$R_i = R_K \parallel R_G$$

$$R_o = R_D \parallel R_y$$

$$F_{\text{üst}} = \min(f_i, f_o)$$

$$f_i = \frac{1}{2\pi \cdot C_i \cdot R_i}$$

$$f_o = \frac{1}{2\pi \cdot R_o \cdot C_o}$$

① $Z_i \rightarrow Z_B$ bulunur

② $V_i = Z_B i_b$ bulunur ($G \rightarrow S$ arası gerilim)

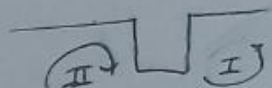
③ V_o bulunur $\rightarrow$ (Yükten aşağı +
Aşağıdan yukarı -)

④ A_v bulunur A_{vs} bulunur.

$$\rightarrow A_{vs} = A_v \cdot \frac{Z_i}{Z_i + R_s}$$

⑤ A_i bulunur $\rightarrow A_i = -A_v \cdot \frac{Z_i}{R_y}$

⑥ Z_o bulunur



Önemli Not: hoe veya rd varken kaynak dönüşümü ile hesap sadece Emitör ve Source köprülenmediğinde (∞) yapılır.

hoe var, Emitör köprülenmemiş $\rightarrow Z_i$ bulunurken I. çevre
(Kaynak dönüşüm yapıldığında artık Collectörden gelen akıma $hfe \cdot i_b$ demeyiz - I_c deriz.)
 Z_o' bulunurken $= \frac{V_o}{I_c}$ II. çevre.
 $i_b = \beta I_c$

rd var, Source köprülenmemiş $\rightarrow A_v$ bulunurken V_o ve V_i deki $I_d = \frac{V_{gs}}{r_{ds}}$ $\rightarrow$ I. çevre
($g_m V_{gs} \rightarrow I_d$)

$$\rightarrow Z_o' \text{ bulunurken } \frac{V_o}{I_d} = Z_o'$$

Önemli Not! Emitör ve Source çıkışlarında yani yük Eveya S_y bağlandığında sadece II. çevre kullanılır.

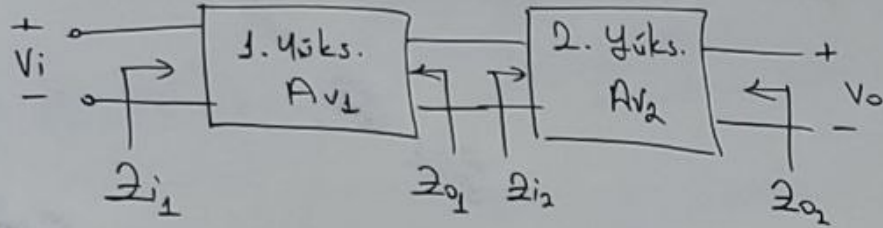
$V_o = -I_d \cdot r_{ds}$ V_{gs} ~~gözetiriz~~ $V_{gs} = \dots I_d$ yapmak için $V_i = 0$ II. çevreyi ~~gözetiriz~~ kullanırız.

$$\text{Emitör} \rightarrow Z_o = \frac{V_o}{I_o} \rightarrow V_o = R_E \dots I_o + I_b$$

$I_b = \dots I_o$ yardımı. Bunun için II. çevreyi kullanırız.

$$\text{Source çıkışı} \rightarrow Z_o = \frac{R_s}{1 + g_m R_s}$$

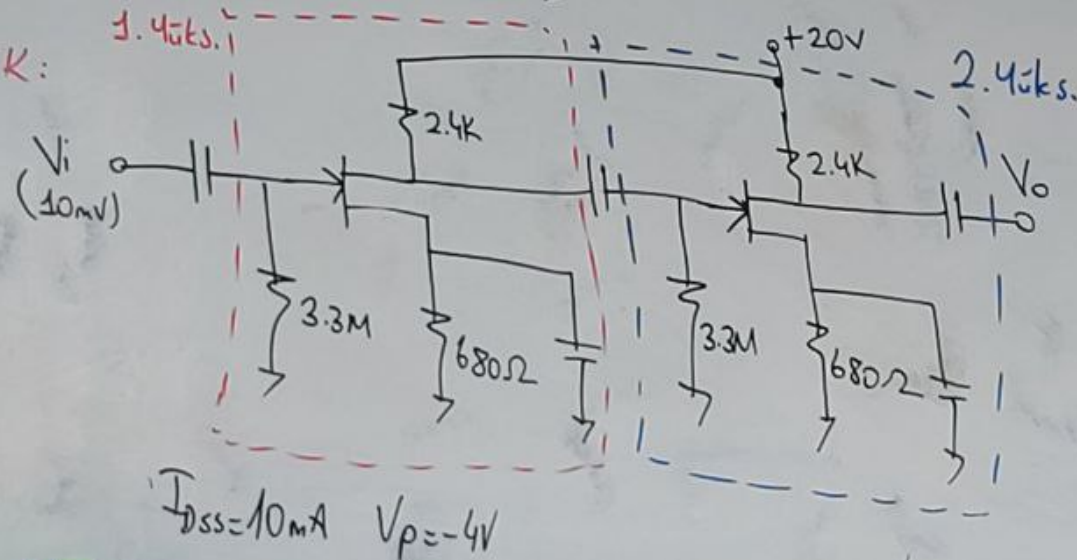
Kaskad Yükselteçler



$$A_v = A_{v1} \cdot A_{v2}$$

$$Z_i = Z_{i1} \quad Z_o = Z_{o2}$$

ÖRNEK:



1. ve 2. yükselteç özdesi;

DC analizden;

$$V_{GSQ} = -1.9V \quad I_{DQ} = 2.8mA$$

$$g_m = \frac{2 I_{DSS}}{|V_p|} \left(1 - \frac{V_{GS}}{V_p}\right) = 2.6mS$$

$$A_{v1} = -g_m (R_D \parallel R_L) = -2.6mS (2.4k \parallel 3.3k)$$

$$A_{v1} = -6.23$$

$$A_{v2} = -g_m R_D = -2.6mS (2.4k)$$

$$A_{v2} = -6.24$$

$$A_v = A_{v1} \cdot A_{v2} = 38.87$$

$$Z_i = Z_{i1} = 3.3M$$

$$Z_o = R_o = 2.4k$$

$$V_i = 10mV \text{ ise}$$

$$V_o = (38.87)(10mV)$$

$$V_o = 388.7mV$$

1. ve 2. yükseltilen b2des;

DC analizden;

$$V_{GSQ} = -1.9V \quad I_{DQ} = 2.8mA$$

$$g_m = \frac{2 I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = 2.6mS$$

$$A_{V1} = -g_m (R_D \parallel R_f) = -2.6mS (2.4k \parallel 3.3M)$$

$$A_{V1} = -6.23$$

$$A_{V2} = -g_m \cdot R_D = -2.6mS (2.4k)$$

$$A_{V2} = -6.24$$

Yüks.

$$A_V = A_{V1} \cdot A_{V2} = 38.87$$

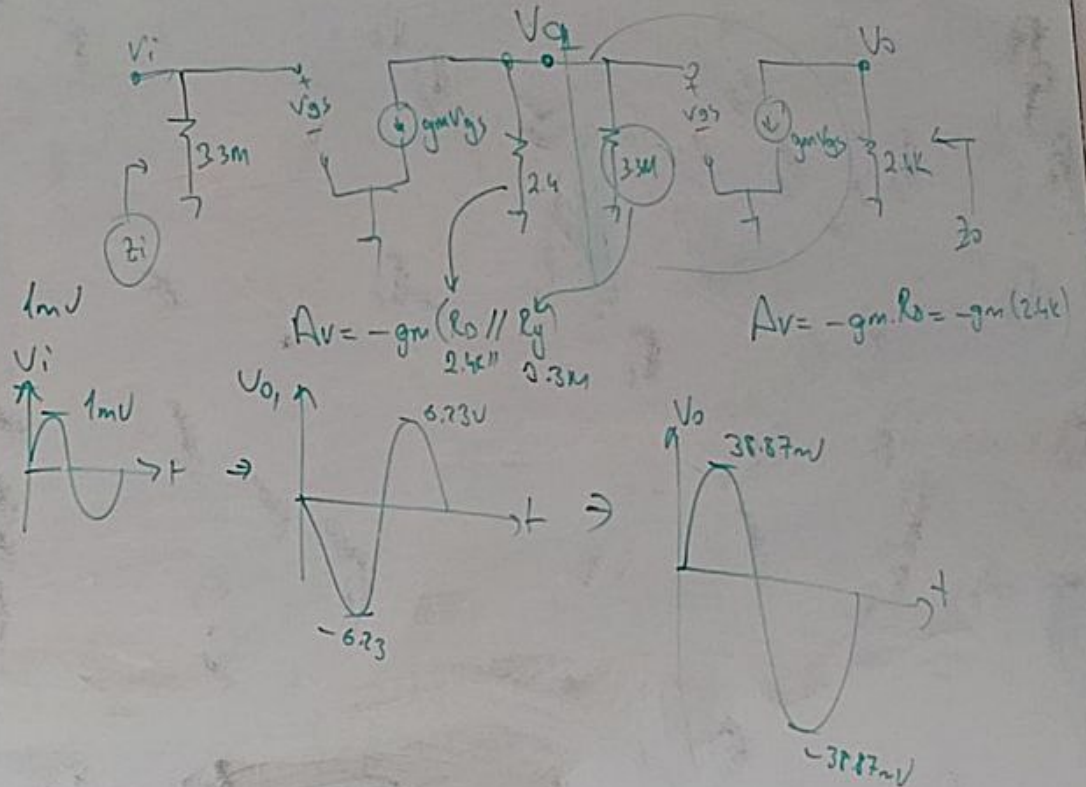
$$Z_i = Z_{i1} = 3.3M$$

$$Z_o = R_D = 2.4k$$

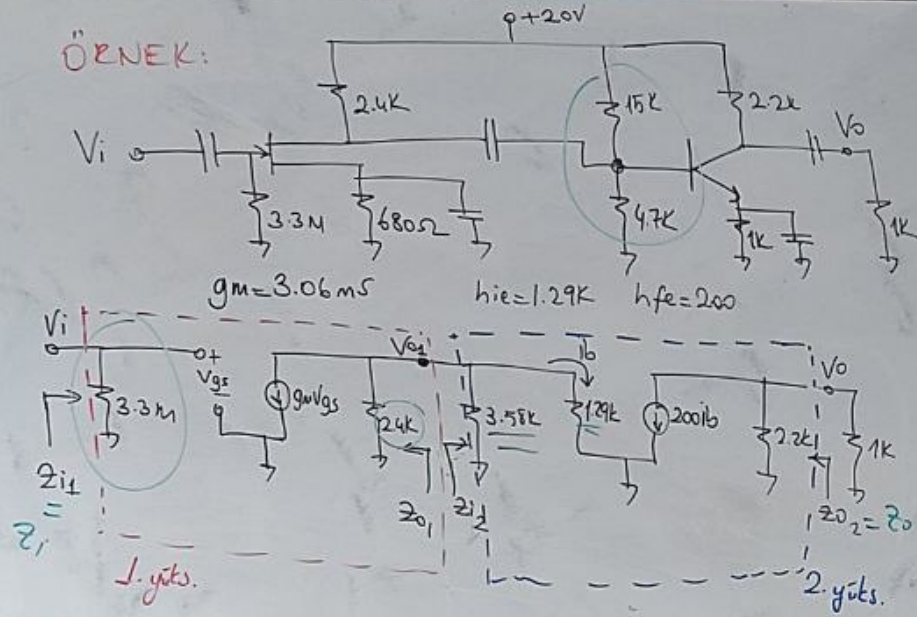
$$V_i = 10mV \text{ ise}$$

$$V_o = (38.87)(10mV)$$

$$V_o = 388.7mV$$



ÖRNEK:



$$A_{V1} = -g_m(2.4 \text{ k} \parallel Z_{i2})$$

$$Z_{i2} = 3.58 \text{ k} \parallel 1.29 \text{ k} = 0.95 \text{ k}$$

$$A_{V1} = -3.06 \text{ mS}(2.4 \text{ k} \parallel 0.95 \text{ k})$$

$$A_{V1} = -2.08$$

$$Z_{i1} = Z_i = 3.3 \text{ M}$$

$$Z_{o1} = 2.4 \text{ k}$$

$$Z_{o2} = Z_o = 2.2 \text{ k}$$

$$A_{V2} = -\frac{h_{fe}}{h_{ie}}(R_c \parallel R_L)$$

$$A_{V2} = -\frac{200}{1.29 \text{ k}}(2.2 \text{ k} \parallel 1 \text{ k})$$

$$A_{V2} = -106.6$$

$$A_V = A_{V1} \cdot A_{V2} = 222$$

$$A_i = -A_V \cdot \frac{Z_i}{R_y} = -222 \left(\frac{3.3 \text{ M}}{1 \text{ k}} \right)$$

$$A_i = -732.6 \text{ k}$$

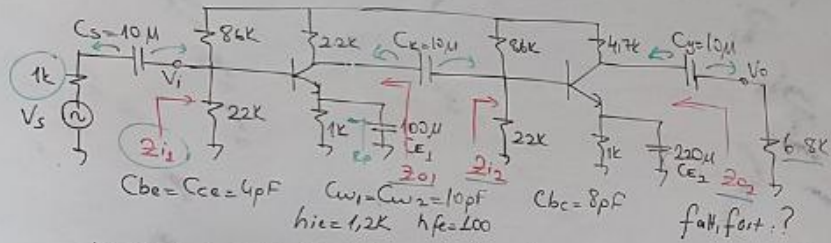
$$V_i = 10 \mu\text{V} \text{ olsaydı } V_o = 2.22 \text{ mV (faz farkı yok)}$$

$$I_i = 1 \mu\text{A} \text{ olsaydı } I_o = -732.6 \text{ mA (180° faz farkı)}$$

1. yüksek FET olduğu için FET'inde giriş Empedansı çok yüksek olduğu için akım kazancı çok yüksek oldu. $Z_i \uparrow A_i \uparrow$



Kaskad Yükselteçlerde Frekans Cevabı



Her iki katın orta frk bölgesindeki kazançları;

$$A_{v1} = -\frac{h_{fe}}{h_{ie}} (R_c \parallel Z_{i2}) \quad Z_{i2} = (84k \parallel 22k) \parallel 1.2k$$

$$A_{v1} = -\frac{100}{1.2k} (22k \parallel 1.123k) \quad Z_{i2} = 17.52k \parallel 1.2k = 1.123k$$

$$A_{v1} = -61.95 \quad Z_{o1} = 2.2k$$

$$A_{v2} = -\frac{100}{1.2k} (4.7k \parallel 6.8k) \Rightarrow A_{v2} = -231.6$$

Alt Kesim Freq.

C_s için;

$$f_{cs} = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} \cdot (1k \parallel 1.2k)}$$

$$f_{cs} = 7.5 \text{ Hz}$$

C_{ce} için;

$$f_{ce} = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} \cdot (1.123k \parallel 2.2k)}$$

$$f_{ce} = 4.79 \text{ Hz}$$

C_y için;

$$f_{cy} = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} \cdot (4.7k \parallel 6.8k)}$$

$$f_{cy} = 1.38 \text{ Hz}$$

C_{E2} için

$$R_{e2} = 1k \parallel \frac{R_{A1}}{101}$$

$$R_{A1} = (1k \parallel 17.52k) + 6.8k$$

$$R_{A1} = 2.146k$$

$$R_{e2} = 20.80 \Omega$$

$$f_{ce1} = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} \cdot (20.8 \Omega)}$$

$$f_{ce1} = 76.55 \text{ Hz}$$

C_{E2} için;

$$R_{e2} = 1k \parallel \frac{R_{A2}}{101}$$

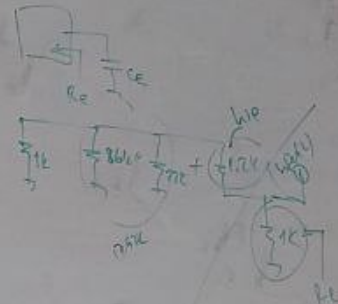
$$R_{A2} = (20.1 \parallel 17.52k) + 12k$$

$$R_{A2} = 3.15k \quad R_{e2} = 30.24 \Omega$$

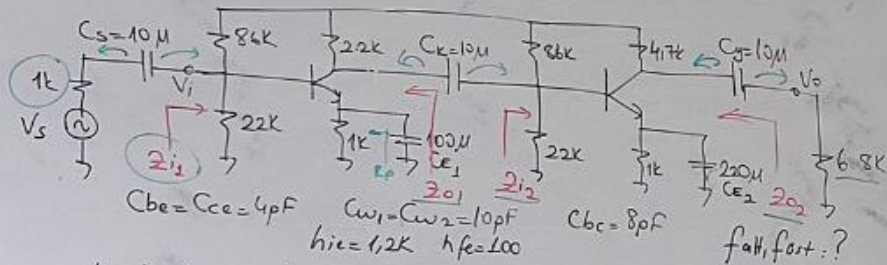
$$f_{ce2} = \frac{1}{2\pi \cdot 20 \cdot 10^{-6} \cdot (30.24 \Omega)}$$

$$f_{ce2} = 239.34 \text{ Hz}$$

$$f_{alt} = 76.55 \text{ Hz}$$



Kaskad Yükselteçlerde Frekans Cevabı



Her iki katın orta frk. bölgesindeki kazançları;

$$A_{v1} = -\frac{h_{fe}}{h_{ie}} (R_c \parallel Z_{i2}) \quad Z_{i2} = (86k \parallel 22k) \parallel 1.2k$$

$$A_{v1} = -\frac{100}{1.2k} (22k \parallel 1.13k)$$

$$A_{v1} = -6195$$

$$A_{v2} = -\frac{100}{1.2k} (4.7k \parallel 6.8k) \Rightarrow A_{v2} = -231.6$$



$$Z_{i2} = 1.123k$$

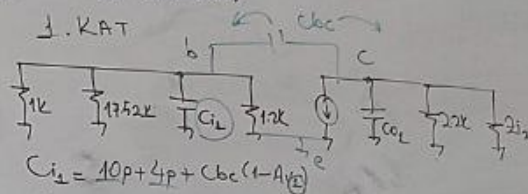
$$Z_{i2} = 1.123k$$

$$Z_{o1} = 2.2k$$

$$Z_{o2} = 4.7k$$

Üst Kesim Frk. İcin;

1. KAT



$$C_{i2} = 10p + 4p + C_{bc}(1 - A_{v2})$$

$$C_{i2} = 517pF$$

$$R_{i2} = 1k \parallel 17.52k \parallel 1.2k = 0.527k$$

$$f_{i1} = \frac{1}{2\pi(517pF)(0.527k)} = 584kHz$$

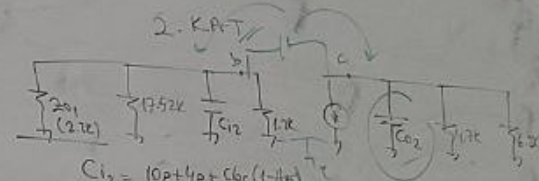
$$C_{o1} = 10p + 4p + C_{bc}(1 - \frac{1}{A_{v1}})$$

$$C_{o1} = 22pF$$

$$R_{o1} = 2.2k \parallel 4.7k = 0.74k$$

$$f_{o1} = \frac{1}{2\pi(22pF)(0.74k)} = 938kHz$$

2. KAT



$$C_{i2} = 10p + 4p + C_{bc}(1 - A_{v2})$$

$$C_{i2} = 1370pF$$

$$R_{i2} = 2.2k \parallel 17.52k \parallel 1.2k = 0.74k$$

$$f_{i2} = \frac{1}{2\pi(1370pF)(0.74k)} = 115kHz$$

$$f_{üst} = 115kHz$$

$$C_{o2} = 10p + 4p + C_{bc}(1 - \frac{1}{A_{v2}})$$

$$C_{o2} = 22pF$$

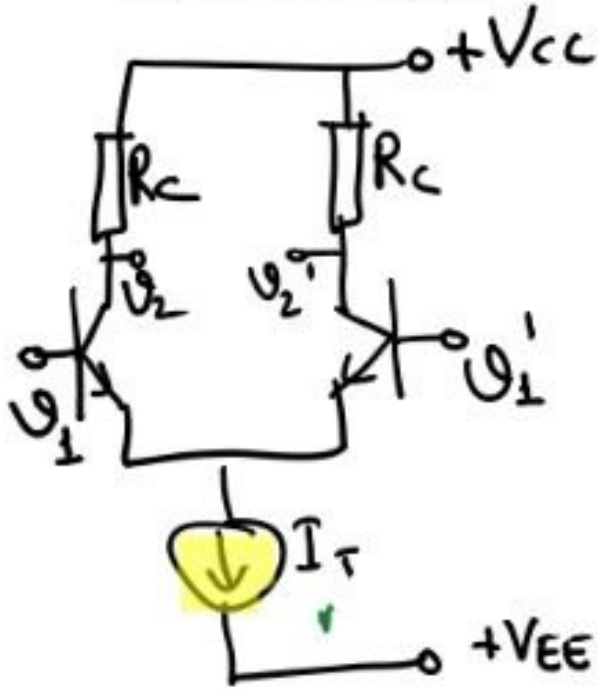
$$R_{o2} = 4.7k \parallel 6.8k = 2.8k$$

$$f_{o2} = \frac{1}{2\pi(22pF)(2.8k)} = 258kHz$$

ÖDEV: Kaskad kw. ders notundaki tüm sorular

Bölüm-7 : Farksal Yükselteç ve OpAmp Devreleri

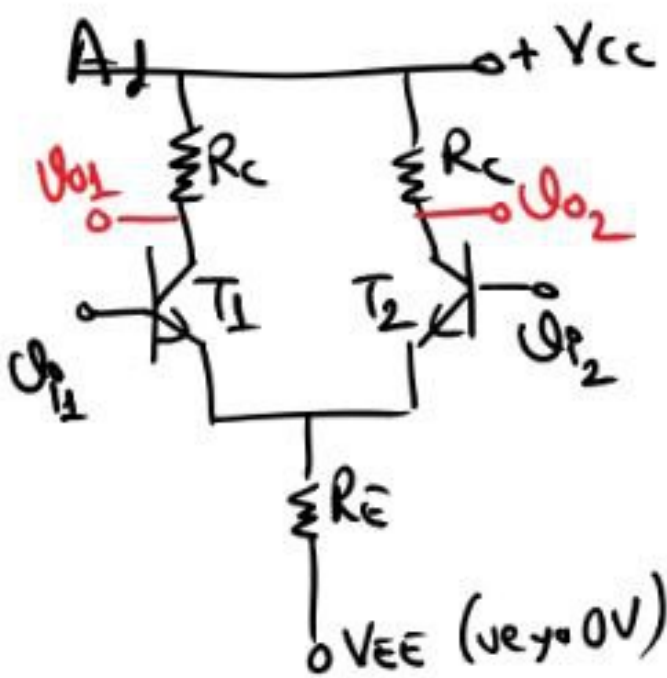
7.1 Farksal Yükselteç Devresi (Uzun Kuyuklu Devre)



Ayarajları:

- * Köprülene kondansatörü gerektirmeden kat bazına yükler gerilim kazancı sağlar.
- * Bağlama kondansatörü gerektirmeden çok katlı yapılar oluşturulabilir.
- * Sıcaklığı karşı duyarlılığı düşüktür.
- * İki giriş sinyali vardır ve çıktı bu iki giriş arasındaki gerilim farkını kuvvetlendirir (2it 1a 2da) iki çıktı elde
- * Çıkışta birbirine eşit etme imkanı vardır.
- * Sıfır freksten (dc-seviye) yükler frekanslara kadar düzgün frekans karakteriğine sahip.

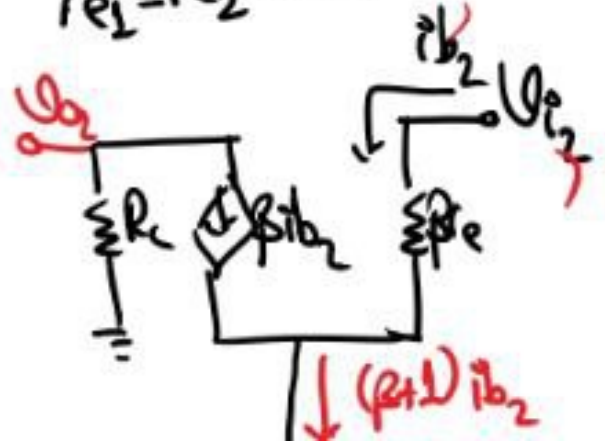
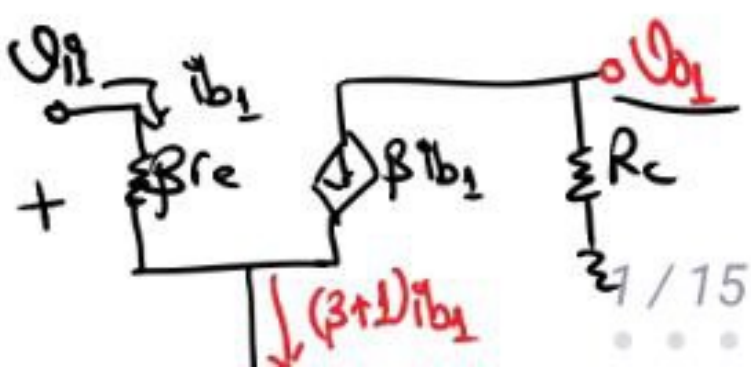
Devrede sükunet akımının sıcaklığı karşı duyarlılığının az olması için I_T akım kaynağı kullanılır. Fakat bu mümkün değilse bunun yerine çok yüksek bir R_E direnci kullanılabilir. Bu durumda:



$$T_1 \equiv T_2 \text{ ise}$$

$$\beta_1 \equiv \beta_2 \equiv \beta$$

$$r_{e1} \equiv r_{e2} \equiv r_e$$





$$(\beta+1)ib_1 = (\beta+1)ib_2$$

$$ib_1 = -ib_2$$

$$A_d = \frac{V_{o1} - V_{o2}}{V_{i1} - V_{i2}}$$

$$\begin{aligned} V_{o1} &= -\beta ib_2 R_c \\ V_{o2} &= -\beta ib_2 R_c \end{aligned}$$

$$V_{i1} - V_{i2} = ib_2 \beta r_e - ib_2 \beta r_e$$

$$A_d = \frac{-\beta ib_2 R_c + \beta ib_2 R_c}{ib_2 \beta r_e - ib_2 \beta r_e} \leftarrow ib_2 = -ib_1$$

$$A_d = \frac{-\beta ib_1 R_c - \beta ib_2 R_c}{ib_2 \beta r_e + ib_1 \beta r_e} = \frac{-2\beta ib_1 R_c}{2ib_1 \beta r_e}$$

$$A_d = -\frac{R_c}{r_e}$$

$$A_c = \frac{V_{o1} + V_{o2}}{V_{i1} + V_{i2}}$$

$$ib_1 = ib_2$$

$$V_{o1} = -\beta ib_2 R_c \quad V_{o2} = -\beta ib_2 R_c$$

$$V_{i1} = ib_2 \beta r_e + R_E (\beta+1) (ib_1 + ib_2)$$

$$V_{i1} = ib_2 \beta r_e + 2R_E ib_2 (\beta+1)$$

$$V_{i2} = V_{i1}$$

$$A_c = \frac{-2\beta ib_2 R_c}{2(ib_2 \beta r_e + 2R_E ib_2 (\beta+1))}$$

$$A_c = -\frac{R_c \beta}{\beta r_e + 2R_E (\beta+1)}$$

$$A_c = - \frac{R_c}{r_e + 2R_E} \quad \checkmark \quad r_e \ll R_E$$

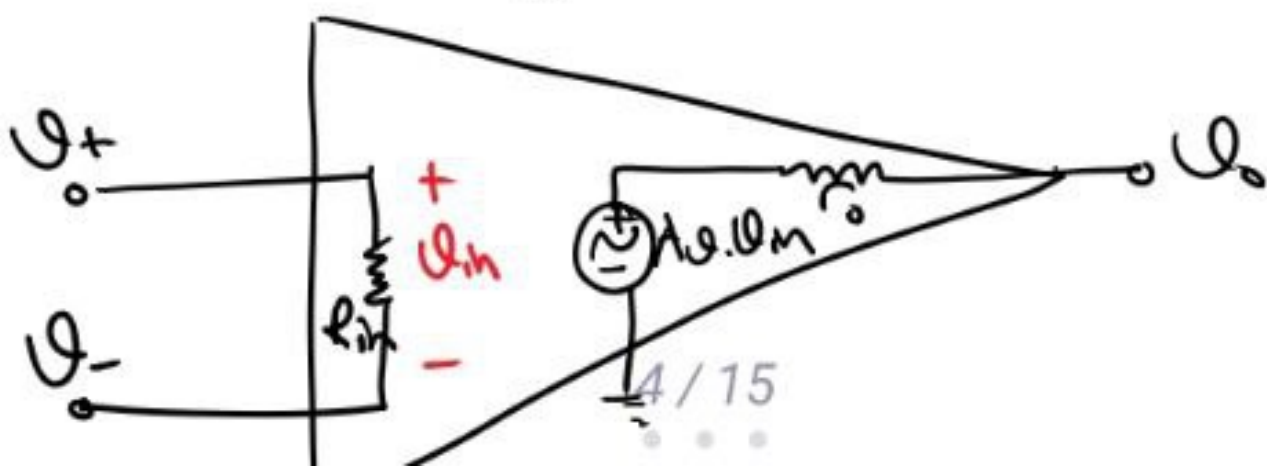
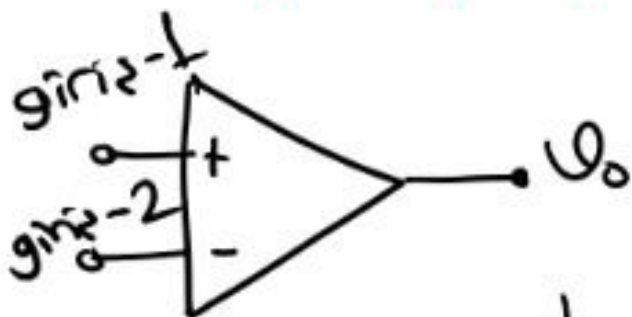
$$A_c \approx - \frac{R_c}{2R_E}$$

CMRR $\Rightarrow$ Common mode Rejection Ratio

$$CMRR = 20 \log \left| \frac{A_d}{A_c} \right| \quad \checkmark \downarrow$$

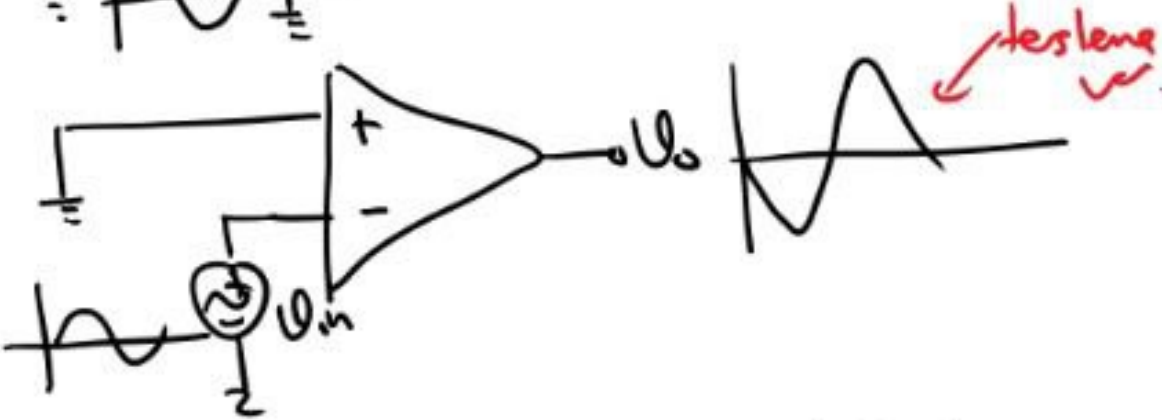
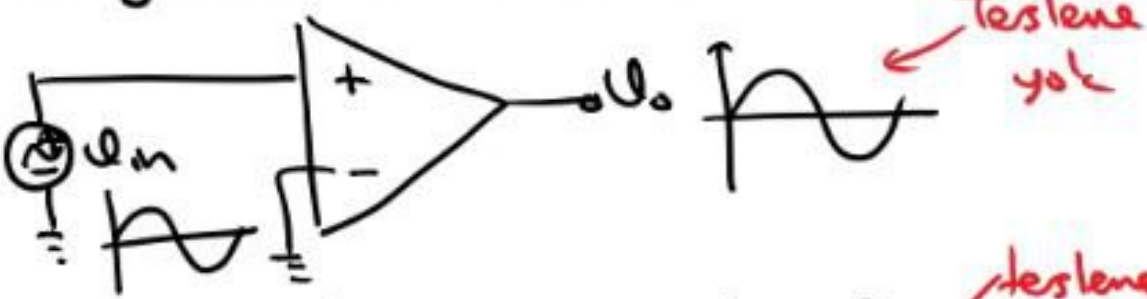
$$CMRR = 20 \log \left| \frac{r_e + 2R_E}{r_e} \right|$$

7.2 Op-Amp (Operational Amplifiers)



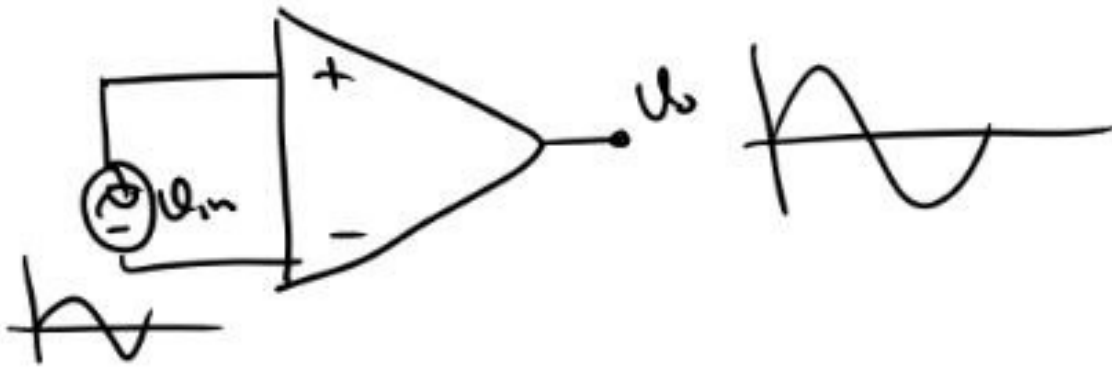


- Tek giriş kullanılması halinde:



Not: Op-Amp tek çıkış verdiği gibi bazı uygulamalarda çift çıkış da üretebilir. Op-Amp fark sinyallerini yükseltirken, ortak mod sinyallerinde çıkış "0" olur.

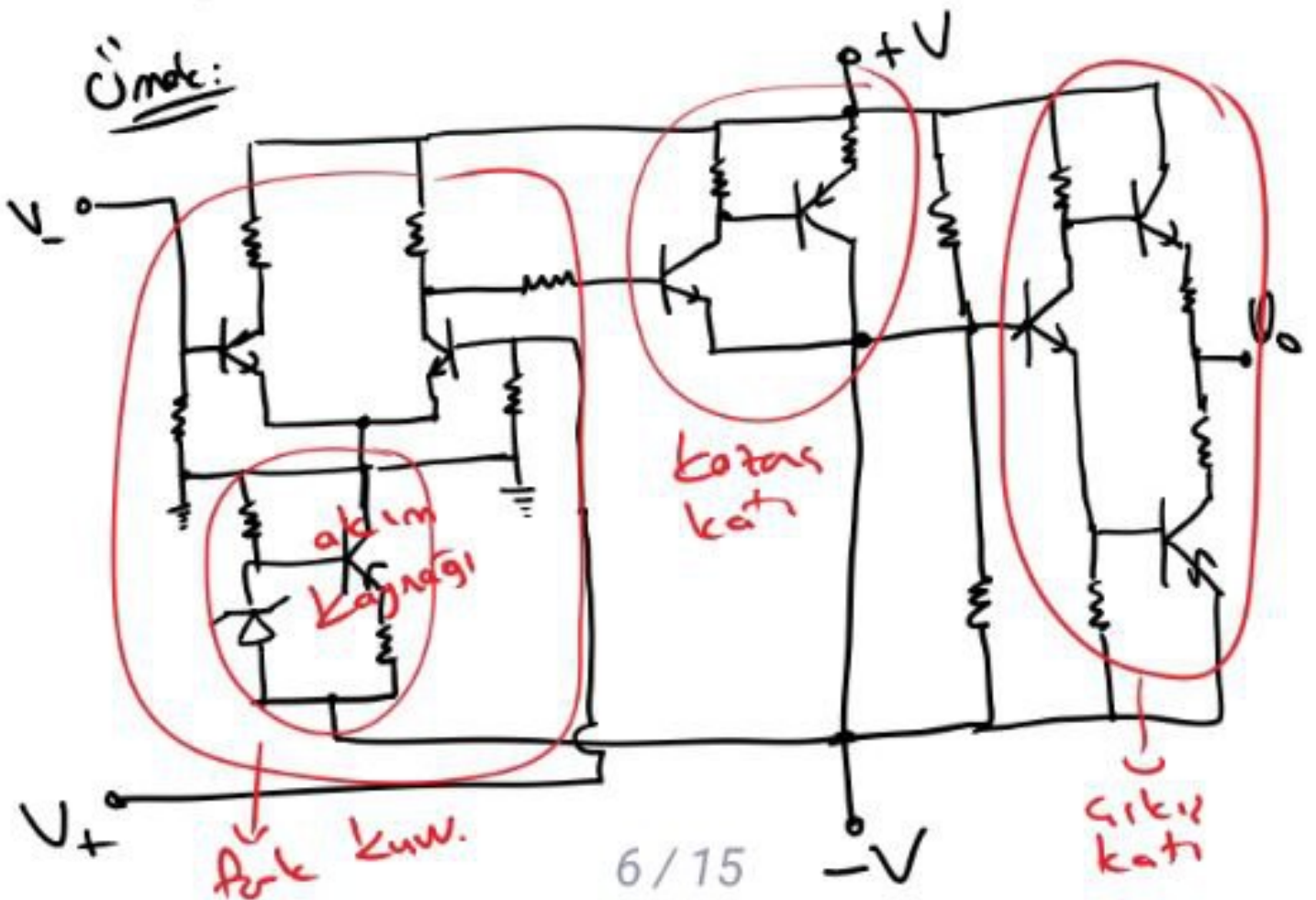
⌞ Çift giriş kullanılması halinde:



- Op-Amp iç yapısı:



Örnek:

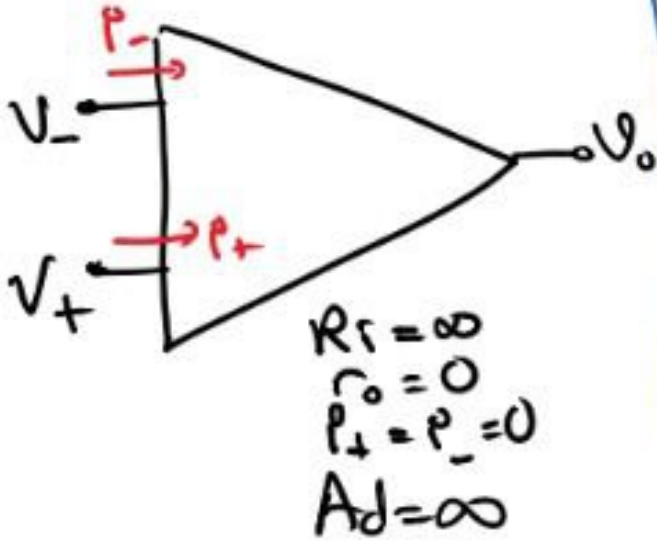


Diferansiyel ve Ortak-Mod Kazanma:

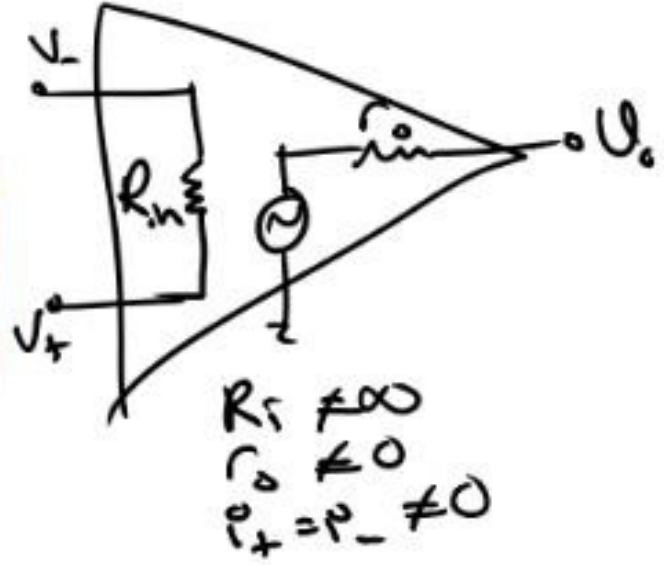
$$CMRR = 20 \log_{10} \frac{A_d}{A_c}$$

Op-Amp'ın İlgili Temel Bilgiler:

İDEAL Op-Amp

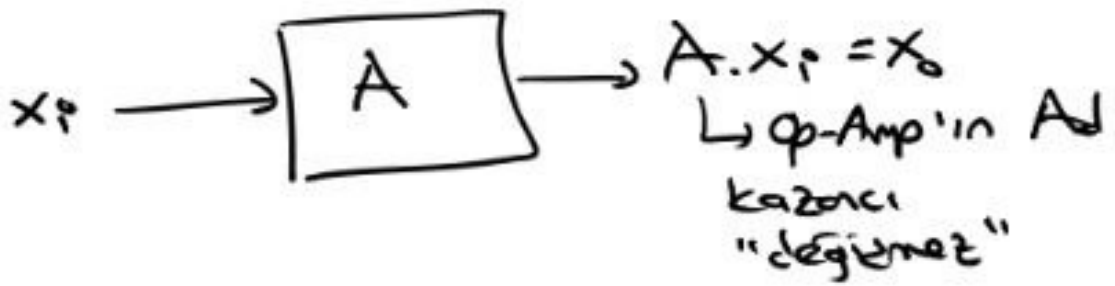


İDEAL Olmayan Op-Amp

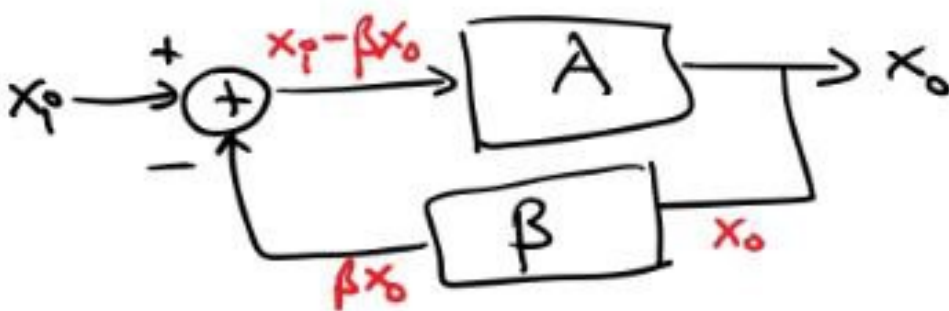


Op-Amp'lar Yüksek Farklı Kazan Sağlama-
larına rağmen daha çok farklı yükselter olarak
değil, geri besleneli yükselter olarak kullanılır.

Geri beslenmelidir:



Geri beslenmeli:



$$x_o = A (x_i - \beta x_o)$$

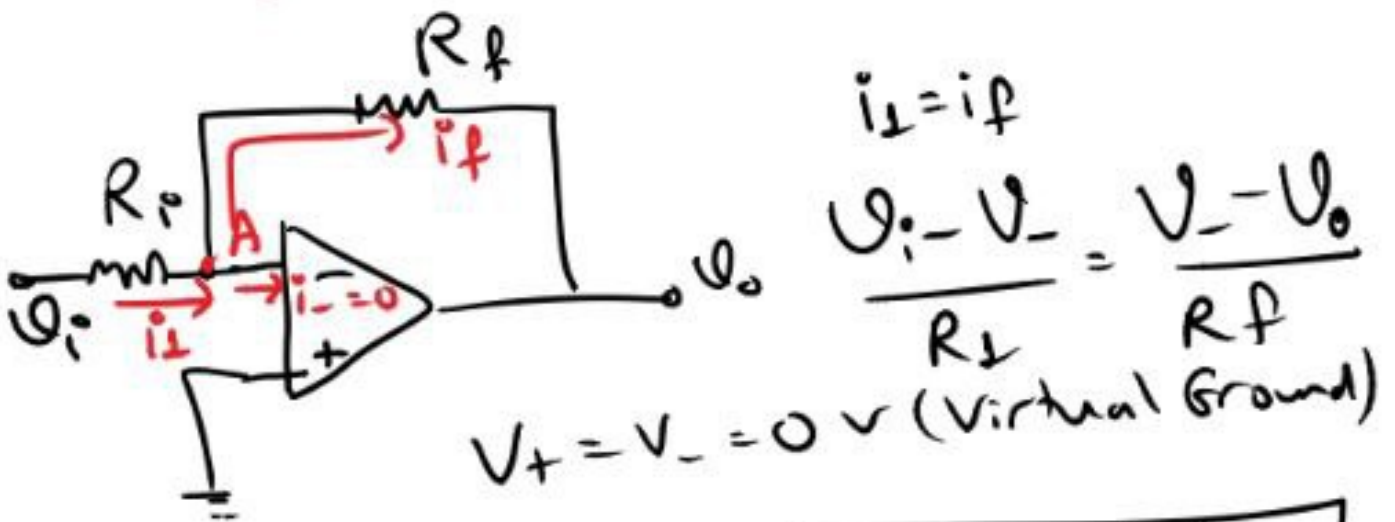
$$x_o (1 + A\beta) = A x_i \Rightarrow \frac{x_o}{x_i} = \frac{A}{1 + A\beta}$$

Geri beslenmeli: Pratik Op-Amp Devreleri:

7/15
 (1) Tersleyen Yükselter (Inverting Amplifier)

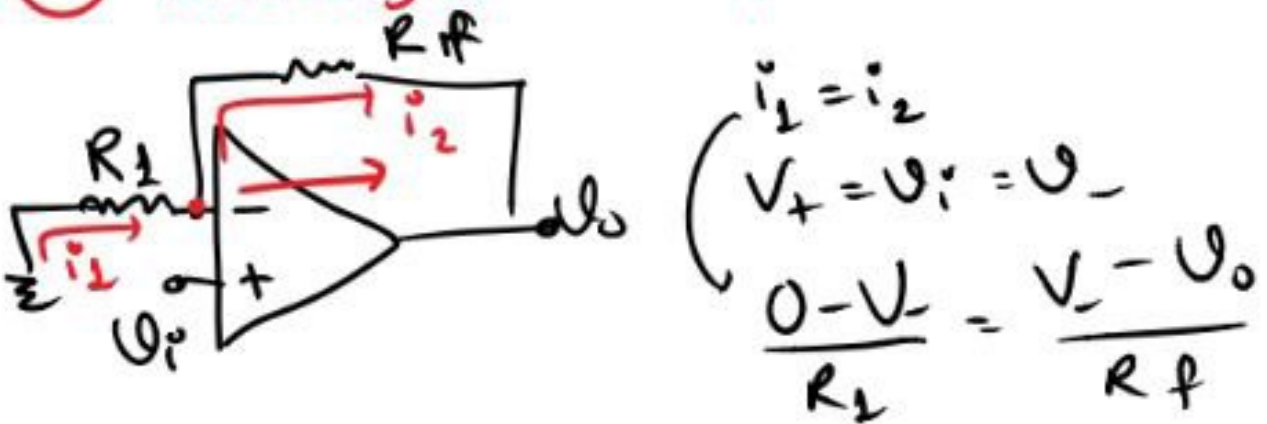
Gör beşleneli: Pratik Op-Amp Devreleri:

① Tersleyen Yükçetler (Inverting Amplifier)



$$\frac{V_i}{R_i} = -\frac{V_o}{R_f} \Rightarrow \boxed{\frac{V_o}{V_i} = -\frac{R_f}{R_i}}$$

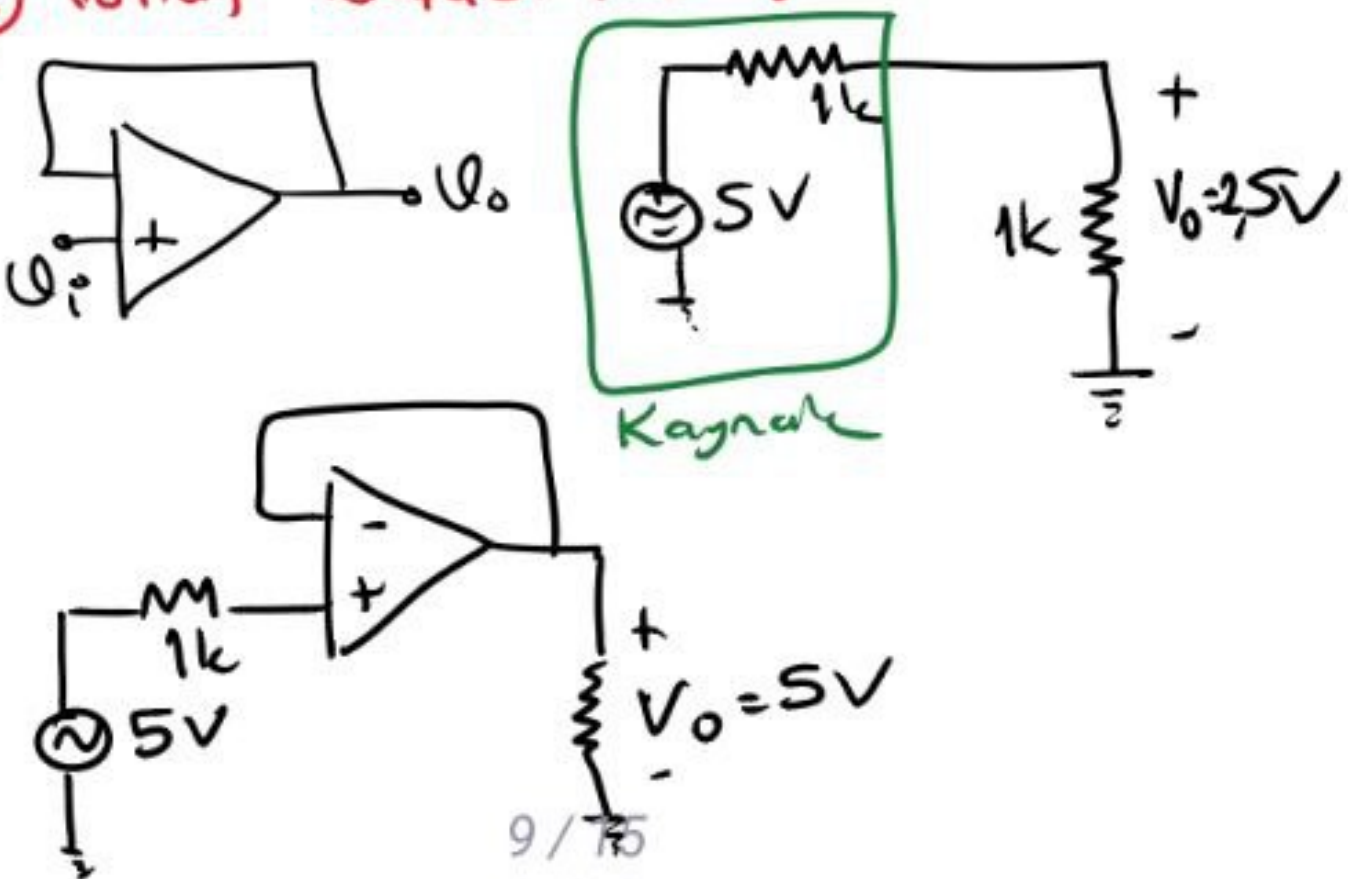
② Terslemeyen Yükçetler (Non-Inverting Amp)



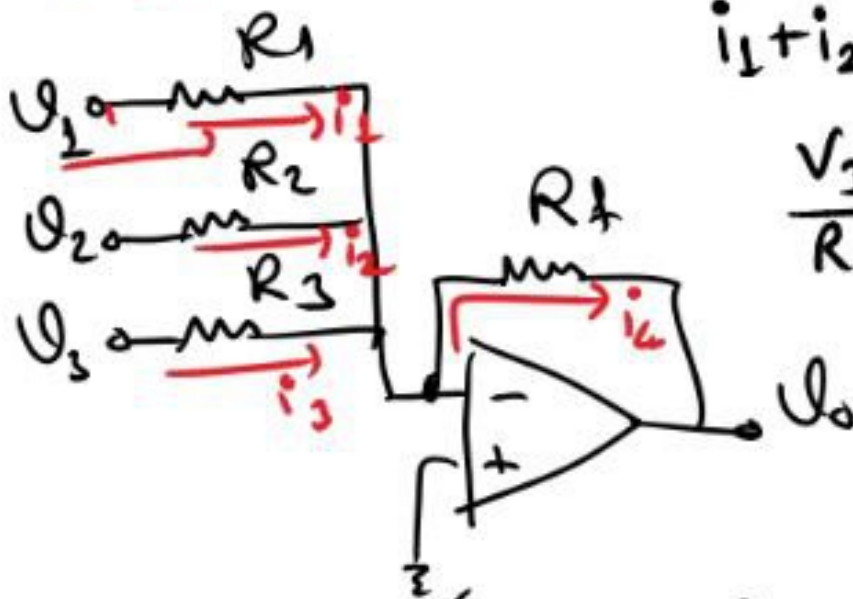
$V_- = V_i$ ise:

$$-\frac{V_i}{R_i} = \frac{V_i - V_o}{R_f} \Rightarrow \boxed{\frac{V_o}{V_i} = 1 + \frac{R_f}{R_i}}$$

③ Voltaj Takipçisi (Voltage Follower - Buffer)



④ Toplayıcı Yükselteç:

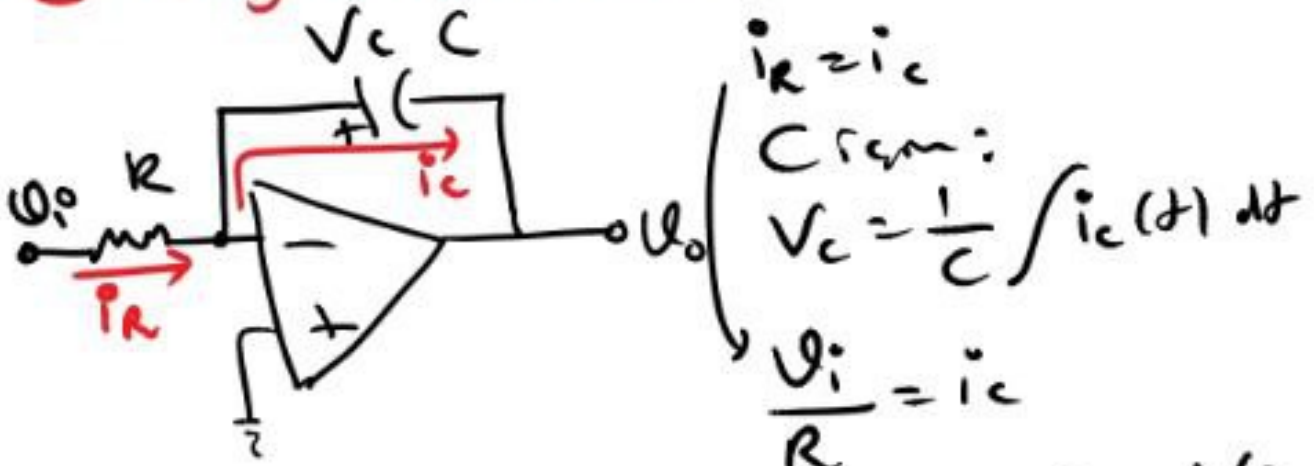


$$i_1 + i_2 + i_3 = i_L$$

$$\frac{V_3}{R_3} + \frac{V_2}{R_2} + \frac{V_1}{R_1} = \frac{-V_o}{R_f}$$

$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right)$$

⑤ İntegral Alıcı Devre:



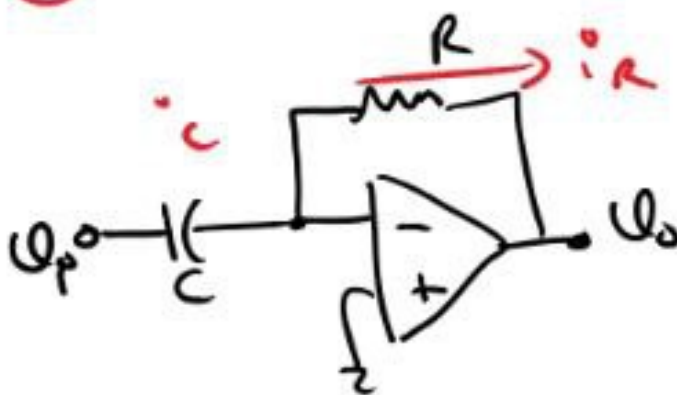
$$\begin{aligned} i_R &= i_C \\ C \text{ için:} \\ V_C &= \frac{1}{C} \int i_C(t) dt \\ \frac{V_i}{R} &= i_C \end{aligned}$$

$$V_C = -V_o = \frac{1}{C} \int i_C(t) dt \Rightarrow i_C = \frac{V_i}{R}$$

$$V_o = -\frac{1}{C} \int \frac{V_i(t)}{R} dt$$

$$\Rightarrow V_o = -\frac{1}{RC} \int V_i(t) dt$$

⑥ Türev Alıcı Devre:



$$\begin{aligned} i_C &= i_R \\ i_C &= -\frac{V_o}{R} \end{aligned}$$

$$V_C = V_i$$

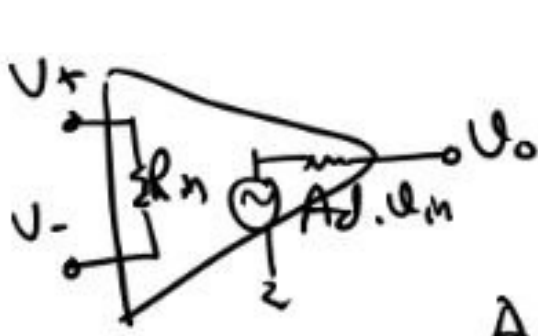
$$V_C = V_i = \frac{1}{C} \int -\frac{V_o}{R} dt$$

$$V_i(t) = -\frac{1}{RC} \int V_o(t) dt$$

$$V_o(t) = -RC \frac{dV_i(t)}{dt}$$

Türev alımı

Virtual Ground (Serel Toprak)



$$V_{in} = V_+ - V_-$$

ideal op-amp'ta:
 $A_d = \infty$ olur.

$$A_d = \frac{V_o}{V_{in}} = \infty$$

Yani $V_{in} \equiv 0$ olmalıdır

$\Rightarrow V_+ - V_- = 0$ ise $V_+ = V_-$ alınır.

Devre çözümlerinde ideal op-amp yaklaşımı benzersiz daha pratik bir şekilde çözüme ulaştırılabilir. İdeal op-amp için:

$$\textcircled{*} V_+ = V_- \quad (\text{virtual ground})$$

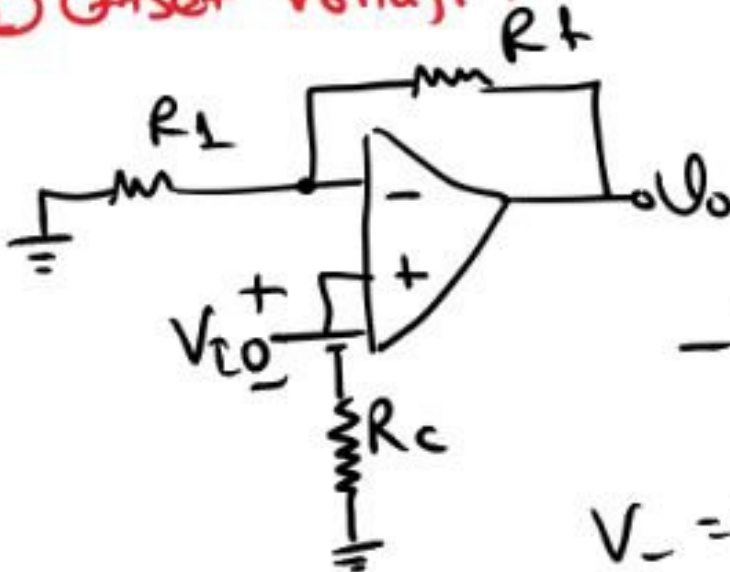
$$\textcircled{*} P_+ = P_- = 0 \quad (R_n = \infty)$$

olarak çözüm yapılır

Op-Amp Parametreleri

İdeal olmayan op-amp'ler.

① Offset Voltajı:



$$\frac{0 - V_-}{R_1} = \frac{V_- - V_o}{R_f}$$

$$-\frac{V_-}{R_1} = \frac{V_-}{R_f} - \frac{V_o}{R_f}$$

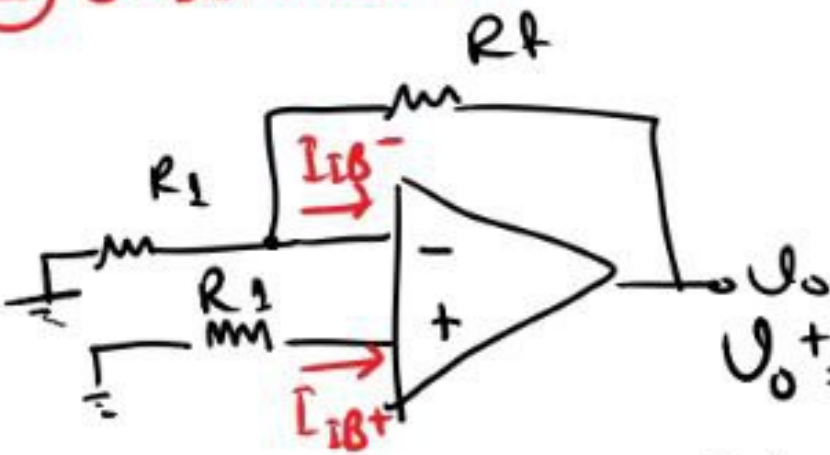
$$V_- = V_+ = V_{io}$$

$$\frac{V_o}{R_f} = \frac{V_-}{R_f} + \frac{V_-}{R_1} \Rightarrow$$

$$V_o = \frac{(R_f + R_1)}{R_1} V_{io}$$

off-set'ten kaynaklanan
 girer

② Offset Akımı:



$$\bar{I}_{IO} = \bar{I}_{IB+} - \bar{I}_{IB-}$$

Süperpozisyon

$$V_o^+ = \bar{I}_{IB+} \cdot R_1 \left(1 + \frac{R_f}{R_1}\right)$$

$$V_o^- = \bar{I}_{IB-} \cdot R_1 \left(-\frac{R_f}{R_1}\right)$$

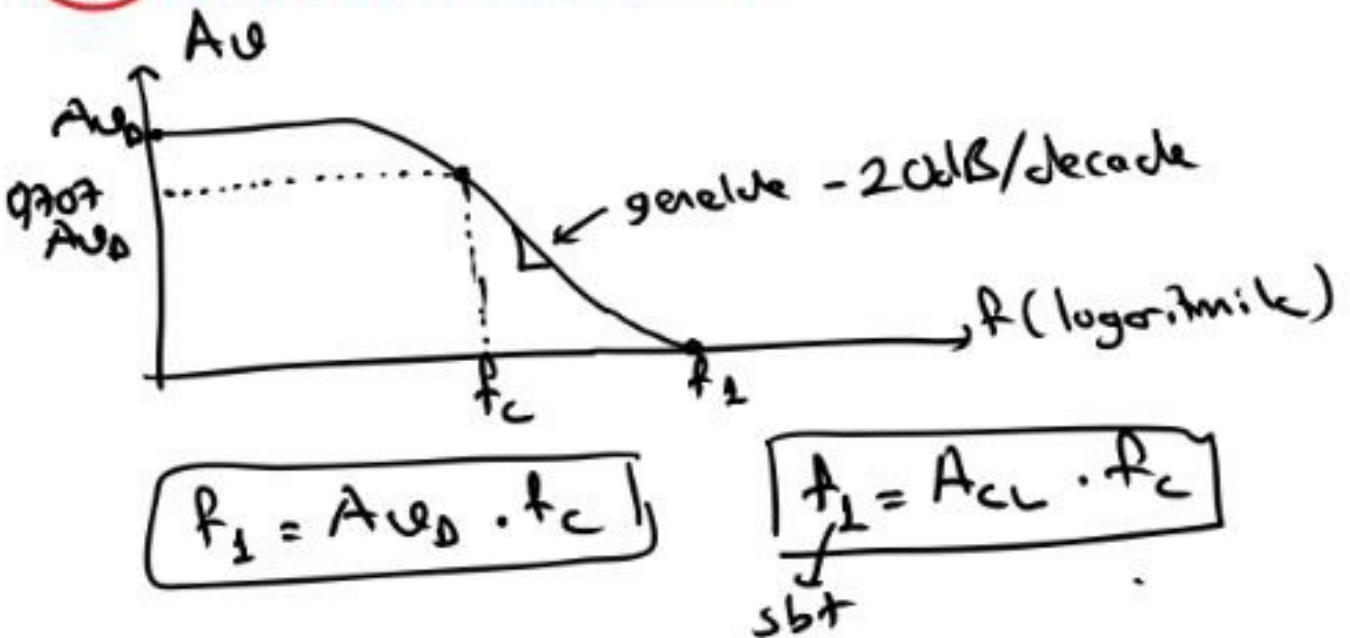
$$V_o = \bar{I}_{IB+} \cdot R_1 \left(1 + \frac{R_f}{R_1}\right) - \bar{I}_{IB-} \cdot R_1 \cdot \frac{R_f}{R_1}$$

$$V_o = \bar{I}_{IB+} (R_1 + R_f) - \bar{I}_{IB-} \cdot R_f \quad \leftarrow R_1 \ll R_f \text{ seçilir}$$

$$V_o = R_f (\bar{I}_{IB+} - \bar{I}_{IB-}) = \bar{I}_{IO} \cdot R_f$$

Toplam $\Rightarrow V_{offset} = V_{IO} \frac{R_1 + R_f}{R_1} + \bar{I}_{IO} R_f$

③ Frekans Parametreleri:



SLEW RATE (SR)

$$SR = \frac{\Delta V_o}{\Delta t} \quad \left(\frac{V}{\mu s}\right) \rightarrow \text{çıkıp 1 } \mu s \text{ içinde en fazla kaç volt değişir?}$$

Örnek:

$SR = 2V/\mu s$ olan bir op-amp'ta giriş sinyali: $10\mu s$ içinde $0.5V$ değişim gösteriyor. Buna göre kapalı devre kazancı maksimum kaç olabilir?

$$V_o = A_{cl} \cdot V_i \Rightarrow \frac{\Delta V_o}{\Delta t} = A_{cl} \cdot \frac{\Delta V_i}{\Delta t}$$

$$A_{cl} = \frac{\Delta V_o / \Delta t}{\Delta V_i / \Delta t} = \frac{SR}{\Delta V_i / \Delta t} = \frac{2V/\mu s}{0.5V/10\mu s}$$

$$A_{cl} = 40$$

$$A_{cl} = 401$$

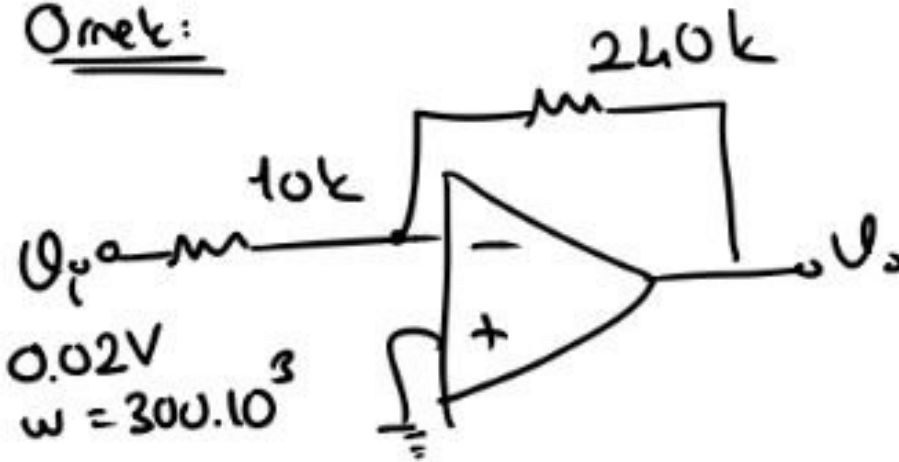
Maximum Signal Frekansı:

$$U_o = K \cdot \sin(2\pi ft) \text{ ise;}$$

$$f \leq \frac{SR}{2\pi K} \text{ Hz veya } \omega \leq \frac{SR}{K} \text{ rad/s}$$

olmalıdır.

Örnek:



$SR = 0.5 \text{ V}/\mu\text{s}$
ise
 U_o 'da
bozulma
olur mu?

$$SR = 0.5 \text{ V}/\mu\text{s}$$

$$A_{cl} = \left| \frac{R_f}{R_i} \right| = 24$$

$$K = A_{cl} \cdot U_i = 0.48 \text{ V}$$

$$\omega_s \leq \frac{SR}{K} \text{ olmalı (bozulma olmaması için)}$$

$$\omega_s \leq \frac{0.5 \text{ V}/\mu\text{s}}{0.48} \rightarrow 1.1 \times 10^6 \text{ rad/s}$$

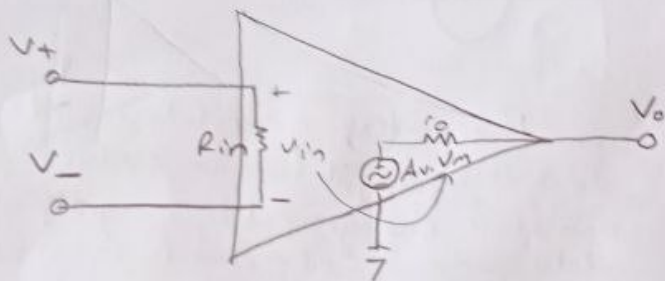
$$300 \cdot 10^3 \leq 1100 \cdot 10^3 \text{ old için bozulma olmaz}$$

Opampın Temel Kazancı Formülleri

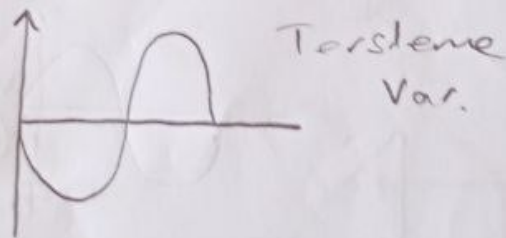
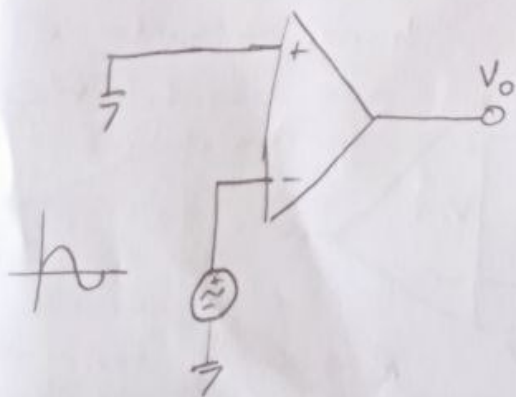
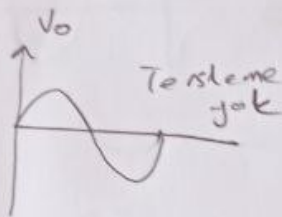
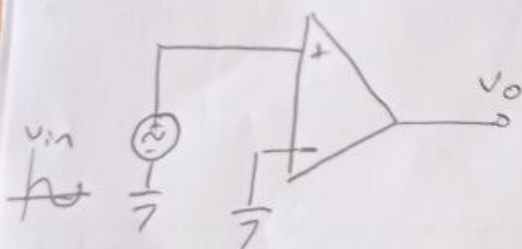
$$A_d = K_d = \frac{V_{o1} - V_{o2}}{V_{i1} - V_{i2}} \rightarrow \text{Farklısal Kazancı (Genellikle bu kazanç 65dB kullanılır.)}$$

$$A_c = K_c = \frac{V_{o1} + V_{o2}}{V_{i1} + V_{i2}} \rightarrow \text{Ortak Kazancı}$$

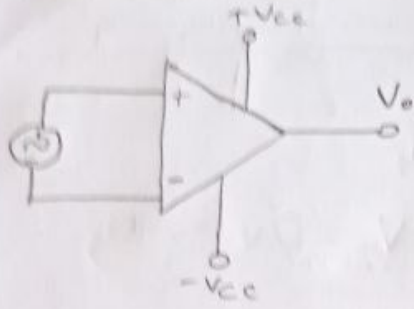
Opamp Temel Yapısı ve Çalışması



Tek girişin kullanılması halinde:

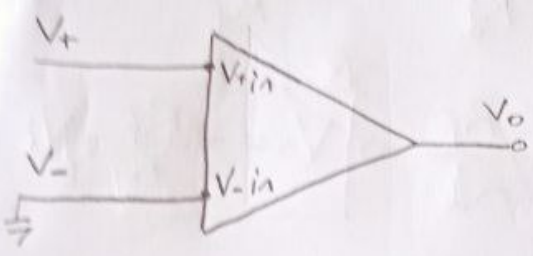


Opampın Temel Formülleri



1) Her zaman $V_o \leq V_{cc}$ 'dir.

2) $V_o = A_v \cdot V_i \rightarrow$ Aak aevrilm
kazanma kchp
(Komparator vb)
devreler için
bu formül
önemli.

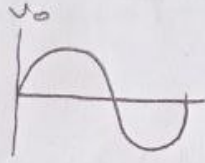
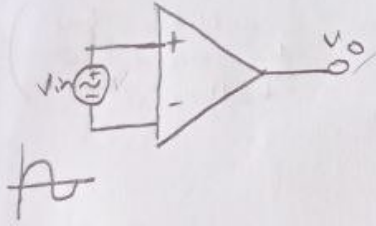


★3) $i_+ = i_- = 0 \rightarrow$ Her soruyu
Bu iki formül
ile çözülmz.

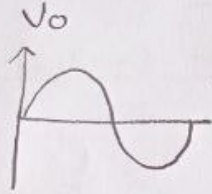
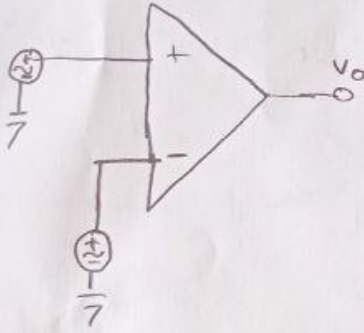
★4) $V_{+in} = V_{-in}$

Geribeslemeli Pratik Opamp Devre

Gift Girişin Kullanılması Halinde

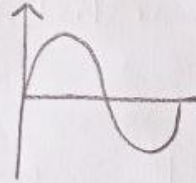
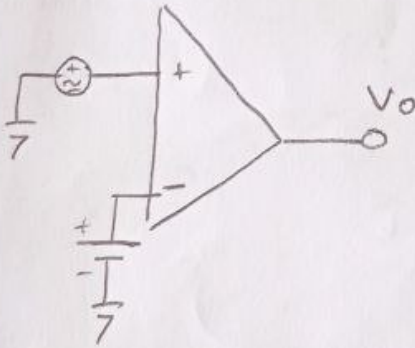


→ Tersleme Yok



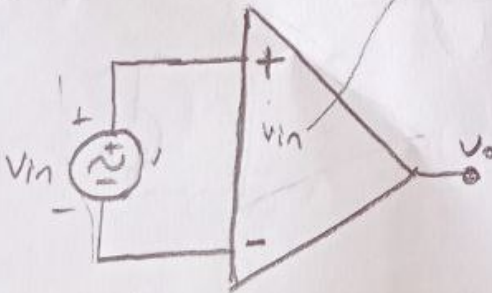
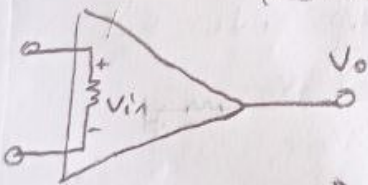
→ Tersleme olabilir mi?

Bu çıkış grafikleri olağüstü değildir. Bşşklöük ve şekil farklı olabilir. Bu grafikler aynı fazda olduğunu göstermek için çizilmiştir.

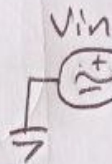


→ Tersleme Olabilir mi?

Konudan Bağımsız



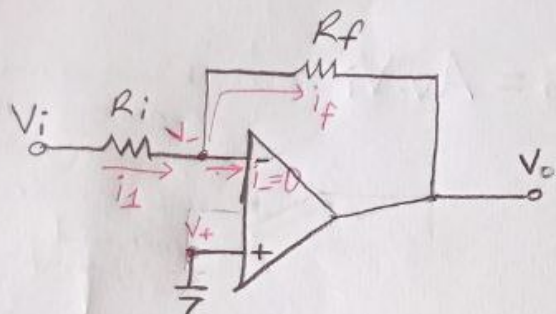
==



Önemli olan içerdeki R_{in} 'den önce üzerine hangi polaritede ve bşşklökte gerilim düştüğüdür. Eğer polarite ve bşşklökte aynı ise R_{in} direnci üzerinden 0 zaman aynı değerdir.

Geribeslemeli Pratik Opamp Devreleri

① Tersleyen Yükselteç (Inverting Amp)



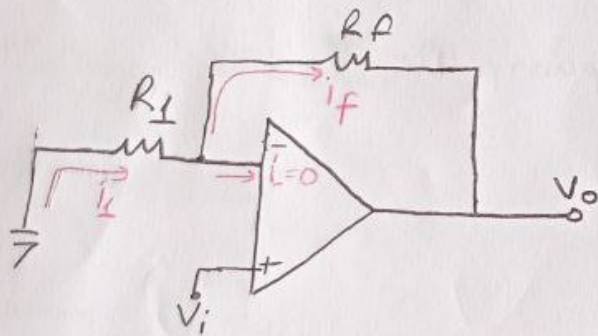
$$i_1 = i_f$$

$$V_+ = V_- = 0V \text{ (Toprak)}$$

$$\frac{V_i - V_-}{R_1} = \frac{V_- - V_o}{R_f}$$

$$\frac{V_i}{R_1} = \frac{-V_o}{R_f} \Rightarrow \frac{V_o}{V_i} = -\frac{R_f}{R_1} \Rightarrow V_o = -\frac{R_f}{R_1} \cdot V_i$$

② Terslemeyen Yükselteç (Non-Inv. Amp)



$$i_1 = i_f$$

$$V_+ = V_{in} = V_-$$

$$\frac{0 - V_-}{R_1} = \frac{V_- - V_o}{R_f}$$

$$V_- = V_i \text{ ise}$$

$$\frac{-V_i}{R_1} = \frac{V_i - V_o}{R_f} \Rightarrow \frac{V_o}{V_i} = 1 + \frac{R_f}{R_1}$$

$$V_o = \left(1 + \frac{R_f}{R_1}\right) \cdot V_i$$

⑤ Integral Alıcı Devresi



$$I_A = I_C \Rightarrow \frac{V_i(t)}{R} = i_C$$

C iletir

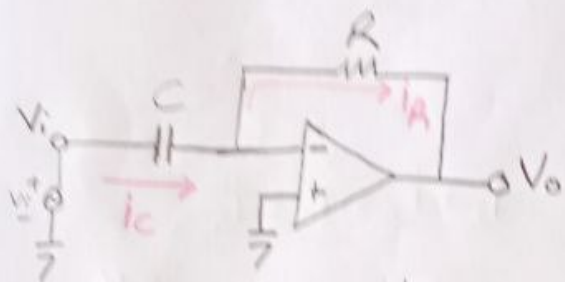
$$V_C = \frac{1}{C} \int i_C(t) \cdot dt$$

$$V_+ = 0 = V_-$$

$$0 - V_o = V_C \Rightarrow V_C = -V_o = \frac{1}{C} \int \frac{V_i(t)}{R} \cdot dt$$

$$V_o = -\frac{1}{C} \int \frac{V_i(t)}{R} \cdot dt \Rightarrow V_o = -\frac{1}{R \cdot C} \int V_i(t) \cdot dt$$

⑥ Tersin Alıcı Devre



$$I_C = I_R \Rightarrow i_C = -\frac{V_o}{R}$$

$$V_+ = 0V = V_-$$

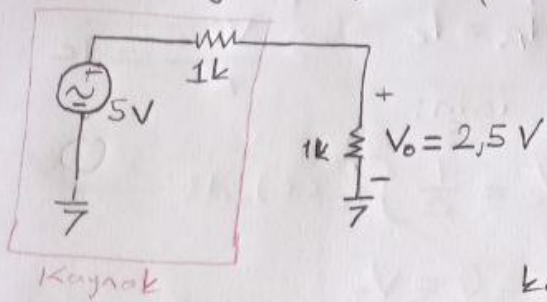
$$V_i - 0 = V_C \Rightarrow V_i = V_C$$

$$V_C = V_i = -\frac{1}{C} \int -\frac{V_o}{R} \cdot dt$$

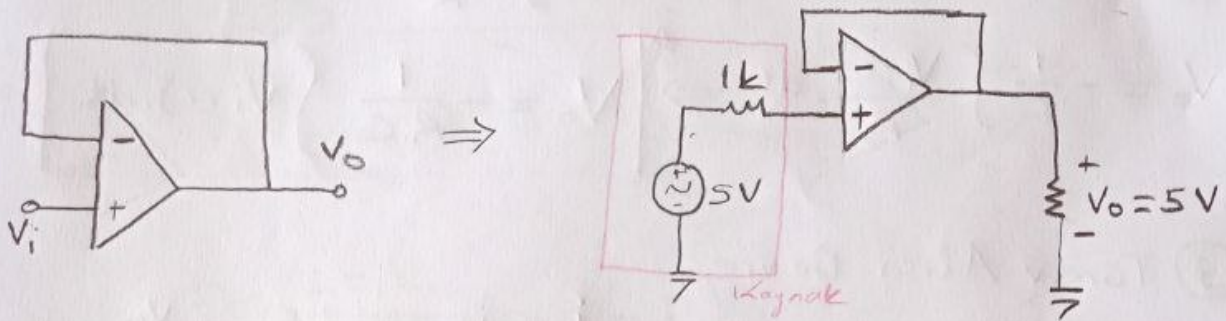
$$V_i(t) = -\frac{1}{RC} \int V_o(t) \cdot dt \leftarrow \text{Tersin alıcı devre}$$

$$V_o(t) = -RC \cdot \frac{dV_i(t)}{dt}$$

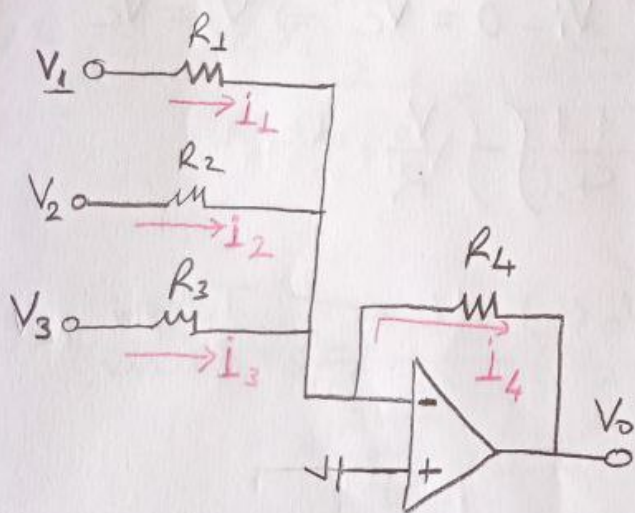
③ Voltaj Takipçisi (Voltage Follower-Buffer)



⇒ Kaynağın iç direnci olduğu için kaynak voltajının tamamı yük direnci üzerindeki düşmez. Eğer yük direncine kaynak voltajının tamamını düşürmek istiyorsak Voltaj takipçisi devresi yaparız.



④ Toplayan Yükseltici



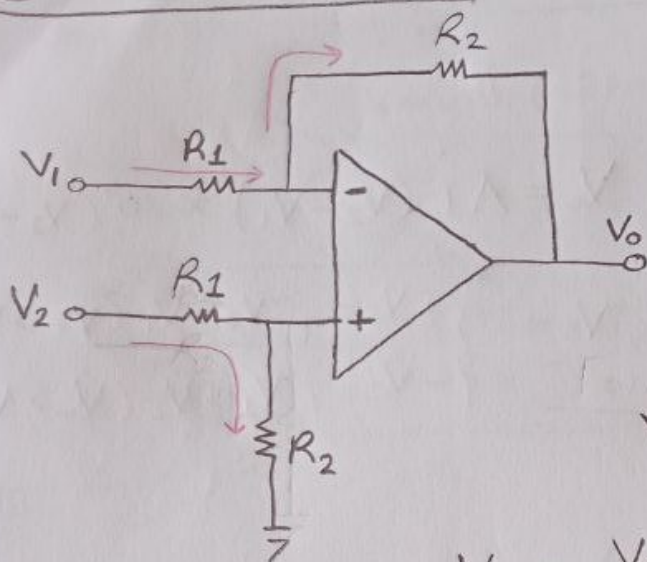
$$i_1 + i_2 + i_3 = i_4$$

$$\frac{V_3}{R_3} + \frac{V_2}{R_2} + \frac{V_1}{R_1} = \frac{-V_0}{R_f}$$

$$V_0 = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right)$$

$$V_0 = - \left(\frac{R_f}{R_1} \cdot V_1 + \frac{R_f}{R_2} \cdot V_2 + \frac{R_f}{R_3} \cdot V_3 \right)$$

7 Çıkarma Devresi



Formül

$$V_o = \frac{R_2}{R_1} (V_2 - V_1)$$

Formülün Çıkarılması

$$V_- = V_+ \quad i_- = i_+ = 0$$

$$V_+ = \frac{V_2}{R_1 + R_2} \cdot R_2 = V_-$$

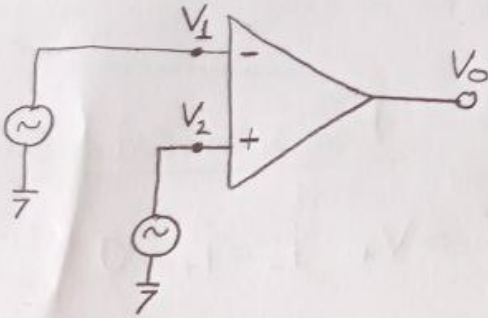
$$\frac{V_1 - V_-}{R_1} = \frac{V_- - V_o}{R_2}$$

$$\frac{V_1 - \frac{V_2}{R_1 + R_2} \cdot R_2}{R_1} = \frac{\frac{V_2}{R_1 + R_2} \cdot R_2 - V_o}{R_2}$$

$$\star \quad V_o = \frac{R_2}{R_1} (V_2 - V_1)$$

Enstrümantasyon Devreleri

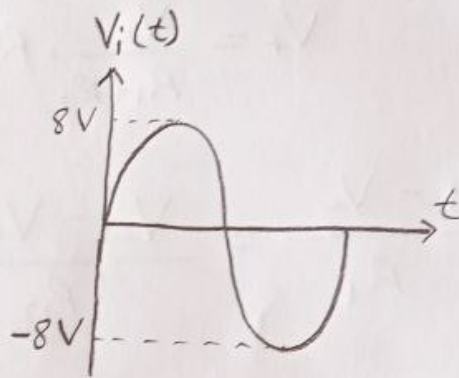
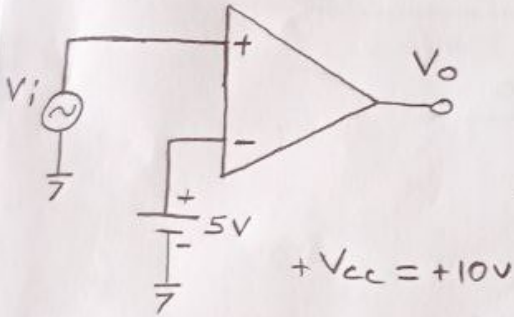
① - Karşılaştırmalı (Komparatör) Devresi



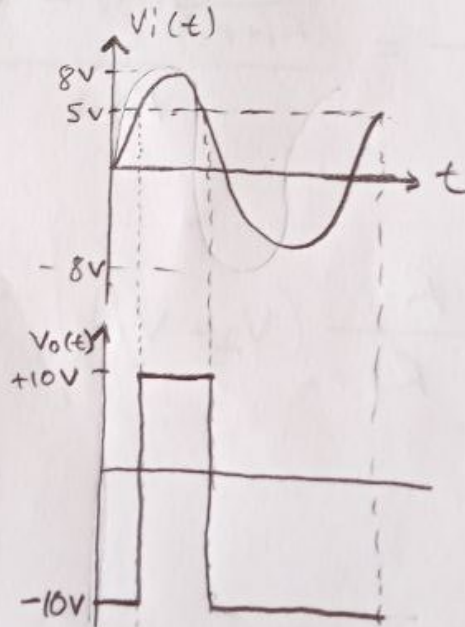
$$V_o = A_d \cdot (V_2 - V_1) \approx 10^5 (V_2 - V_1)$$

$$V_o = \begin{cases} +V_{cc}, & V_2 > V_1 \quad (V_+ > V_-) \\ -V_{cc}, & V_1 > V_2 \quad (V_- > V_+) \end{cases}$$

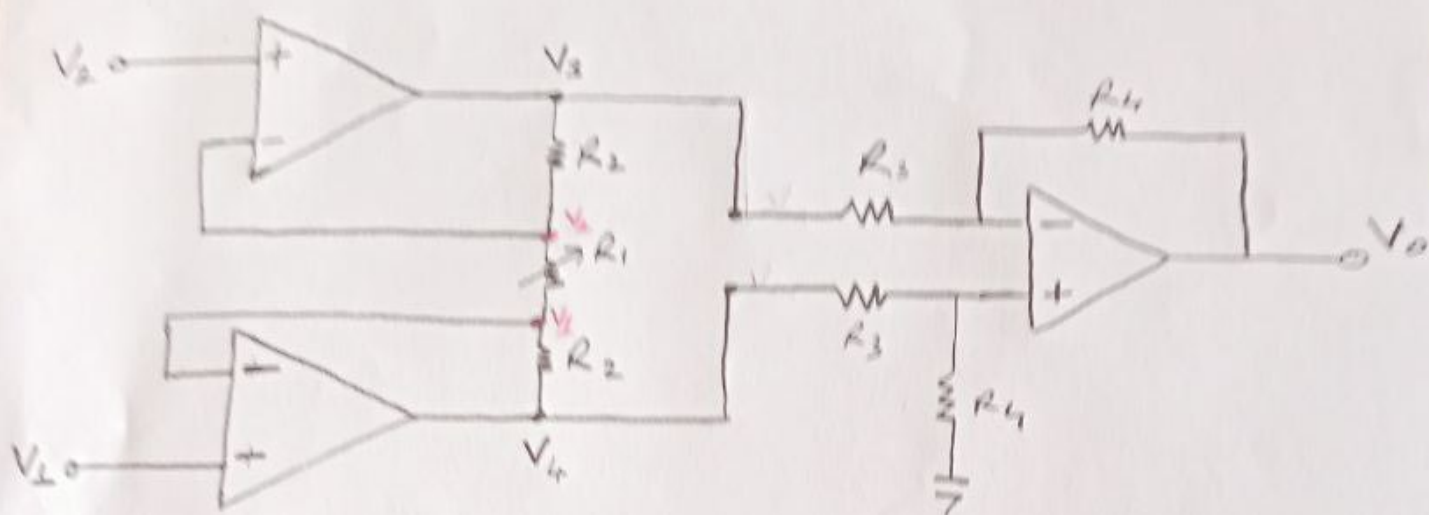
Örnek ($V_{cc} = \pm 10V$)



Çözüm



② Enstrümantasyon Yükseltici



$$\frac{V_2 - V_1}{R_1} = \frac{V_3 - V_4}{2R_2 + R_1}$$

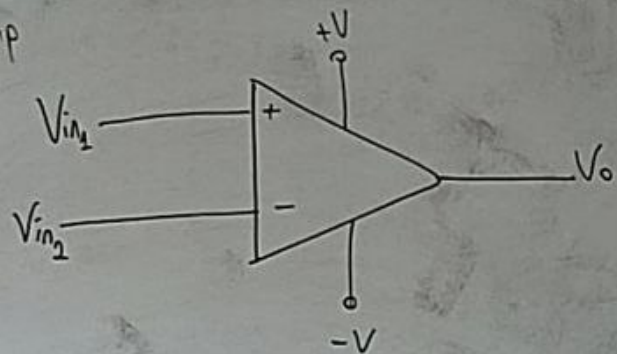
$$V_0 = -\frac{R_4}{R_3} (V_4 - V_3)$$

$$\frac{R_3 \cdot V_0}{R_4} = V_2 - V_1$$

$$V_0 = \left[\left(\frac{2R_2 + R_1}{R_1} \right) \left(\frac{R_4}{R_3} \right) \right] (V_1 - V_2)$$

PSNR

Opamp



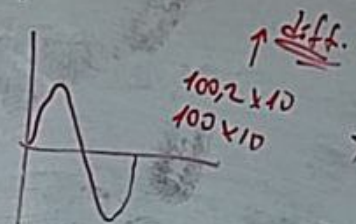
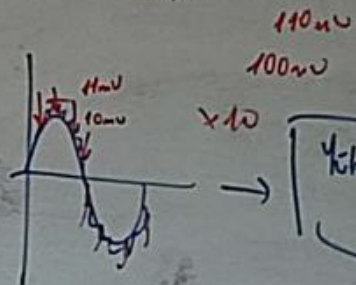
$$A_c = K_c = \frac{V_{o1} + V_{o2}}{V_{i1} + V_{i2}} = \text{Ortak Kazanç}$$

common

$$A_d = K_d = \frac{V_{o1} - V_{o2}}{V_{i1} - V_{i2}} = \text{Farklı Kazanç}$$

$$h_{ie} = h_{fe} \cdot (r_e)$$

$$r_e = \frac{26 \text{ mV}}{I_e}$$



$$\frac{120}{80}$$

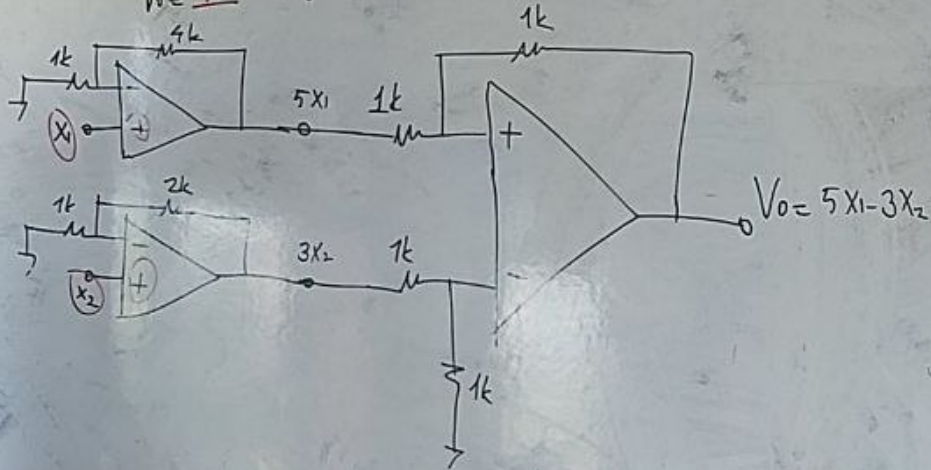
$$\frac{122}{86}$$

$$\frac{120.3}{79.9}$$

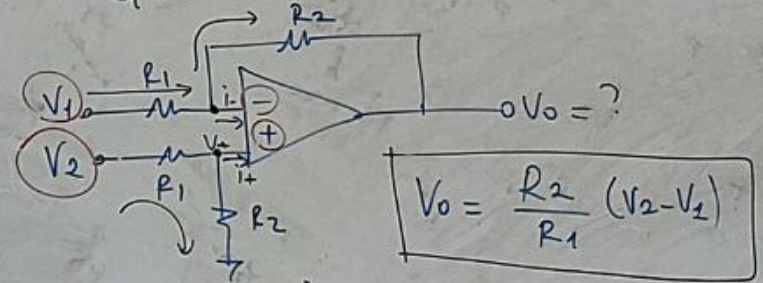
Opamp Parametreleri

OPAMP UYGULAMALARI

Örnek: X_1 ve X_2 girişler olmak üzere;
 $V_0 = 5X_1 - 3X_2$ 'yi opamp ile gerçekleştir.



Çıkartıcı Devresi



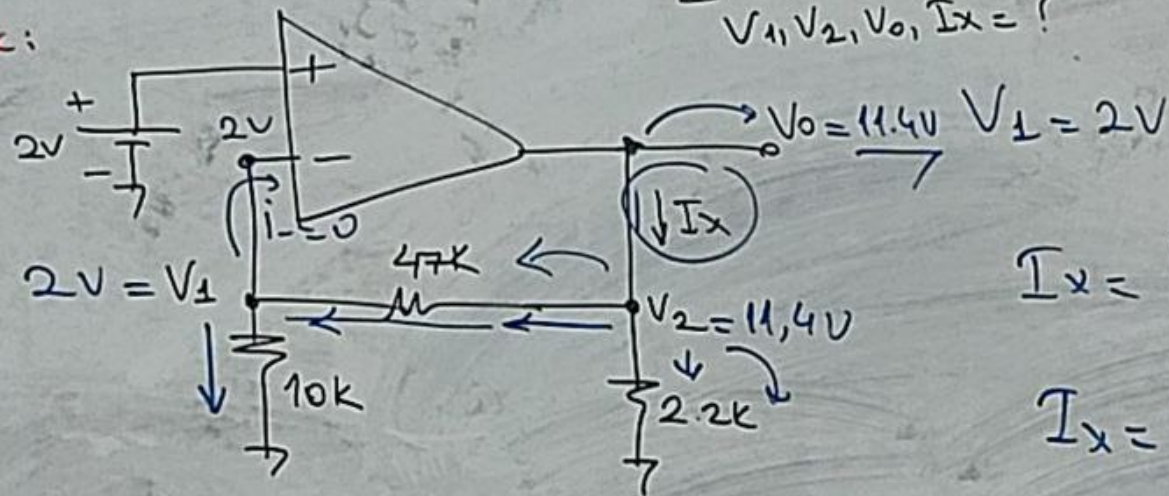
$$V_- = V_+ \quad i_- = i_+ = 0$$

$$V_+ = \frac{V_2}{(R_1 + R_2)} \cdot R_2 = V_-$$

$$\frac{V_2 - V_-}{R_1} = \frac{V_- - V_0}{R_2}$$

$$\frac{V_1 - \frac{V_2}{(R_1 + R_2)} \cdot R_2}{R_1} = \frac{\frac{V_2}{(R_1 + R_2)} \cdot R_2 - V_0}{R_2}$$

Örnek:



$$I_x = \frac{V_2 - V_1}{47k} + \frac{V_2 - 0}{2.2k}$$

$$I_x = \frac{11.4 - 2}{47k} + \frac{11.4 - 0}{2.2k} = 5.38mA$$

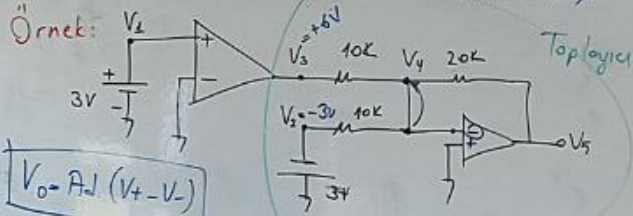
$$\frac{V_2 - 2V}{47k} = \frac{2 - 0}{10k} \Rightarrow \boxed{V_2 = 11.4V}$$

X₂



Opamp Parametreleri

OPAMP UYGULAMALARI



$$V_o = A_d (V_+ - V_-)$$

$$V_o = 10^5 (3V - 0V)$$

$$V_o = 3 \cdot 10^5 V$$

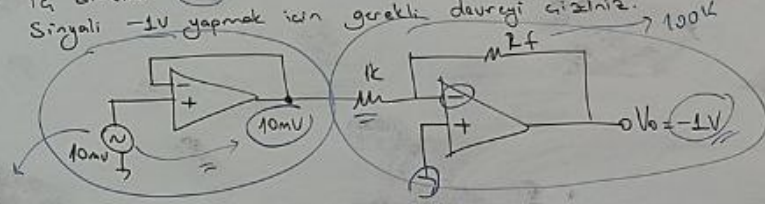
$$V_o = +6V$$

$$V_o = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 \right)$$

$$\Rightarrow V_5 = - \left(\frac{20k}{10k} \cdot (6V) + \frac{20k}{10k} \cdot (-3V) \right)$$

$$V_5 = -6V$$

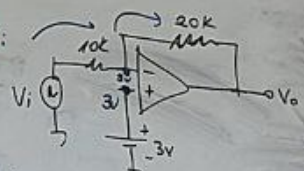
Örnek: İki direnci 50k olan bir transducer'den alınan 10mV'lık sinüsoidal Singali -1V yapmak için gerekli devreyi çiziniz.



$$10mV \rightarrow -1V$$

$$- \frac{R_f}{R_i} = \frac{-1V}{10mV} \Rightarrow R_f = 100k$$

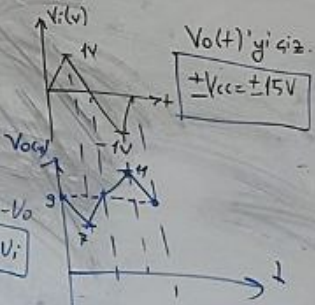
Örnek:



$$\frac{V_i - 3}{10k} = \frac{3 - V_o}{20k}$$

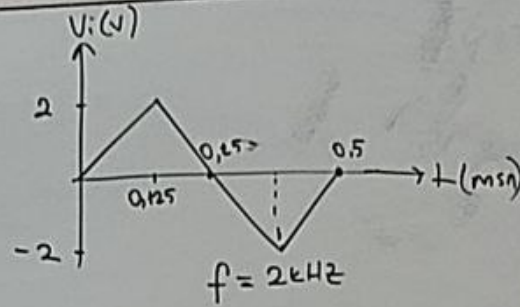
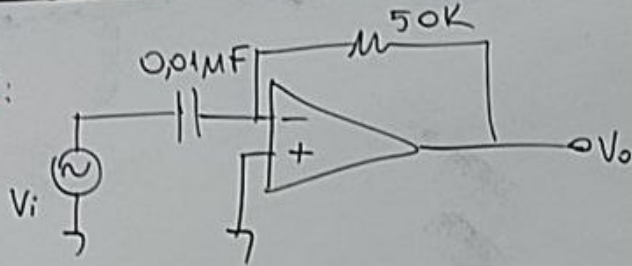
$$2V_i - 6 = 3 - V_o$$

$$V_o = 9 - 2V_i$$



$$\pm V_{cc} = \pm 15V$$

ÖRNEK:



$$V_o = -R.C. \frac{dV_i(t)}{dt}$$

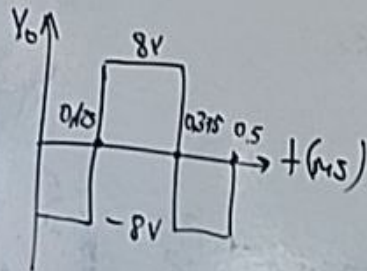
Giriş sinyalinin değişim aralığı dikkate alınır;

$$0 < t < 0,125 \text{ ms} \quad V_i(t) = \frac{2}{0,125 \cdot 10^{-3}} t = 16000t$$

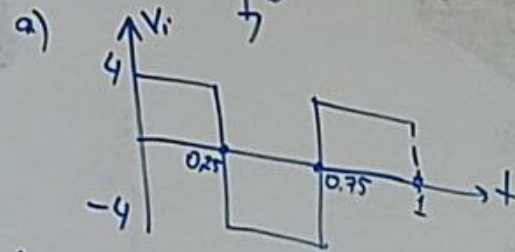
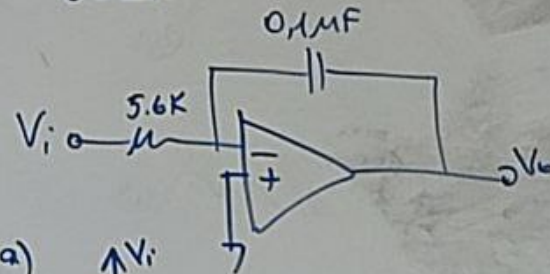
$$V_o = -(50 \cdot 10^3)(0,01 \cdot 10^{-6}) \cdot \frac{d}{dt}(16000t) \Rightarrow \boxed{V_o = -8V}$$

$$0,125 \text{ ms} < t < 0,375 \text{ ms} \quad V_i(t) = -16000t + 4 \Rightarrow \boxed{V_o = 8V}$$

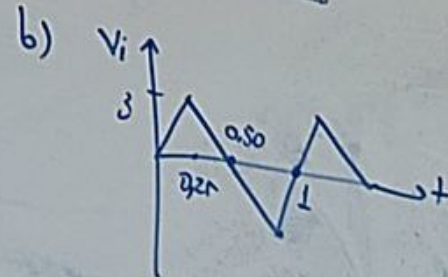
0,375 ms < t < 0,5 ms için $V_o = -8V$ olur.



ÖDEV:



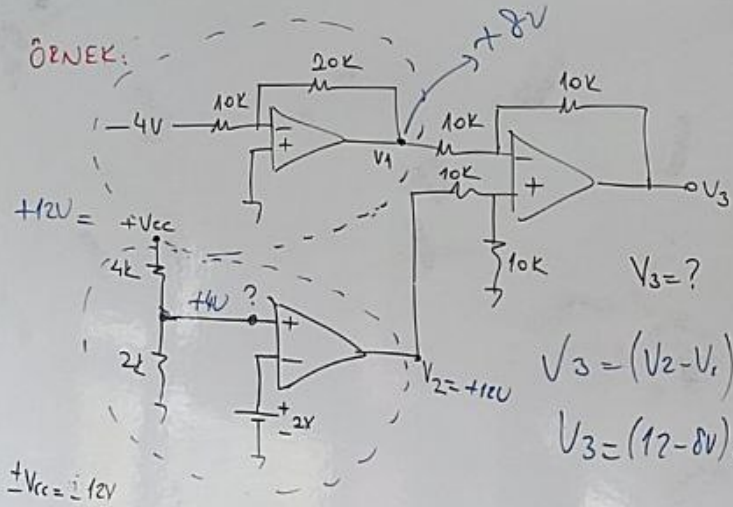
$V_o(t)$ 'yi çiz.



$V_o(t)$ 'yi çiz.



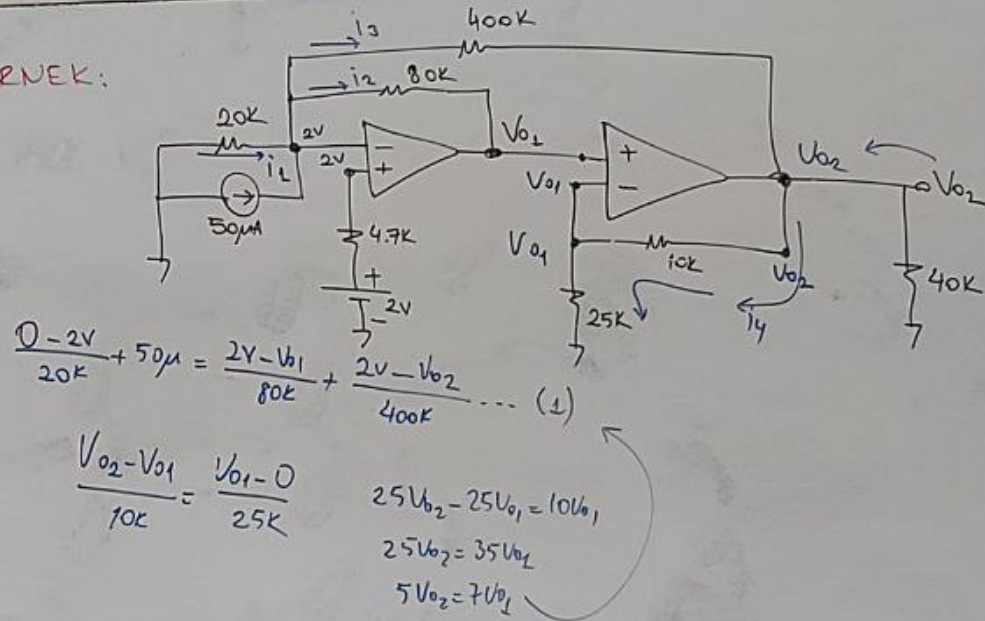
ÖRNEK:



$$V_3 = (V_2 - V_1)$$

$$V_3 = (12 - 8) = 4V$$

ÖRNEK:



$$\frac{0 - 2V}{20K} + 50\mu = \frac{2V - V_{01}}{80K} + \frac{2V - V_{02}}{400K} \dots (1)$$

$$\frac{V_{02} - V_{01}}{10K} = \frac{V_{01} - 0}{25K}$$

$$25V_{02} - 25V_{01} = 10V_{01}$$

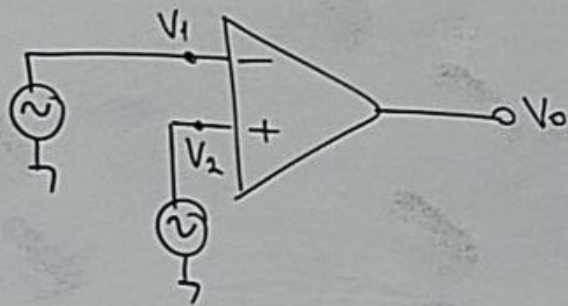
$$25V_{02} = 35V_{01}$$

$$5V_{02} = 7V_{01}$$

$V_{01}, V_{02} = ?$

Enstrümantasyon Devreleri

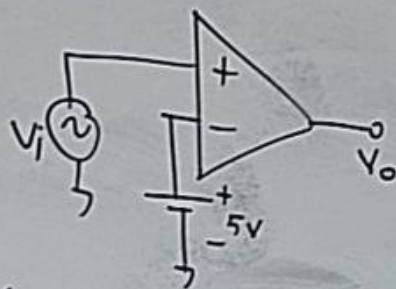
1 - Karşılaştırıcı (Komparatör)



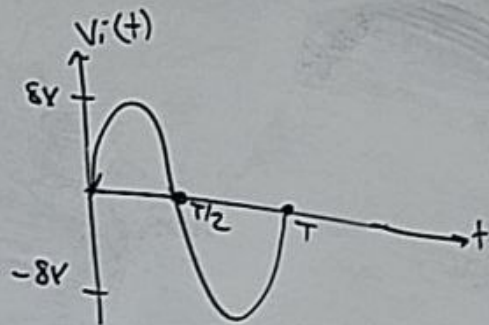
$$V_o = A \cdot (V_2 - V_1) \approx 10^5 (V_2 - V_1)$$

$$V_o = \begin{cases} +V_{cc} & V_2 > V_1 \quad (V_+ > V_-) \\ -V_{cc} & V_1 > V_2 \quad (V_- > V_+) \end{cases}$$

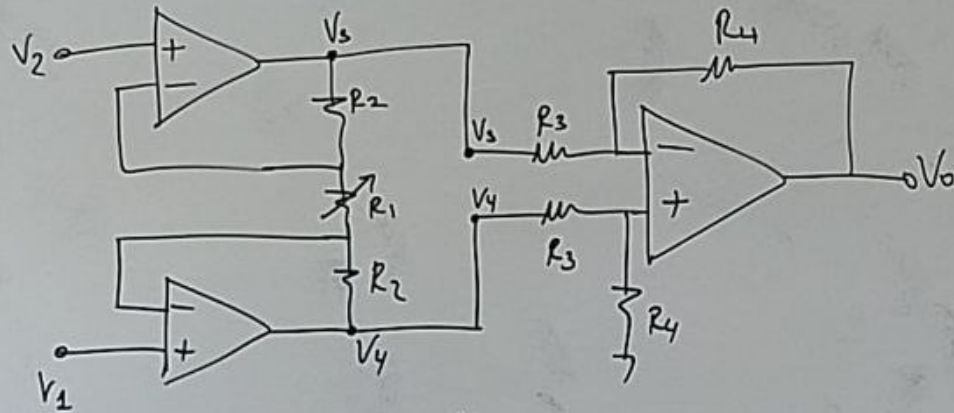
ÖRNEK:



$$V_{cc} = \pm 10V$$



2. Enstrümantasyon Yükselteci

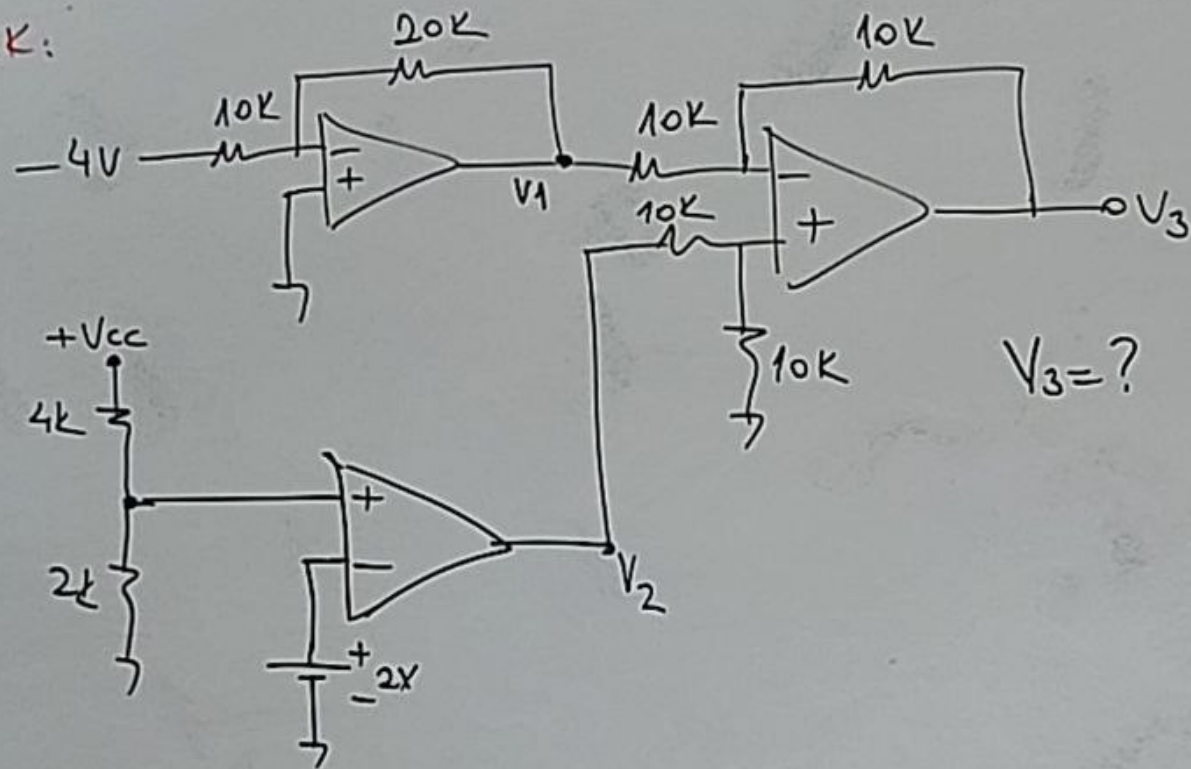


$$\frac{V_2 - V_1}{R_1} = \frac{V_3 - V_4}{2R_2 + R_1}$$

$$V_0 = -\frac{R_4}{R_3} (V_4 - V_3)$$

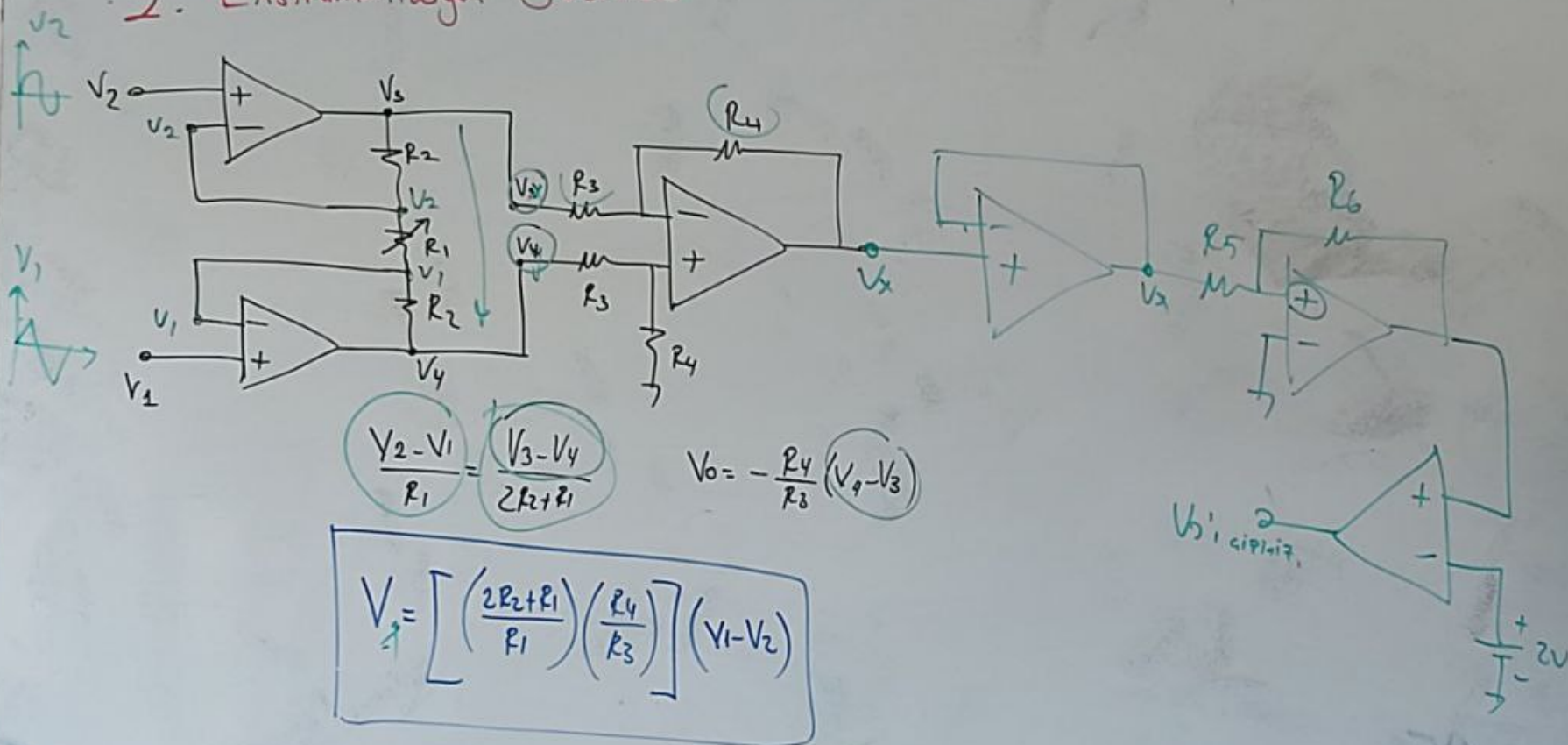
$$V_0 = \left[\left(\frac{2R_2 + R_1}{R_1} \right) \left(\frac{R_4}{R_3} \right) \right] (V_1 - V_2)$$

ÖRNEK:

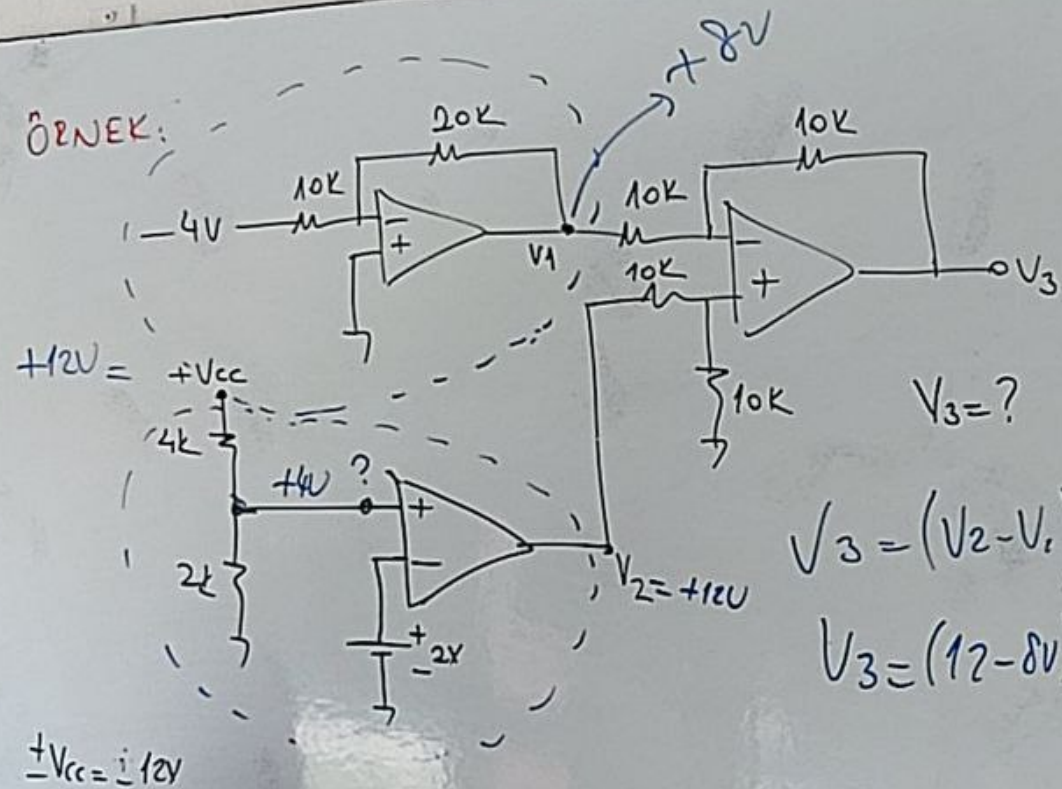


$$\pm V_{CC} = \pm 12V$$

2. Enstrümantasyon Yükselteci



ÖRNEK:

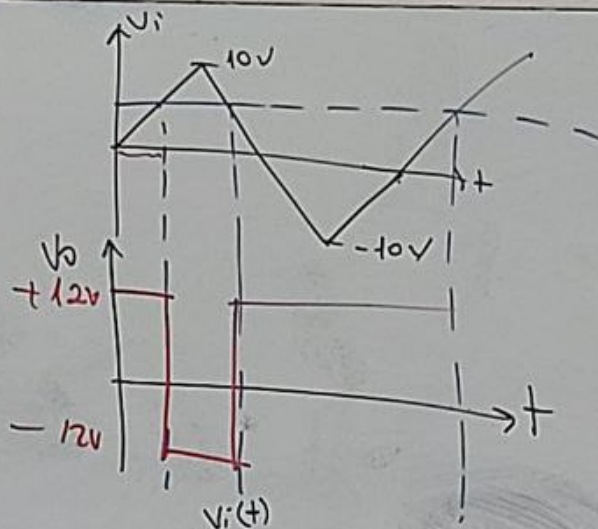
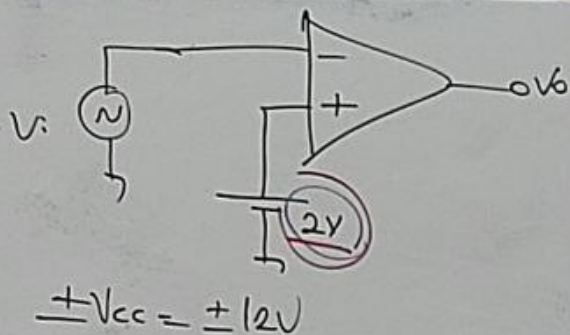


$$V_1 = -\frac{R_f}{R_1} (-4) = \left(-10 \cdot \frac{20}{10}\right) = 8V$$

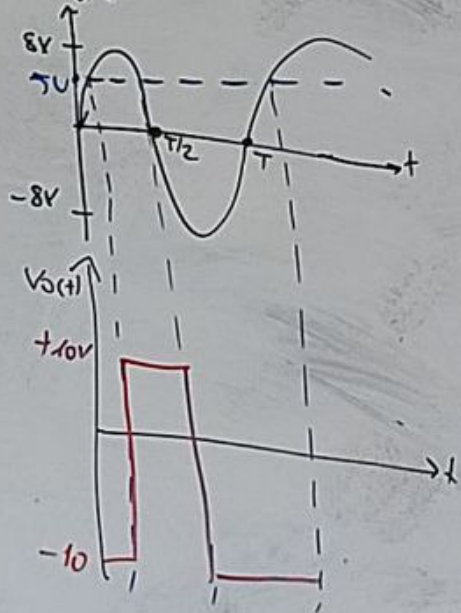
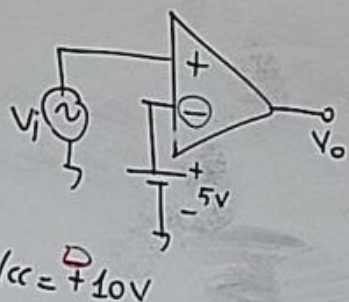
$V_3 = ?$

$$V_3 = (V_2 - V_1)$$

$$V_3 = (12 - 8V) = 4V$$



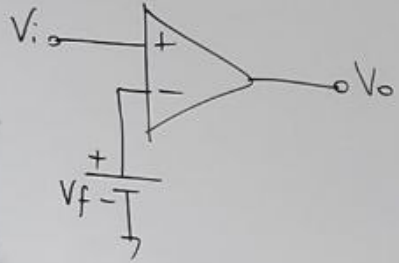
ÖRNEK:



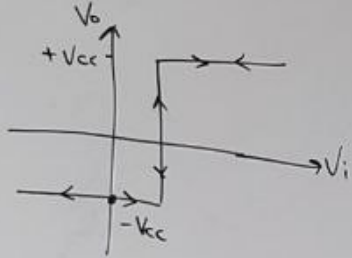
Schmitt Trigger (Tetikleyici)



Simbolü

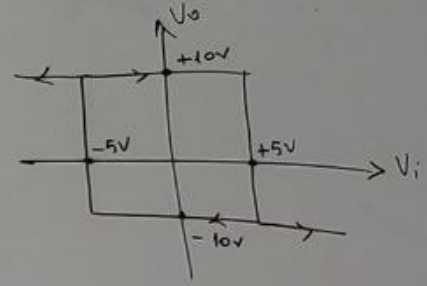
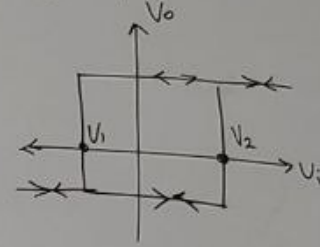
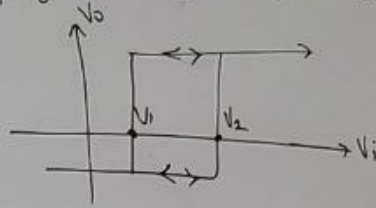


$V_i > V_f$ ise $+V_{cc}$
 $V_i < V_f$ ise $-V_{cc}$

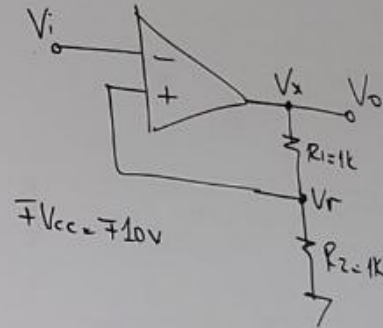


Sayısal devrelerde gürültü engellemek için kullanılır.

V_f gerilimi $[V_1 - V_2]$ aralığında değişirse;



Örnek:



$$V_r = V_x \cdot \frac{R_2}{R_1 + R_2}$$

* $V_x = +V_{cc}$ ise;

$$V_r = V_{cc} \cdot \frac{1k}{2k} = \frac{V_{cc}}{2}$$

$$V_i > V_r \Rightarrow V_x = -V_{cc}$$

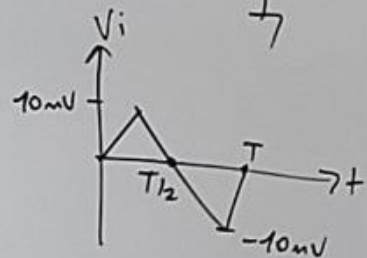
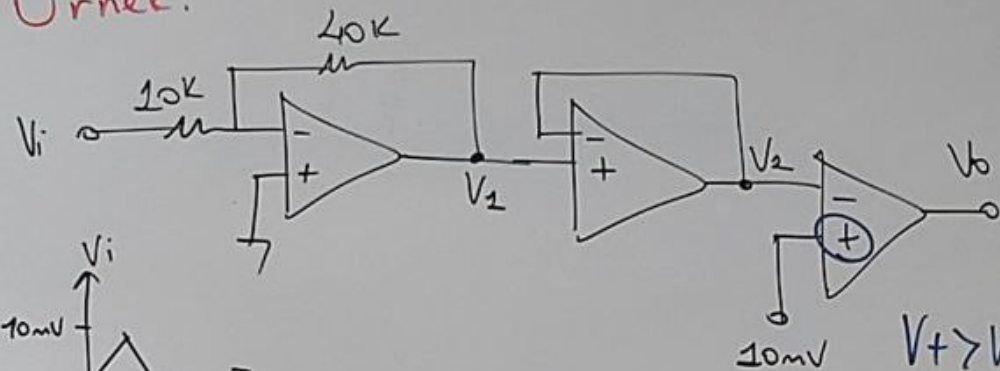
$$V_i < V_r \Rightarrow V_x = +V_{cc}$$

* $V_x = -V_{cc}$ ise;

$$V_r = -\frac{V_{cc}}{2}$$

Örnek:

$$\pm V_{CC} = \pm 10V$$

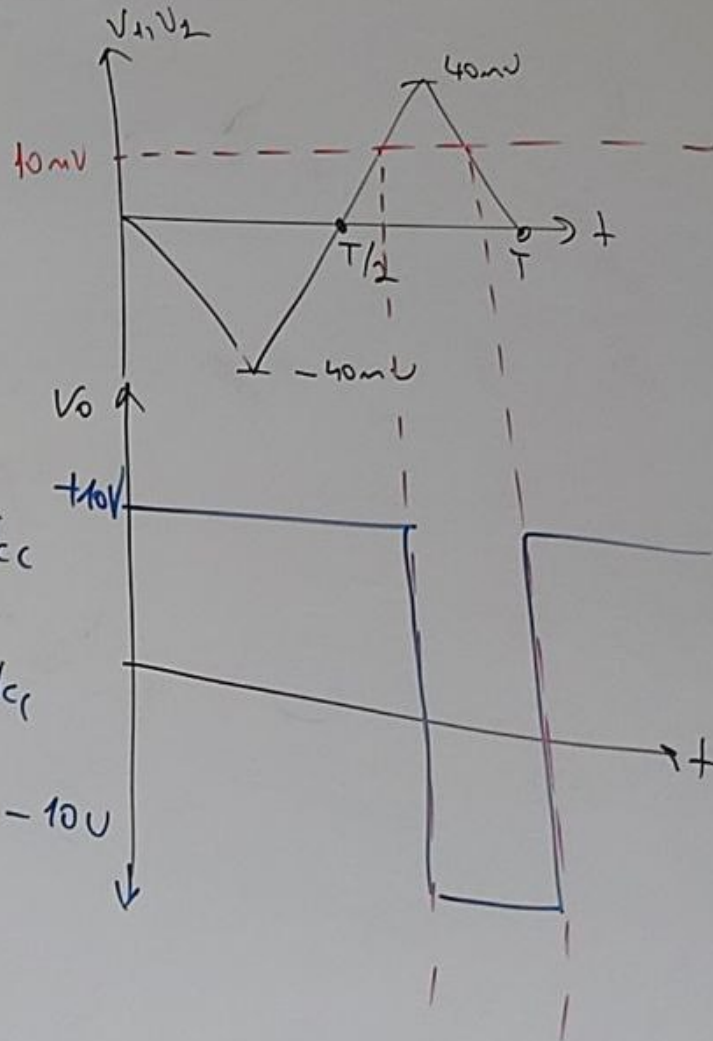
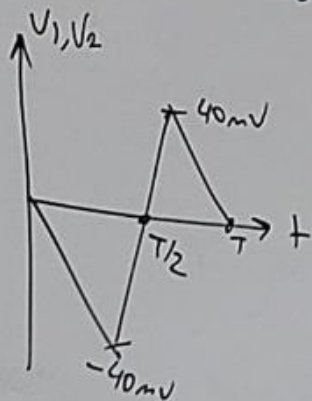


V_1, V_2, V_0 'ı çiz.

$V_+ > V_- \Rightarrow +V_{CC}$
 $V_- > V_+ \Rightarrow -V_{CC}$

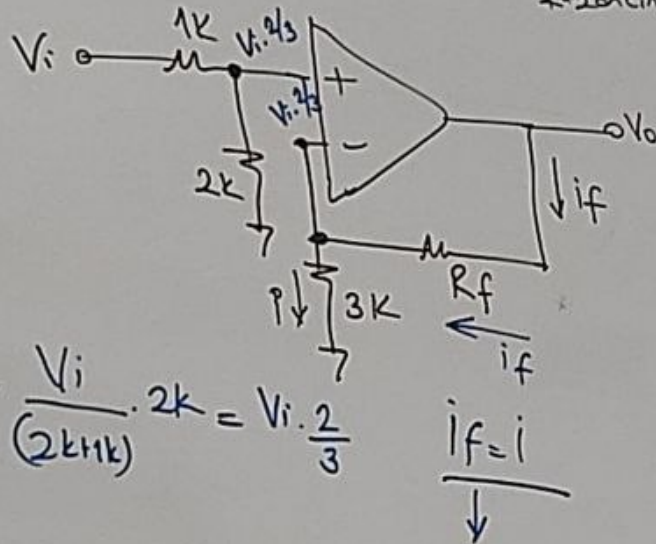
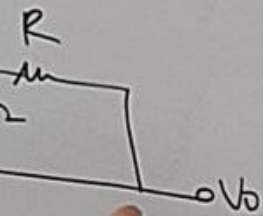
$$V_1 = - \left(\frac{40k}{10k} \right) V_i = -4 \cdot V_i$$

$$V_2 = V_1$$



Sınav Sorusu:

Aşağıda verilen ideal opampın kazancının 5 olması için $R_f = ?$



$$\boxed{\frac{V_o}{V_i} = 5} \quad V_o = 5V_i$$

$$V_+ = V_- \\ i_+ = i_- = 0$$

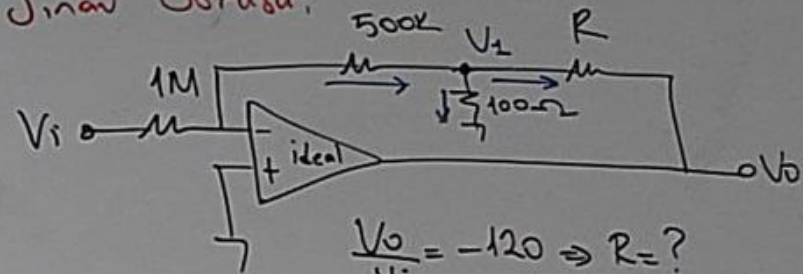
$$V_+ = \frac{V_i}{(2k + 1k)} \cdot 2k = V_i \cdot \frac{2}{3}$$

$$\frac{V_o - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3k}$$

$$\frac{\frac{13V_i}{3}}{R_f} = \frac{\frac{2}{3}V_i R_f}{3k}$$

$$\frac{39}{2} = R_f \Rightarrow \boxed{R_f = 19.5k}$$

Sınav Sorusu:



$$\frac{V_o}{V_i} = -120 \Rightarrow R = ?$$

$$\frac{V_i - 0}{\frac{1M}{2}} = \frac{0 - V_1}{500k}$$

$$\boxed{V_i = -2V_1}$$

$$V_1 = -\frac{V_i}{2}$$

$$\frac{0 - V_1}{500k} = \frac{V_i - 0}{100\Omega} + \frac{V_1 - V_o}{R}$$

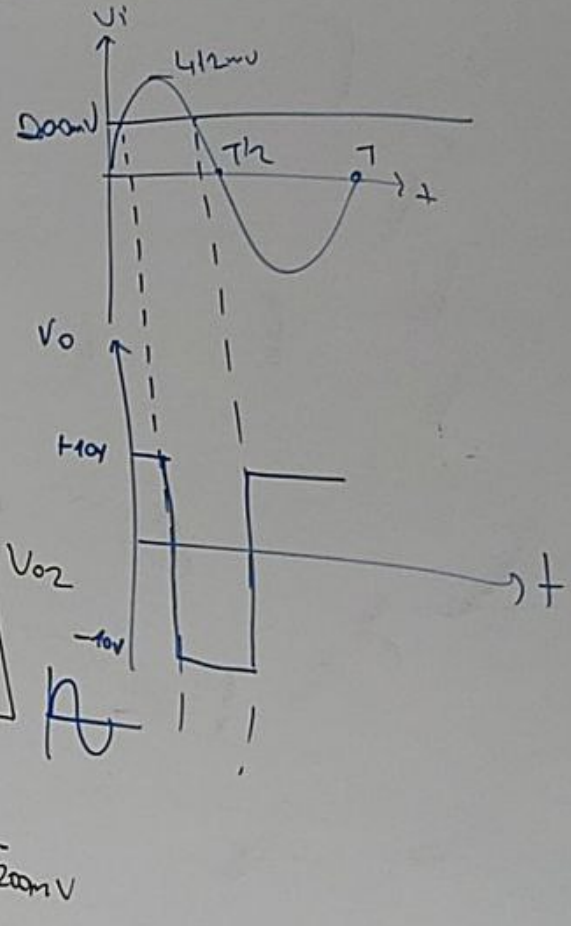
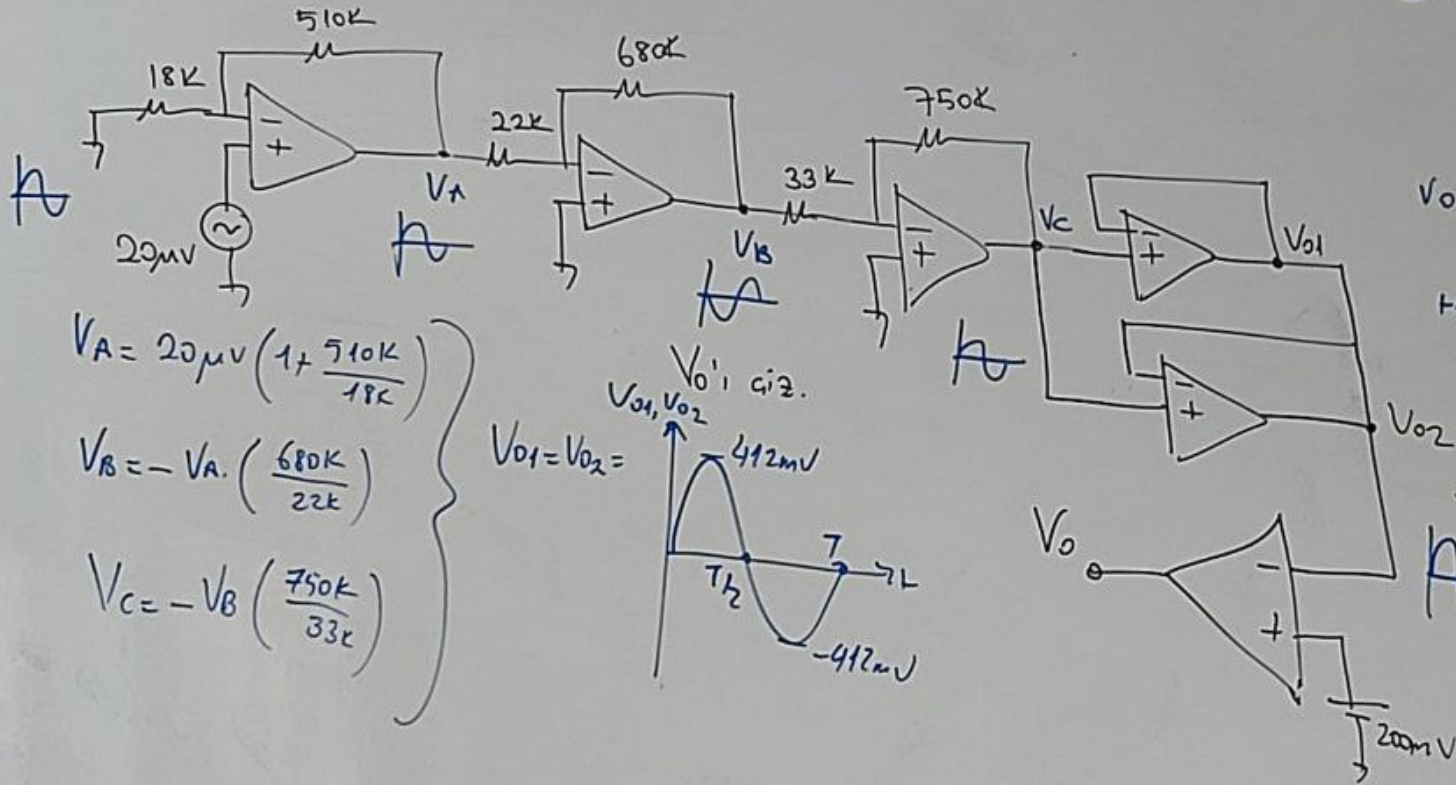
$$\frac{-V_1}{500k} = \frac{V_1}{100\Omega} + \frac{V_1 - V_o}{R}$$

$$\frac{\frac{V_i}{2}}{500k} = \frac{-\frac{V_i}{2}}{100\Omega} + \frac{\left(-\frac{V_i}{2}\right) - V_o}{R}$$

$$V_o = -120V_i \Rightarrow R = ?$$

Sinav Sorusu:

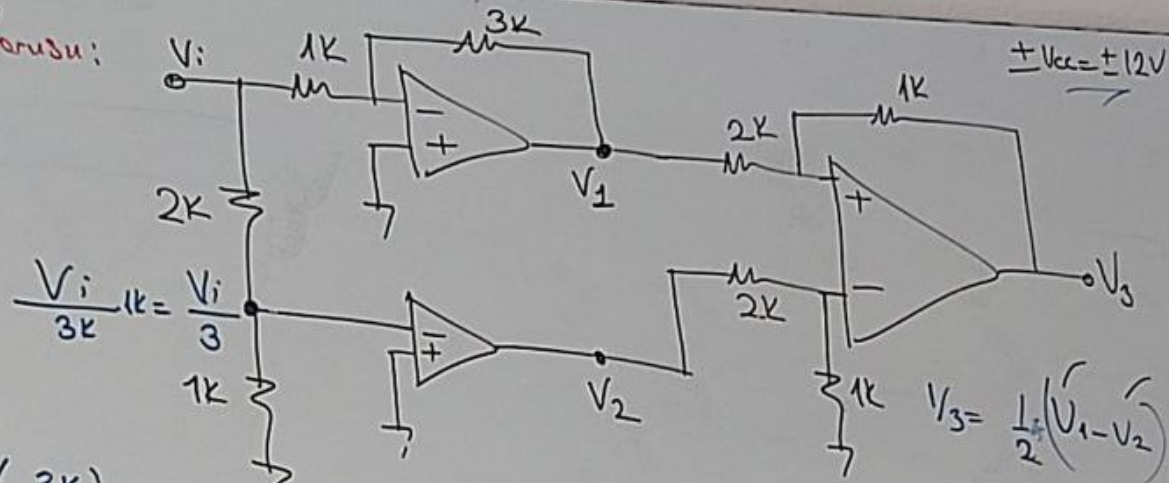
$$\pm V_{CC} = \pm 10V$$



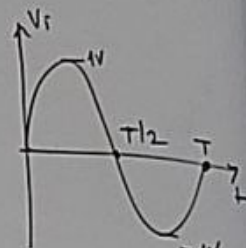
$\pm 15V$

$V_2 = ?$

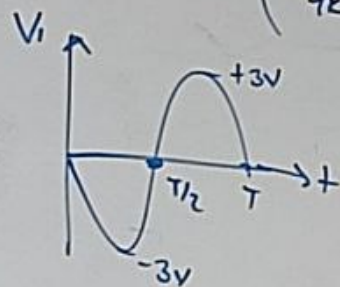
Sınav Sorusu:



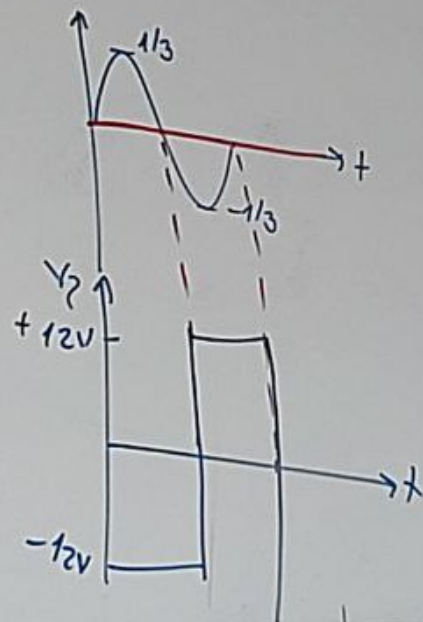
V_3 'ü çiziniz.



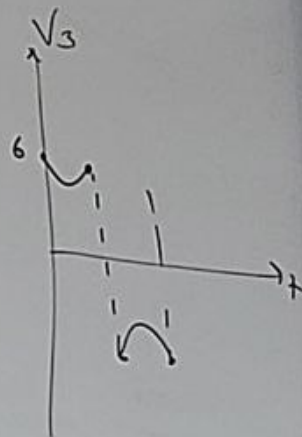
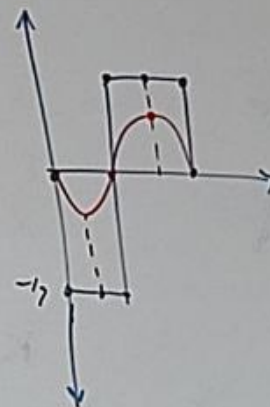
$$V_1 = -V_i \cdot \left(\frac{3k}{1k} \right) = -3V_i$$



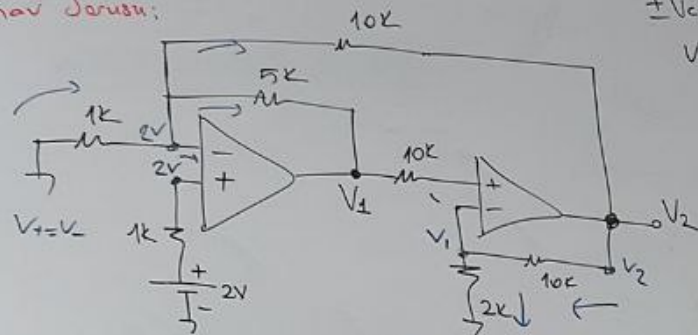
$V_2 \Rightarrow$



$$V_3 = \frac{1}{2}(V_1 - V_2)$$



Sinar Sarusu:



$$\pm V_{cc} = \pm 15V$$

$$V_1 = ? \quad V_2 = ?$$

$$\frac{0-2}{\frac{1k}{10}} = \frac{2-V_1}{\frac{5k}{1}} + \frac{2-V_2}{\frac{10k}{1}}$$

$$-20 = 4 - 2V_1 + 2 - 2V_2$$

$$2V_1 + 2V_2 = 26$$

$$8V_1 = 26$$

$$V_1 = 3.25V$$

$$\frac{V_2 - V_1}{\frac{10k}{5}} = \frac{V_1}{\frac{2k}{1}}$$

$$V_2 - V_1 = 5V_1$$

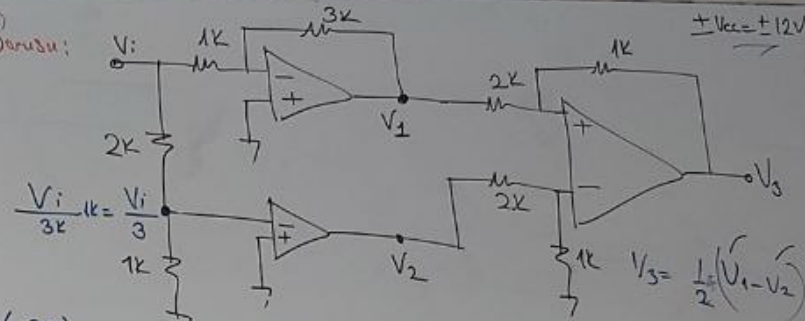
$$V_2 = 6V_1$$

$$V_2 = 2.25 \times 6$$

$$V_2 = 13.5$$

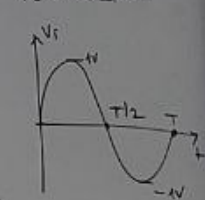
$$V_2 = +15V$$

Sinar Sarusu:

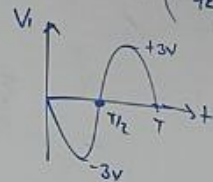


$$\pm V_{cc} = \pm 12V$$

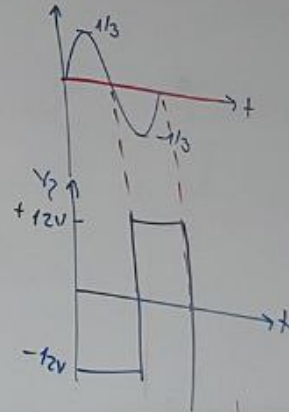
V_3 is a sine wave.



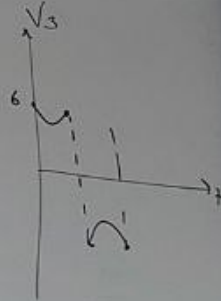
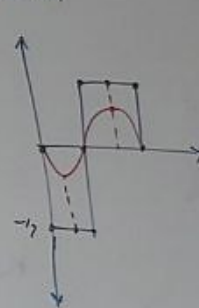
$$V_1 = -V_i \left(\frac{3k}{1k} \right) = -3V_i$$



$V_2 \Rightarrow$



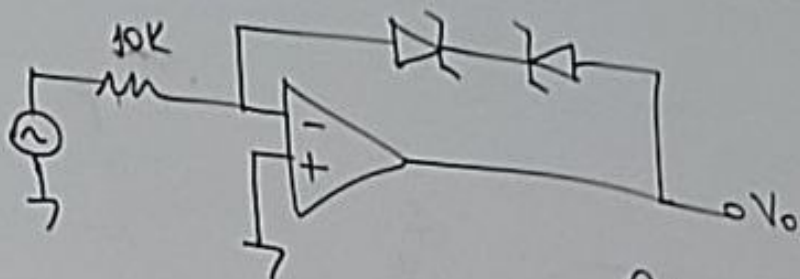
$$V_3 = \frac{1}{2}(V_1 - V_2)$$



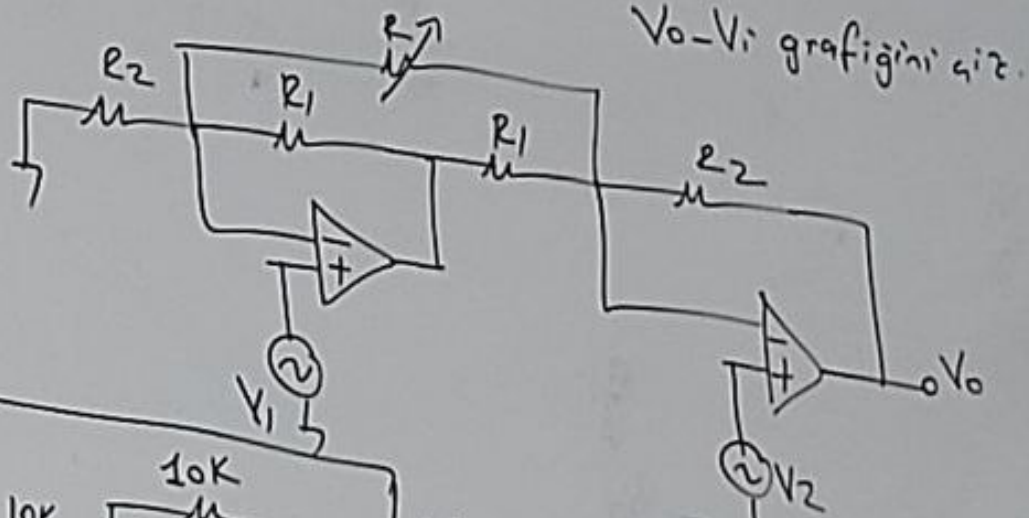
Çalışma Sorusu

$$V_{z1} = V_{z2} = 10V \quad (V_D = 0.7)$$

1

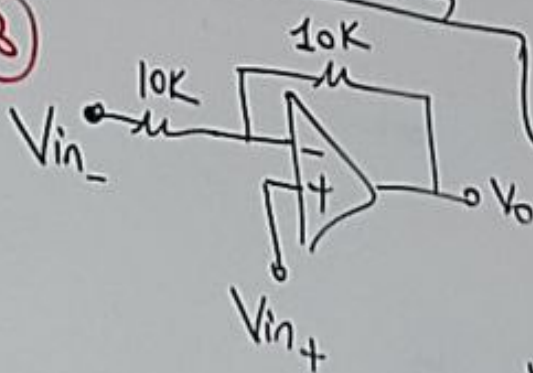


2



$A_v = \infty$
 $V_o - V_i$ grafigini çiz.

3



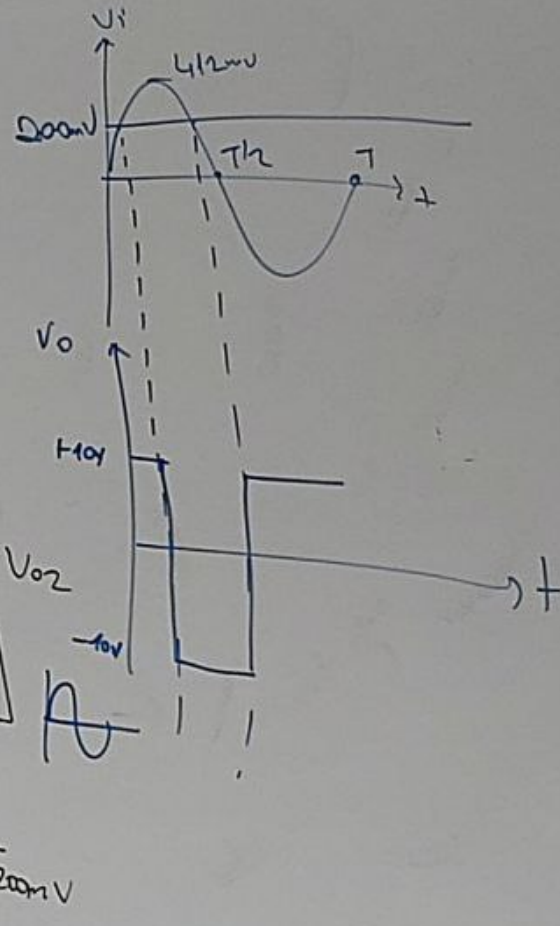
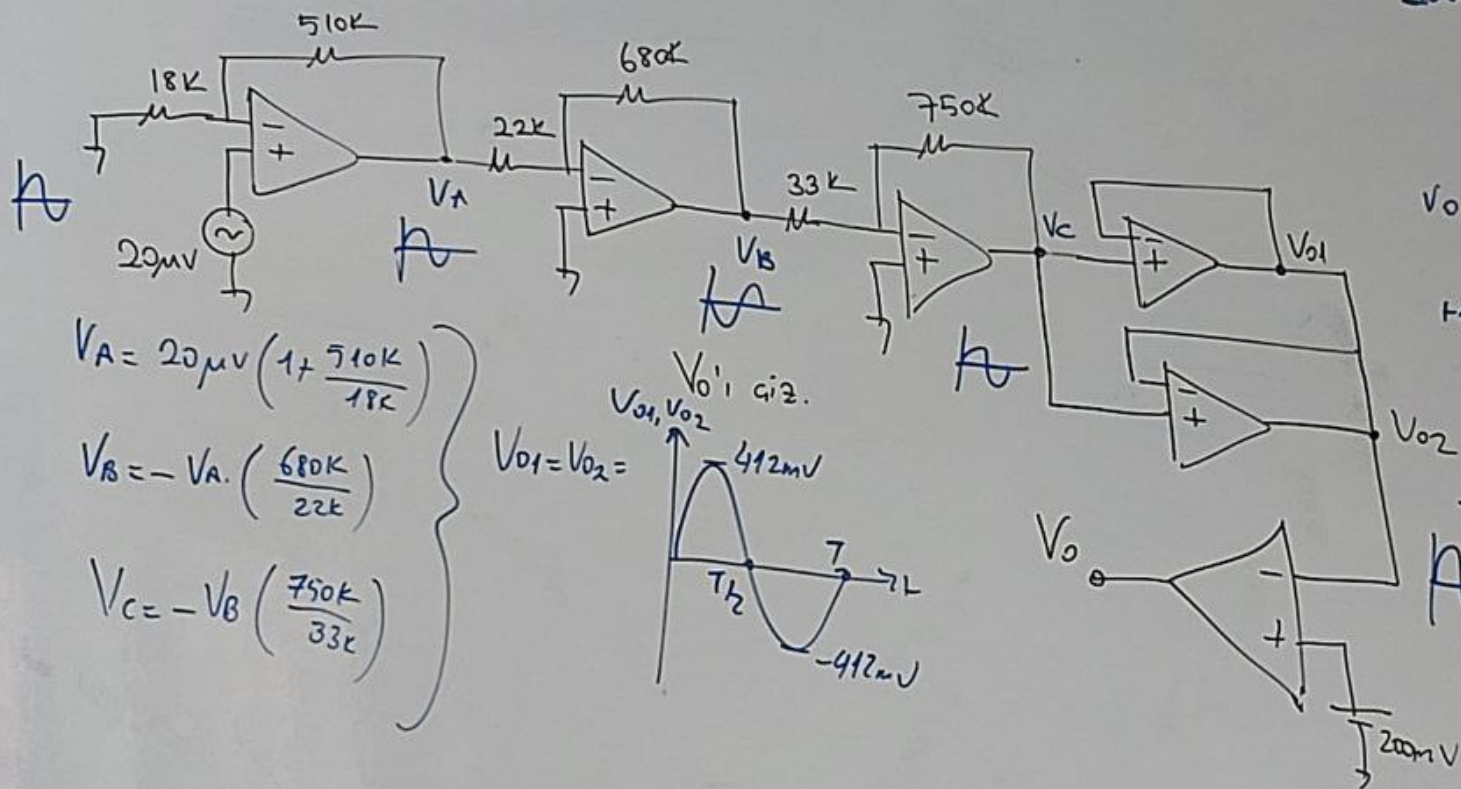
V_o ifadesini elde et.

$$V_{in-} = 5V_{pp} \sin$$

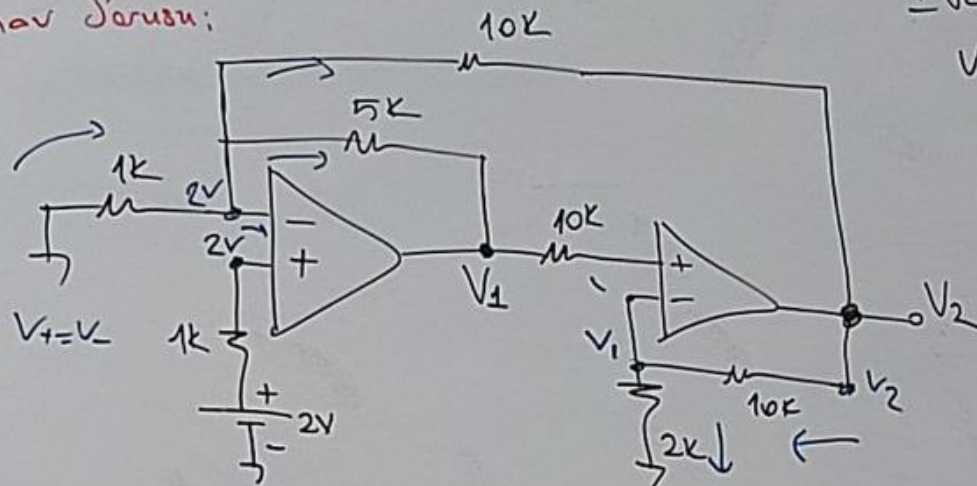
$$\left. \begin{array}{l} V_{in+} = 5V \\ V_{in-} = -5V \end{array} \right\} \text{ olursa } V_o - V_{in-} \text{ graf. çiz.}$$

Sinar Sarusu:

$$\pm V_{CC} = \pm 10V$$



Sinar Saru:



$$\pm V_{cc} = \pm 15V$$

$$V_1 = ? \quad V_2 = ?$$

$$\frac{0-2}{\left(\frac{1k}{10}\right)} = \frac{2-V_1}{5k} + \frac{2-V_2}{10k}$$

$$-20 = 4 - 2V_1 + 2 - V_2$$

$$2V_1 + V_2 = 26$$

$$6V_1$$

$$8V_1 = 26$$

$$V_1 = 3.25V$$

$$\frac{V_2 - V_1}{\frac{10k}{5}} = \frac{V_1}{2k}$$

$$V_2 - V_1 = 5V_1$$

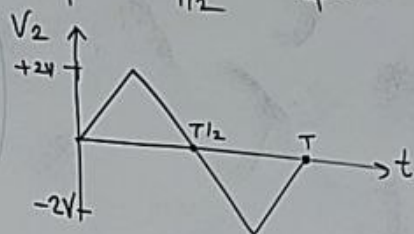
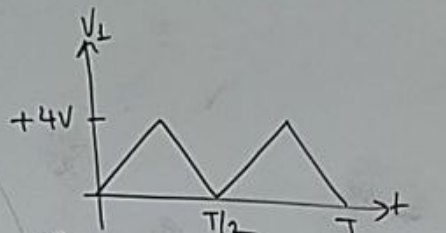
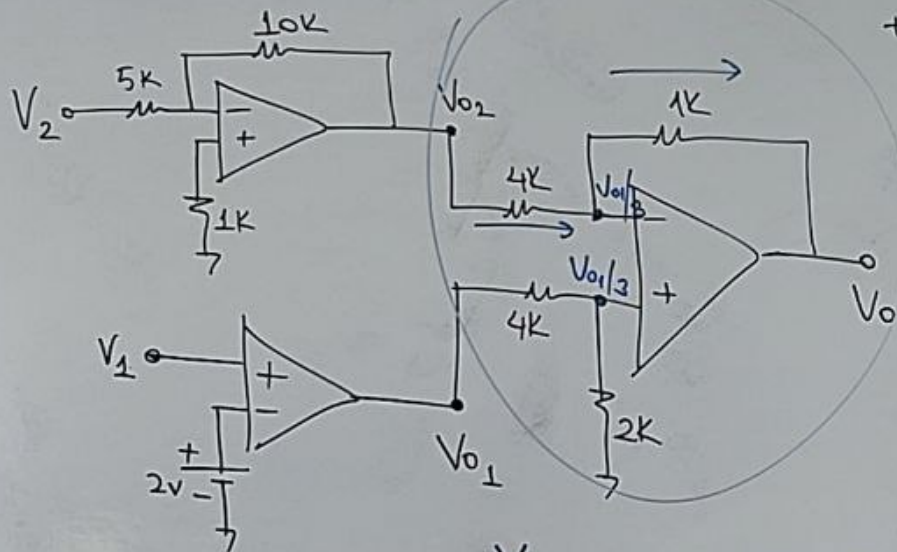
$$V_2 = 6V_1$$

$$V_2 = 2.25 \times 6$$

$$V_2 = 13.5$$

$$V_2 = +15V$$

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$$\pm V_{CC} = \pm 12V$$

$$T = 16ms$$

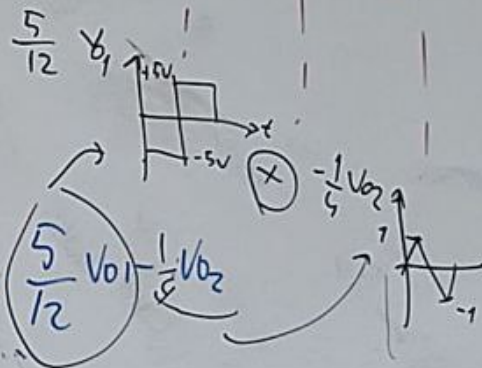
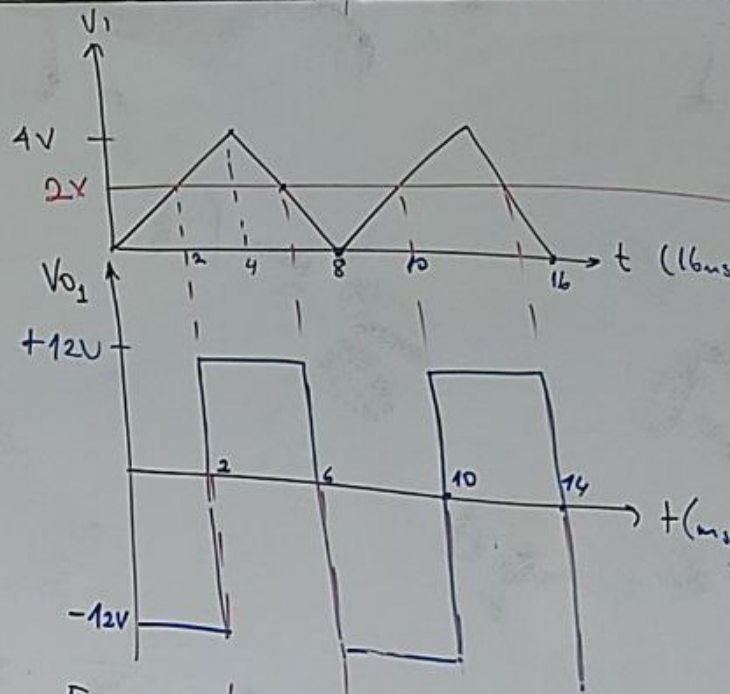
V_{01}, V_{02}, V_0 'ı özetli olarak çiz.

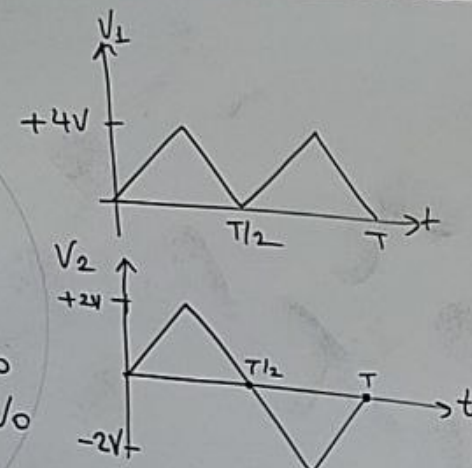
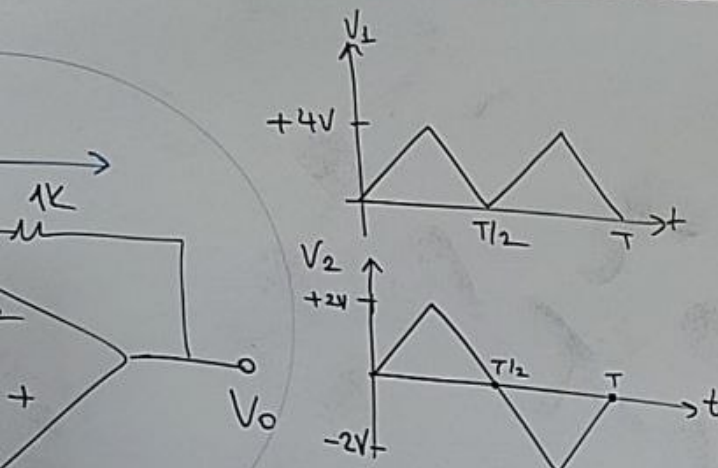
$$\frac{V_{01}}{6K} \cdot 2K = V_+$$

$$\frac{V_{02} - V_{01}/3}{4K} = \frac{V_{01}/3 - V_0}{1K}$$

$$V_{02} - \frac{V_{01}}{3} = \frac{4}{3}V_{01} - 4V_0$$

$$4V_0 = \frac{5}{3}V_{01} - V_{02} \Rightarrow V_0 = \frac{1}{4} \left(\frac{5}{3}V_{01} - V_{02} \right) = \left(\frac{5}{12}V_{01} - \frac{1}{4}V_{02} \right)$$





$\pm V_{cc} = \pm 12V$
 $T = 16ms$

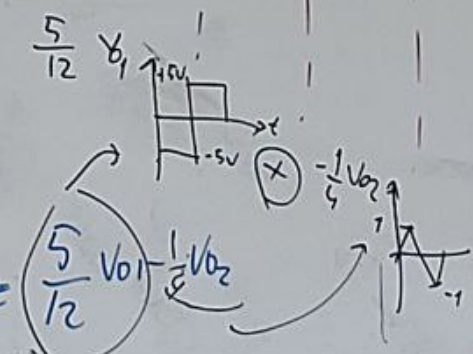
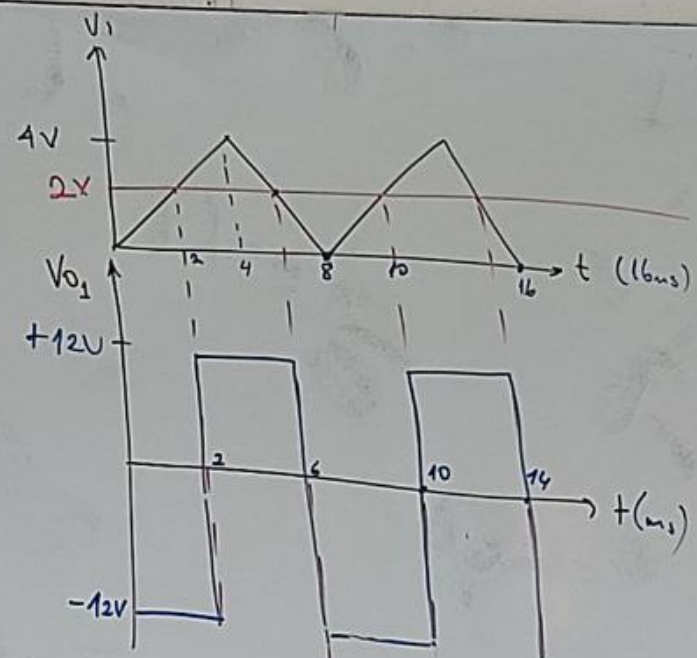
V_{01}, V_{02}, V_0 i özetli olarak çiz.

$$\frac{V_{02} - V_{01}/3}{4K} = \frac{V_{01}/3 - V_0}{1K}$$

$$V_{02} - \frac{V_{01}}{3} = \frac{4}{3}V_{01} - 4V_0$$

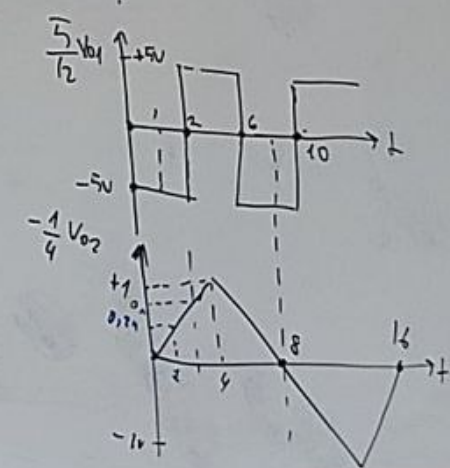
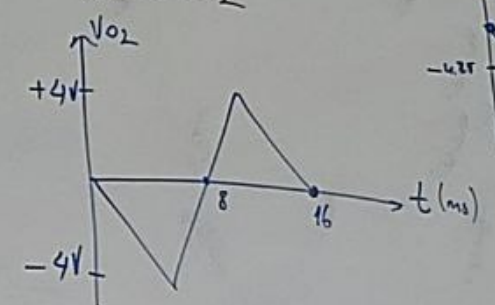
$$4V_0 = \frac{5}{3}V_{01} - V_{02}$$

$$V_0 = \frac{1}{4} \left(\frac{5}{3}V_{01} - V_{02} \right) = \left(\frac{5}{12}V_{01} - \frac{1}{4}V_{02} \right)$$

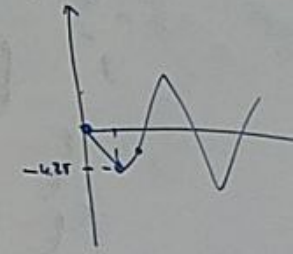


$$V_{02} = -V_2 \left(\frac{10K}{5K} \right)$$

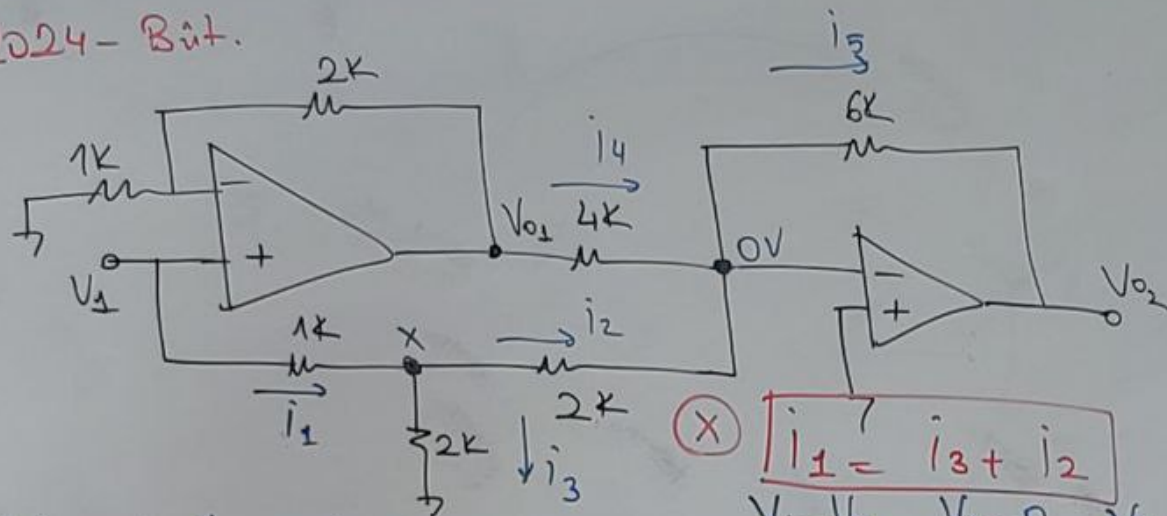
$$V_{02} = -2V_2$$



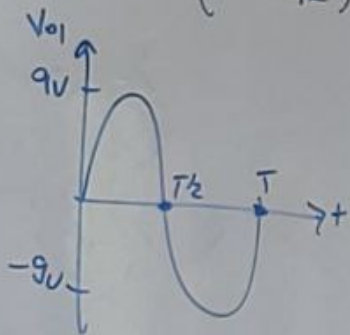
$$V_0 = \frac{5}{12}V_{01} - \frac{1}{4}V_{02}$$



2024 - Büt.



$$V_{01} = V_1 \cdot \left(1 + \frac{2k}{1k}\right) \Rightarrow \boxed{V_{01} = 3V_1}$$



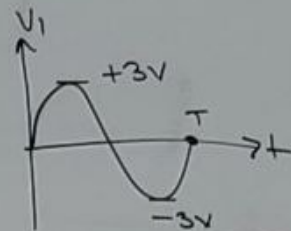
$$\boxed{i_1 = i_3 + i_2}$$

$$\frac{V_1 - V_x}{1k} = \frac{V_x - 0}{2k} + \frac{V_x - 0}{2k}$$

$$V_1 - V_x = V_x$$

$$\boxed{V_1 = 2V_x}$$

$$V_x = \frac{V_1}{2}$$



$$\pm V_{CC} = \pm 15V$$

V_{01}, V_{02} gain.

$$i_2 + i_4 = i_5$$

$$\frac{V_{01} - 0}{4k} + \frac{V_x - 0}{2k} = \frac{0 - V_{02}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1/2}{2k} = -\frac{V_{02}}{6k}$$

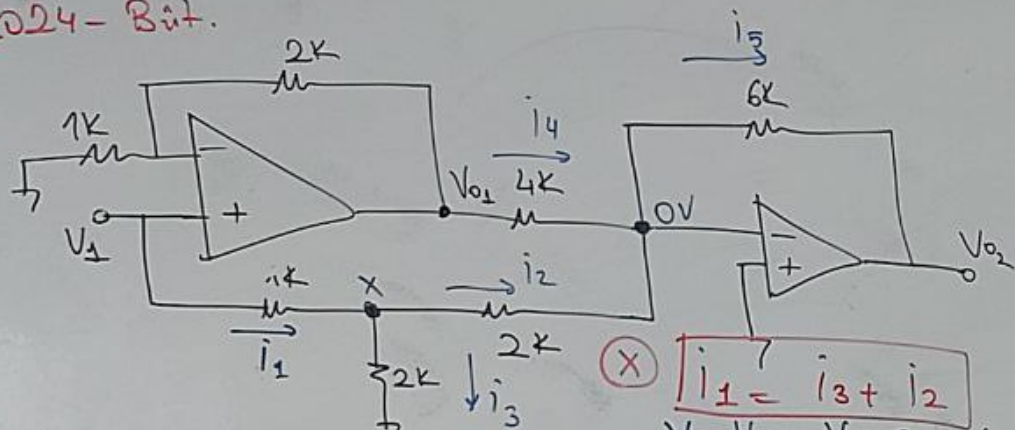
$$\frac{3V_1}{4k} + \frac{V_1}{4k} = -\frac{V_{02}}{6k}$$

$$\frac{4V_1}{4k} = -\frac{V_{02}}{6k}$$

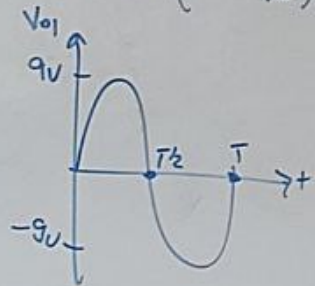
$$\boxed{V_{02} = -6V_1}$$



2024-Bit.



$$V_{01} = V_1 \cdot \left(1 + \frac{2k}{1k}\right) \Rightarrow \boxed{V_{01} = 3V_1}$$



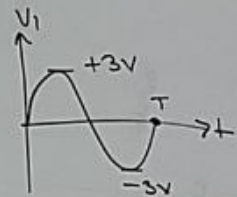
$$\textcircled{X} \quad i_1 = i_3 + i_2$$

$$\frac{V_1 - V_x}{1k} = \frac{V_x - 0}{2k} + \frac{V_x - 0}{2k}$$

$$V_1 - V_x = V_x$$

$$\boxed{V_1 = 2V_x}$$

$$V_x = \frac{V_1}{2}$$



$$\pm V_{CC} = \pm 15V$$

V_{01}, V_{02} gain.

$$i_2 + i_4 = i_5$$

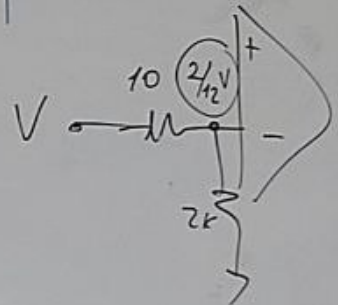
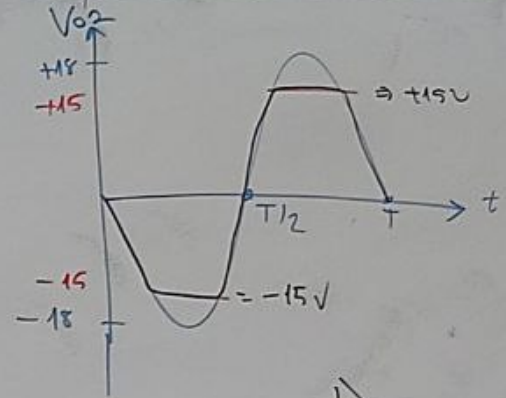
$$\frac{V_{01} - 0}{4k} + \frac{V_x - 0}{2k} = \frac{0 - V_{02}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1/2}{2k} = -\frac{V_{02}}{6k}$$

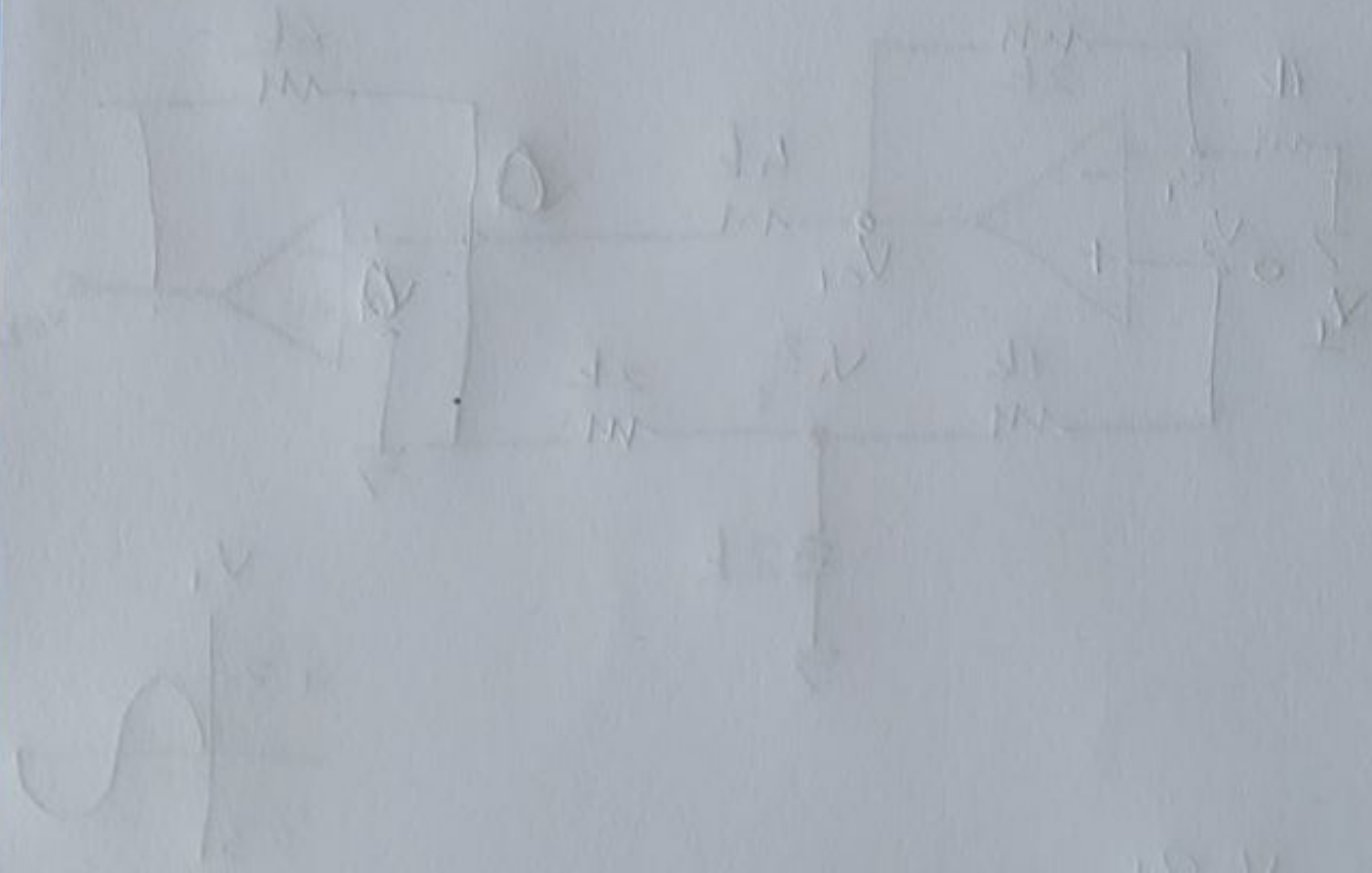
$$\frac{3V_1}{4k} + \frac{V_1}{4k} = -\frac{V_{02}}{6k}$$

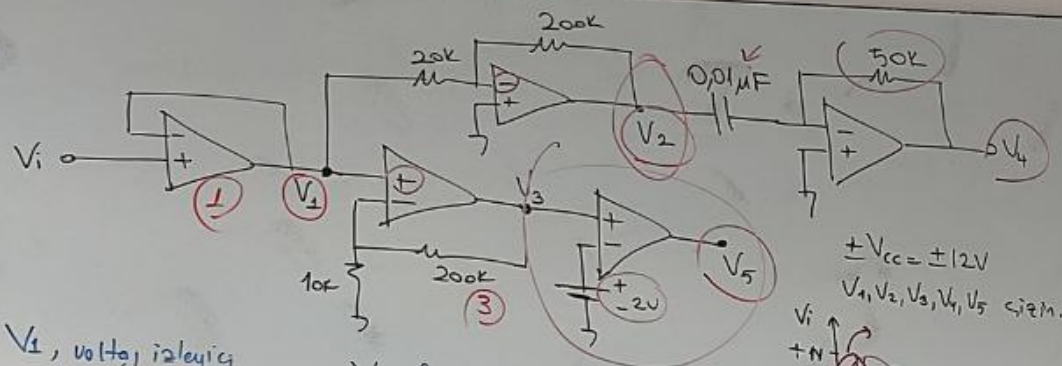
$$\frac{4V_1}{4k} = -\frac{V_{02}}{6k}$$

$$\boxed{V_{02} = -6V_1}$$



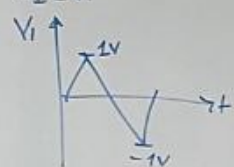
Akım tek bir uçureden iletliyorsa
gerilim bölücü yapabiliriz.





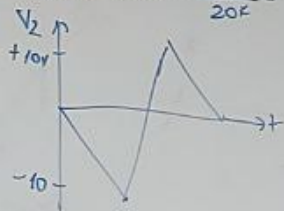
V_1 , voltaj izleyici

$$V_1 = V_i$$



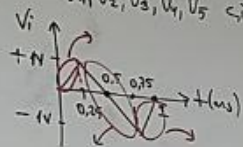
V_2 , faz çeviren kuv.

$$V_2 = -V_1 \cdot \frac{200K}{20K} = -10V_1$$



$$\pm V_{cc} = \pm 12V$$

V_1, V_2, V_3, V_4, V_5 çizim.

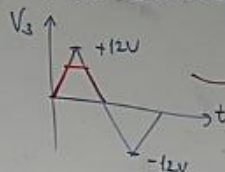


V_3 , faz çevirmeyen kuv.

$$V_3 = \left(1 + \frac{200K}{10K}\right) \cdot V_1$$

$$V_3 = 21V_1$$

$\pm V_{cc} = \pm 12V$ old. göre



V_5 , karşılaştırıcı



V_4 , türev alma

$$V_4(t) = -R \cdot C \cdot \frac{dV_3(t)}{dt} = -(50 \cdot 10^{-3}) \cdot \frac{dV_3}{dt}$$

$$0.2 < 0.25ms \Rightarrow V_4(t) = -40 \cdot 10^3 \cdot t$$

$$V_4(t) = +20$$

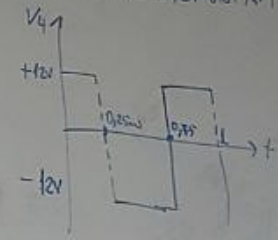
$$0.25ms < 0.75ms \Rightarrow V_4(t) = 40 \cdot 10^3 \cdot t - 20$$

$$V_4(t) = -20$$

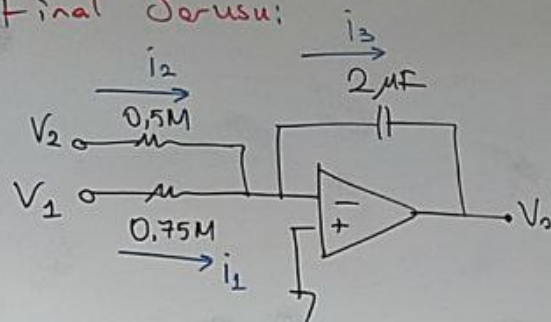
$$0.75ms < 1ms \Rightarrow V_4(t) = -40 \cdot 10^3 \cdot t + 40$$

$$V_4(t) = +20$$

$\pm V_{cc} = \pm 12V$ old. için



Final Sorusu:



$$\left. \begin{array}{l} V_1 = 10 \sin(200t) \\ V_2 = 15 \sin(200t) \\ \pm V_{cc} = \pm 15V \end{array} \right\} \Rightarrow V_0 \text{ 'i çizin.} \\ (V_0(t) \text{ 'yi elde ediniz})$$

$$i_1 + i_2 = i_3$$

$$\frac{V_2 - 0}{R_1} + \frac{V_2 - 0}{R_2} = \frac{0 - V_0}{X_c}$$

$$\text{NOT: } \sin(200t) = \cos(200t - 90^\circ) \\ = 1 \angle -90^\circ = -j$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} = \frac{-V_0}{1/j\omega c}$$

$$V_0 = -\frac{V_2}{j\omega c R_2} - \frac{V_1}{j\omega c R_1}$$

$$V_0 = -\frac{-j15}{j(200)(2 \cdot 10^{-6})(0.5 \cdot 10^6)} - \frac{-j10}{j(200)(2 \cdot 10^{-6})(0.75 \cdot 10^6)}$$

$$V_0 = \frac{3}{40} + \frac{1}{30}$$

$V_0(t)$ ise;

$$V_0(t) = \left(\frac{3}{40} + \frac{1}{30} \right) \cos(200t)$$